

Probing Axion-Like Particles with Black Holes, Galaxies, and Gravitational Waves

Thesis submitted for the degree of
Doctor of Philosophy (Sc.)

In

Physics

By

Pratick Sarkar

Registration No.: 2023 03 06 01 02 020



School of Physical Sciences

Indian Association for the Cultivation of Science

(Deemed to be University Under Section 3 of UGC Act, 1956)

2026

Declaration

I declare that the thesis entitled “Probing Axion-Like Particles with Black Holes, Galaxies, and Gravitational Waves”, submitted for the PhD degree, is the outcome of research work carried out by me at the **School of Physical Sciences**, Indian Association for the Cultivation of Science, Kolkata, India, under the supervision of **Professor Sourov Roy** and that neither any part of the thesis nor the whole of the thesis has been submitted to any University or Institution for obtaining any degree / diploma / academic award. I declare that during writing this thesis, I have adhered to the principles of academic integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission and the thesis does not contain any plagiarized contents. In keeping with the general practice in reporting scientific observations, due acknowledgements and citations have been made and copyright permission has been obtained (wherever applicable), whenever the work described based on the findings of other investigators. I shall be solely responsible for any dispute and plagiarism issue arising out of this doctoral thesis.

Date & Place:

May 29, 2026
Kolkata

Signature of the Candidate



PRATICK SARKAR

Registration Number: 2023 03 06 01 02 020

Certificate from the Supervisor

I hereby certify that the research work described in the thesis entitled “Probing Axion-Like Particles with Black Holes, Galaxies, and Gravitational Waves”, has been carried out by **Mr. Pratick Sarkar** (Ph.D. Registration No.: 2023 03 06 01 02 020) at the School of Physical Sciences, Indian Association for the Cultivation of Science, Kolkata, India, under my supervision and that neither any part of the thesis nor the whole of the thesis has been submitted to any University or Institution for obtaining any degree/diploma/academic award.

Sourov Roy 27.08.2026
Signature of the thesis supervisor with date & official seal

वरिष्ठ प्राध्यापक / Senior Professor
भौतिक विज्ञान शाखा (एस पी एस) / School of Physical Sciences (SPS)
इण्डियन एसोसिएशन फॉर द कल्चिवेशन ऑफ साइन्स
Indian Association for the Cultivation of Science
जदवपुर, कोलकाता - 700032 / Jadavpur, Kol - 700032, INDIA

Abstract

Axion-like particles (ALPs) are hypothetical pseudoscalar fields that arise in extensions of the Standard Model involving broken global symmetries, and also appear naturally in string-inspired frameworks. Owing to their weak couplings and broad mass range, ALPs are challenging to probe in laboratory experiments, making astrophysical systems powerful complementary laboratories. This thesis investigates how black holes, galaxies, and the stochastic gravitational-wave background can be used to probe ALP properties across multiple mass and coupling regimes.

Photon–ALP conversion in magnetized environments surrounding black holes is shown to occur efficiently when photons propagate along geodesics in strong gravitational fields. For supermassive black holes, this effect becomes significant for ALP masses $(1 - 100)$ neV and photon energies in the X-ray and gamma-ray bands, potentially leading to an observable dimming of the photon ring.

Apart from being a beyond SM particle, the ALP could be an excellent dark matter candidate. Assuming ALPs constitute the whole dark matter content of the observable Universe, their decay into two photons can generate narrow spectral features in the infrared, optical, and ultraviolet bands. Using multiwavelength observations of the central region of the M87 galaxy, robust constraints on the ALP–photon coupling are derived in the eV mass range, improving upon existing bounds in specific mass window.

Ultralight ALPs may also form fuzzy dark matter, characterized by solitonic cores at the centers of galaxies. In the presence of supermassive black holes, these cores modify the inspiral dynamics of black hole binaries through dynamical friction, leading to deviations from the canonical power-law form of the stochastic gravitational-wave background. By modeling these effects and confronting them with pulsar timing array observations, gravitational-wave data are shown to provide a sensitive probe of ultralight ALP phenomenology. Overall, this work demonstrates that a combination of electromagnetic and gravitational-wave observations offers a coherent and powerful framework for testing axion-like particle scenarios and exploring physics beyond the Standard Model.

*To my mother and
in memory of my father*

Acknowledgements

First and foremost, I would like to express my deepest gratitude to my supervisor, Prof. Sourov Roy. It has been an absolute privilege to work under his guidance over the past five years. I have learned immensely from his approach to scientific inquiry and have been continually inspired by his ideas and insights. There were numerous occasions when he could immediately sense that something was amiss in my work, often before any detailed calculation revealed it. This intuition for physics has taught me the importance of rigor and has encouraged me to consistently re-examine the foundations of my results.

Beyond the scientific mentorship, what truly made these years special was the environment of openness and enjoyment that accompanied our collaboration. Our discussions ranged freely from physics to broader reflections on life, and these conversations greatly enriched my experience as a researcher. I am also sincerely grateful for the many opportunities and encouragement he provided to travel for scientific purposes. His support made it possible for me to attend conferences, schools, and workshops in Spain, Portugal, and across India—including Delhi, Hyderabad, Bengaluru, Bhubaneswar, and Prayagraj—experiences that have been both professionally and personally formative. I remain deeply grateful to Professor Roy for his mentorship, encouragement, and unwavering support. These years have been an unforgettable and formative chapter of my life.

I would like to express my sincere gratitude to my collaborator, Professor Soumitra SenGupta. Working with him has been an intellectually enriching experience. His exceptional clarity of thought, deep physical intuition, and ability to view complex problems from a broader and more transparent perspective have greatly influenced my understanding of physics. I feel very fortunate to have had the opportunity to collaborate with him.

I am sincerely grateful to my collaborators, Dr. Arpan Kar and Dr. Subhadip Sau, for their invaluable guidance and insights. Through numerous in-depth discussions with Dr. Kar, I learned the importance of rigorous analysis, careful scrutiny of results, and precise scientific writing, which have profoundly shaped my approach to research. I also benefited greatly from my collaboration with Dr. Sau, whose expertise in gravitational physics and methodical approach to mathematical and numerical problem-solving significantly deepened my understanding of the subject. Their mentorship and collaborative support have played a crucial role in my intellectual development and growth as a researcher.

I would also like to express my sincere gratitude to my course instructors for their lectures, guidance, and insightful comments during departmental seminars, which

provided me with valuable perspectives and repeatedly encouraged me to reassess and refine my work. I am equally thankful to the other faculty members of the department for the many meaningful discussions and interactions throughout my Ph.D. years, which greatly contributed to my academic growth and overall development.

I would also like to thank the present and past members of the SPS office and the academic office for their assistance with various administrative matters throughout my doctoral studies. Finally, I am grateful to my past and present colleagues at IACS for their support and camaraderie in both academic and non-academic aspects. Last but not least, I wish to express my deepest gratitude to my mother, who has been a constant source of strength and support throughout all these years.

Contents

List of Publications	xxiii
Poster Presented	xxv
1 Introduction	1
1.1 The Search for Physics Beyond the Standard Model	1
1.2 Axions : A Brief Theoretical Overview	3
1.3 Motivation for Axion Like Particles	7
1.4 ALP as dark matter candidate	9
1.5 Astrophysical Systems as Laboratories for ALPs	15
1.5.1 Observational Signatures of ALPs through Oscillations and Decays	15
1.5.2 Signatures of Fuzzy ALP Dark Matter in the Gravitational Wave Background	18
1.6 Thesis Outline	20
2 Photon-ALP oscillations in Black Hole atmosphere	23
2.1 Photon-Axion Conversion Mechanism	24
2.1.1 Photon–Axion Conversion Parameters Associated with the SMBH	30
2.2 Photon orbiting-time around the photon sphere	32
2.3 Photons approaching the photon sphere	36
2.3.1 Relation between emission angle and impact parameter	36
2.3.2 Flow into photon sphere from spherical region	37
2.3.3 Sources of photons near galactic black holes	38
2.4 Conversion of photons into axions and relative luminosity	40
2.4.1 Dependence of the conversion probability and conversion fac- tor on spacetime metric	40

2.4.2	Spherically symmetric spacetime in brane-world scenario . . .	41
2.4.3	Dimming of the photon ring luminosity and observation of the axions	43
2.5	Conclusion	49
3	ALP Dark matter Decay signal from Messier 87 Galaxy	53
3.1	Dark Matter Profiles in M87 Galactic Halo	54
3.2	ALP DM decay signal	56
3.3	Observations and Data	57
3.4	Methodology & Results	59
3.4.1	Conservative approach	59
3.4.2	χ^2 fit constraints	61
3.5	Robustness on the results due to different DM halo profile	64
3.6	Systematic Uncertainty	65
3.6.1	Background modeling	65
3.6.2	Detector calibration	66
3.7	Conclusion	66
4	Exploring Self-coupling of Fuzzy ALP DM via GWB	69
4.1	ULDM Soliton around central SMBH	71
4.1.1	Scaling relation	73
4.1.2	Density Profiles	75
4.2	DM influence on Gravitational Wave Background	76
4.3	Results	81
4.3.1	The GW spectrum	81
4.3.2	Typical values of self coupling $\hat{\lambda}$	82
4.3.3	Allowed values of λ	82
4.3.4	Relation between the GW amplitude and the γ_{GW} parameter .	85
4.4	Discussion and Conclusion	86
5	Conclusions and Outlook	89

A	The Effective Field theory for the ALP	111
A.1	ALP-Photon Effective Couplings	113
A.2	Axion decay into two photons	115
B	Numerically produced solutions of Gross-Pitaevskii-Poisson equation	117
B.1	Amount of scaling s	118
B.2	Environmental degeneracy	119
B.3	SMBH binary merger rate	120

List of Figures

1.1	Summary plot of current experimental, astrophysical, and cosmological constraints on axions and ALPs.	8
1.2	Constraints and viable regions in ALP mass–coupling parameter space.	14
1.3	Axion–photon interaction diagrams. Left: effective $a\gamma\gamma$ vertex. Right: Conversion of a photon into an axion in an external magnetic field. .	16
2.1	Direction of magnetic field has been shown in the x - z plane making an angle Θ with the propagation vector of the photon.	26
2.2	Dependence of the conversion probability (without the sinusoidal part) on various parameters. The photon frequency and electron density are varied in the ranges $(1\text{--}10^9)$ eV and $(1\text{--}10^8)$ cm^{-3} , respectively. The magnetic field is fixed at $ \mathbf{B} = 30$ Gauss and the axion–photon coupling at $g_{\Phi\gamma} = 10^{-11}$ GeV^{-1}	32
2.3	The blue curve represents a light ray emitting from a point p_e toward the photon sphere. The point p_e is positioned on a sphere with a radius labeled as r_e and it is centred around the black hole within a spherically symmetric coordinate system. The angle Υ_e is the zenith angle.	36
2.4	Photons coming from a spherical region in between r_{in} and r_{out} towards the black hole photon sphere which is shown in red circle. . . .	37
2.5	Variation of the conversion factor (CF) with $M\Delta_M$ and the extra-dimensional parameter β for efficient conversion, corresponding to the condition $2\Delta_M = \Delta_{osc}$	43
2.6	Relative luminosity of photons before (dashed curves) and after (solid curves) photon–axion conversion. The magnetic field associated with the black hole is taken to be $ \mathbf{B} = 30$ Gauss, the axion–photon coupling $g_{\Phi\gamma} = 10^{-11}$ GeV^{-1} , the electron number density $n_e = 10^4$ cm^{-3} , and the black hole mass $M = 6.2 \times 10^9 M_\odot$. The plots are shown for two different axion mass values.	45

2.7	Relative luminosity spectra of axions. The magnetic field strength is taken to be $ \mathbf{B} = 30$ Gauss, the axion-photon coupling $g_{\Phi\gamma} = 10^{-11} \text{ GeV}^{-1}$, the electron number density $n_e = 10^4 \text{ cm}^{-3}$, and the black-hole mass $M = 6.2 \times 10^9 M_\odot$. Results are shown for two different axion masses.	46
2.8	The plot describes the expected spectral luminosity of axions arising from photons in proximity to the photon sphere, considering different axion masses within the framework of a Schwarzschild black hole. . .	47
2.9	The plot describes the low mass astrophysical bounds on the axion-photon coupling from different observations.	48
2.10	The red plot illustrates the relationship between the charge parameter and the required resolution for observing the dimming effect in a spherically symmetric black hole.	49
3.1	Dark matter density profiles of the M87 galaxy adopted in this chapter.	55
3.2	The observed broadband SED of M87, measured during the April 2018 Event Horizon Telescope (EHT) campaign using various instruments.	58
3.3	Constraints on the axion photon coupling $g_{a\gamma\gamma}$ considering the conservative approach.	60
3.4	Upper-limit on $g_{a\gamma\gamma}$ obtained (at 95% C.L.) considering a background model.	62
3.5	Fitted model to IR-Opt-UV data of the 2018(EHT) observation and corresponding constraints due to this fitted model.	63
3.6	The plot illustrates the variation of the constraints with different assumptions for the DM halo profile of M87.	64
4.1	Numerical solutions to the GPP equation are presented for a halo mass of $10^{13} M_\odot$ and ULDM masses of 10^{-21} eV (Left) and $2 \times 10^{-21} \text{ eV}$ (Right).	74
4.2	This figure illustrates the dimensionless function $C(kr, \beta)$, which enters the expression for the dynamical friction force experienced by a test object of mass μ moving with relative velocity v_{rel} through the ULDM.	78
4.3	The GW strains corresponding to different DM density profiles, characterized by distinct self-coupling parameters, are displayed for an SMBH binary with a mass of $10^{8.5} M_\odot$	80
4.4	The GW strains corresponding to different DM density profiles, characterized by distinct self-coupling parameters, are displayed with the SMBH mass of $10^9 M_\odot$ of the binary.	81
4.5	Qualitative limits on the parameter space $(m, \hat{\lambda})$ and the estimated values of the λ parameter are shown along with the dimensionless $\hat{\lambda}$.	84

4.6	The amplitude of the GW strain with change of γ_{GW} parameter. . . .	85
A.1	Axion–photon interaction mediated by an electron loop.	114
A.2	Some of the one-loop contributions to the $a \rightarrow \gamma\gamma$ process mediated by fermions, charged gauge bosons, goldstone bosons, and self-energy insertions.	115
B.1	Solutions of the coupled, dimensionless Gross-Pitaevskii-Poisson equation.	117
B.2	Amount of scaling with different values of $\hat{\lambda}$ parameter for a fixed halo mass.	119

List of Tables

3.1	This table summarizes the observed data for infrared, optical and ultraviolet emissions from the central region of the M87 galaxy. . . .	57
3.2	The table presents the upper bounds on the photon-ALP coupling $g_{a\gamma\gamma}$ across various mass bins, derived using three distinct background models.	65
B.1	Values of $\hat{a}$ chosen to produce density profiles of halo mass $10^{13}M_{\odot}$ without the λ parameter, along with the dimensionless soliton mass $\hat{M}$ for different ULDM masses.	118

List of Publications

Publications/Preprints included in this thesis:

1. S. Roy, P. Sarkar, S. Sau, S. SenGupta,
“Exploring axions through the photon ring of a spherically symmetric black hole,”
Journal of Cosmology and Astroparticle Physics **JCAP** 11 (2023) 099.
[arXiv:2310.05908 \[gr-qc\]](#).

Presented in Chapter 2.
2. A. Kar, S. Roy, P. Sarkar,
“Constraining eV-scale axion-like particle dark matter: insights from the M87 Galaxy,”
Journal of Cosmology and Astroparticle Physics **JCAP** 05 (2025) 100.
[arXiv:2501.01860 \[hep-ph\]](#).

Presented in Chapter 3.
3. P. Sarkar,
“Exploring Ultralight Dark Matter Self-Coupling via the Gravitational Wave Background,”
J. Phys. G: Nucl. Part. Phys. 53 (2026) 085201
[arXiv:2504.19505 \[gr-qc\]](#).

Presented in Chapter 4.

Publications/Preprints not included in this thesis:

1. T. Poddar, S. Roy, P. Sarkar,
“Photon-dark photon oscillation in M87 and Crab Nebula environments,”
[arXiv:2601.15985 \[hep-ph\]](#). (*under review*)
2. A. Gupta, P. Majumdar, S. Roy, P. Sarkar,
“Constraints on high-frequency gravitational waves from graviton-photon conversion in the M87 galaxy,” *Physical Review D* 114, 043038(2026)
[arXiv:2604.01290 \[hep-ph\]](#).

Poster Presented

During the course of his PhD, the author presented the following posters at international and national conferences:

- **Light Dark World 2025 (LDW2025)**, Instituto de Física Teórica (IFT), Madrid, Spain. Poster titled “*Constraining eV -scale Axion Like Particle Dark Matter: Insights from M87 Galaxy.*” (Also accompanied by a one-minute lightning talk.)
- **International Conference on High Energy Physics and Astrophysics (ICHEPAP 2023)**, Saha Institute of Nuclear Physics (SINP), Kolkata, India. Poster titled “*Exploring Axions through the Photon Ring of a Spherically Symmetric Black Hole.*”
- **SANGAM @ HRI, 2026**, Harish Chandra Research Institute. Poster titled “*Ultralight dark matter self coupling via gravitational wave background.*”

Chapter 1

Introduction

*Where the mind is without fear and the head is held high...
Into that heaven of freedom, my Father, let my country awake.*

Rabindranath Tagore, Poem (35) of *Gitanjali*(1912)

1.1 The Search for Physics Beyond the Standard Model

The dawn of the twentieth century witnessed the rise of particle physics, a field that fundamentally transformed our perception of the underlying structure of matter and the Universe. The properties of known matter and forces at the fundamental level are described with remarkable precision by the well-established Standard Model (SM) of particle physics[1, 2, 3]. The SM is a relativistic quantum field theory that provides a unified description of the strong, weak, and electromagnetic interactions among elementary particles. Its theoretical structure is based on the local gauge symmetry $SU(3)_C \times SU(2)_L \times U(1)_Y$, under which quarks and leptons are organized into chiral representations.

The dynamics of the strong interaction are governed by quantum chromodynamics (QCD), a non-Abelian gauge theory describing the interactions of color-charged quarks and gluons [4, 5, 6]. The electroweak sector unifies the weak and electromagnetic forces and is realized through spontaneous symmetry breaking via the Higgs mechanism [7, 8, 9], which generates masses for the $W^\pm$ and Z bosons as well as for fermions through Yukawa interactions. The experimental discovery of the Higgs boson [10, 11] confirmed this mechanism and completed the particle content predicted by the SM.

In addition to its gauge structure, the Standard Model is fully specified by its particle content. The fermionic sector consists of three generations of quarks and leptons. The quarks (u, d) , (c, s) , and (t, b) carry color charge and participate in all three fundamental interactions, while the leptons (e, ν_e) , (μ, ν_μ) , and (τ, ν_τ) are color-neutral and interact via the electroweak force. These fermions are arranged into left-handed weak doublets and right-handed singlets, reflecting the chiral nature

of the weak interaction [12].

The gauge sector contains twelve spin-1 bosons associated with the generators of the gauge group: eight gluons corresponding to $SU(3)_C$, the weak bosons $W^\pm$ and Z , and the photon γ . The scalar sector consists of a single complex Higgs doublet, whose vacuum expectation value breaks $SU(2)_L \times U(1)_Y$ down to $U(1)_{\text{em}}$, thereby generating masses for gauge bosons and fermions while leaving the photon massless.

Together, these ingredients form a self-consistent, renormalizable, and highly predictive framework [13, 14] that has been tested extensively across a wide range of energies and experimental settings. Precision measurements at collider experiments have confirmed its predictions to an extraordinary degree of accuracy, establishing the SM as one of the most successful theories in the history of physics.

Despite its extraordinary success, however, the Standard Model leaves several fundamental questions unanswered. It does not incorporate gravity in a quantum-consistent manner, and thus cannot be regarded as a complete theory of all fundamental interactions.

It also fails to provide a viable candidate for dark matter, whose existence is strongly supported by a wide range of astrophysical and cosmological observations, including the flat rotation curves of spiral galaxies [15, 16], the anisotropies of the cosmic microwave background (CMB) [17], the abundances of light elements predicted by Big Bang nucleosynthesis (BBN) [18, 19], and direct evidence for the separation of dark and baryonic matter in merging galaxy clusters [20, 21]. Furthermore, the observed matter–antimatter asymmetry of the Universe cannot be explained within the SM [22], as the amount of CP violation present is insufficient to account for the baryon asymmetry [23, 24].

Another important shortcoming arises in the context of neutrino physics. In its minimal formulation, the SM predicts massless neutrinos; however, the observation of neutrino oscillations [25, 26] implies that neutrinos possess small but nonzero masses, requiring an extension of the model. In addition, the SM offers no explanation for the hierarchical pattern of fermion masses and mixing angles, which are introduced as free parameters.

Of particular relevance to this thesis is the so-called strong CP problem. The QCD Lagrangian admits a CP-violating term proportional to the parameter θ [14], yet experimental constraints on the neutron electric dipole moment [27] require $\theta \lesssim 10^{-10}$, an unnaturally small value. The absence of observable strong CP violation therefore calls for a dynamical explanation.

Among the theoretical extensions proposed to address these shortcomings, new light bosonic degrees of freedom such as axions and axion-like particles (ALPs) occupy a prominent role. The Peccei–Quinn mechanism [28, 29] provides an elegant solution to the strong CP problem by promoting the θ parameter to a dynamical field, whose associated pseudo-Nambu–Goldstone boson is the QCD axion [30, 31]. More generally, ALPs arise in a wide class of theories beyond the Standard Model, including string-theoretic compactifications [32, 33] that predict a rich spectrum of light pseudoscalar fields.

Owing to their extremely weak couplings to Standard Model particles, such light

fields are challenging to probe in terrestrial experiments. Astrophysical and cosmological environments, however, offer unique opportunities to detect or constrain them. Extreme magnetic fields, strong gravitational backgrounds, long propagation distances, and plasmas make astrophysical systems natural laboratories for exploring new degrees of freedom beyond the SM.

This thesis aims to exploit such environments to study a broad range of phenomena associated with axions, axion-like particles, and related dark-sector fields, thereby contributing to the ongoing search for physics beyond the Standard Model.

1.2 Axions : A Brief Theoretical Overview

The formulation of QCD in the mid 70s led to a profound revision of earlier views on its symmetry structure. In the limit of vanishing quark masses, the QCD Lagrangian for N flavors possesses a large global symmetry $U(N)_V \times U(N)_A$. For the light quarks u and d , whose masses satisfy $m_{u,d} \ll \Lambda_{\text{QCD}}$, this symmetry is expected to be approximately realized in nature (at least upto 2 quark flavors). Experimentally, the vector subgroup $U(2)_V \simeq SU(2)_I \times U(1)_B$ is indeed a good approximate symmetry, as reflected in the observed hadron spectrum.

However, the axial sector behaves differently. The formation of quark condensates, $\langle \bar{u}u \rangle \sim \langle \bar{d}d \rangle \neq 0$, spontaneously breaks the axial symmetry, leading to the appearance of pseudo-Nambu-Goldstone bosons. While the pions can be identified with (approximate) Goldstone modes of $SU(2)_A$, the absence of an additional light pseudoscalar state associated with the $U(1)_A$ symmetry—despite symmetry expectations—constitutes the so-called $U(1)_A$ problem [34]. While the resolution of the long-standing $U(1)_A$ problem through the discovery of instantons [35] and the non-trivial Yang-Mills vacuum [36, 37] explained the absence of an additional light pseudoscalar meson, it simultaneously revealed a new puzzle: the strong CP problem [28, 30, 31, 38, 39, 40].

The resolution of this $U(1)_A$ puzzle lies in the quantum structure of the theory. Although the classical QCD Lagrangian appears invariant under axial $U(1)_A$ transformations, the associated current J_5^μ is not conserved due to the chiral anomaly at the quantum level [41, 42]. Its divergence is given by

$$\partial_\mu J_5^\mu = \frac{g_s^2 N_f}{32\pi^2} G_{\mu\nu}^a \tilde{G}^{a\mu\nu}, \quad (1.1)$$

where g_s is the strong coupling constant, N_f denotes the number of active quark flavours, $G_{\mu\nu}^a$ is the gluon field-strength tensor associated with the $SU(3)_C$ gauge field, and $\tilde{G}^{a\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\alpha\beta}G_{\alpha\beta}^a$ is its dual.

The pseudoscalar density appearing in the anomaly can be expressed as a total derivative [43],

$$G_{\mu\nu}^a \tilde{G}^{a\mu\nu} = \partial_\mu K^\mu, \quad (1.2)$$

where K^μ is the Chern-Simons current,

$$K^\mu = \epsilon^{\mu\alpha\beta\gamma} \left(A_\alpha^a G_{\beta\gamma}^a - \frac{g_s}{3} f^{abc} A_\alpha^a A_\beta^b A_\gamma^c \right), \quad (1.3)$$

with A_α^a denoting the gluon gauge field and f^{abc} the totally antisymmetric structure constants of the $SU(3)_C$ gauge group.

At first sight, since this is a total derivative, one might expect it to have no physical effect. However, this conclusion depends crucially on boundary conditions. While naive boundary conditions would render the surface term vanishing, the correct requirement is that gauge fields approach a pure gauge configuration at spatial infinity [44, 45]. Under these conditions, gauge field configurations can be classified by an integer winding number n , corresponding to mappings between topologically non-trivial vacuum sectors [46].

The true QCD vacuum is therefore not a single configuration but a superposition of these sectors, forming the so-called θ -vacuum,

$$|\theta\rangle = \sum_n e^{-in\theta} |n\rangle. \quad (1.4)$$

Transitions between different sectors are mediated by instantons and are characterized by the topological charge,

$$\nu = \frac{g_s^2}{32\pi^2} \int d^4x G_{\mu\nu}^a \tilde{G}^{a\mu\nu}. \quad (1.5)$$

This non-trivial vacuum structure leads to the presence of an additional term in the QCD Lagrangian,

$$\mathcal{L}_{\text{QCD}} \supset -\frac{\alpha_s}{8\pi} \theta G_{\mu\nu}^a \tilde{G}^{a\mu\nu}, \quad (1.6)$$

which violates P and T symmetries and thus the CP symmetry.

Furthermore, when electroweak interactions are taken into account, chiral rotations of the quark fields shift the parameter θ . As a result, the physically observable quantity is

$$\bar{\theta} = \theta + \arg \det M_u M_d, \quad (1.7)$$

where $M_{u,d}$ are the quark mass matrices. Experimental limits on the neutron electric dipole moment constrain $\bar{\theta} \lesssim 10^{-10}$, leading to the strong CP problem: why is this parameter so small despite no symmetry enforcing it?

Several solutions have been proposed, including the hypothesis of a massless up quark, but these possibilities have been conclusively ruled out by modern lattice QCD calculations [47, 48, 49]. This tension between theoretical allowance and experimental absence of strong CP violation provides the primary theoretical motivation for the Peccei–Quinn mechanism and the associated axion field.

Among the proposed solutions, the most compelling is the introduction of an additional $U(1)$ global chiral symmetry, the Peccei–Quinn (PQ) symmetry [28]. This symmetry is spontaneously broken at a high energy scale f_A , giving rise to a pseudo Nambu–Goldstone boson, the axion field $A(x)$ ^a.

The axion inherits a shift symmetry,

$$A(x) \rightarrow A(x) + \text{const.}, \quad (1.8)$$

^aThroughout this thesis, we follow the convention adopted in the introduction section, denoting the QCD axion by A and the more general axion-like particle by a , unless otherwise specified.

which is explicitly broken by its anomalous coupling to gluons,

$$\mathcal{L} \supset -\frac{\alpha_s}{8\pi} \frac{A}{f_A} G_{\mu\nu}^a \tilde{G}^{a\mu\nu}. \quad (1.9)$$

A crucial consequence is that the $\bar{\theta}$ parameter becomes dynamical. Through a field redefinition,

$$A + \bar{\theta} f_A \rightarrow A, \quad (1.10)$$

the CP-violating term is absorbed into the axion field. Non-perturbative QCD effects subsequently generate a non-trivial potential^b for the shifted axion field, which is minimized at vanishing vacuum expectation value, $\langle A \rangle = 0$, thus eliminating strong CP violation. Fluctuations around this minimum correspond to a physical massive excitation—the axion—with mass [30, 50]

$$m_A = \frac{m_\pi f_\pi}{f_A} \frac{\sqrt{z}}{1+z} \simeq 6 \text{ meV} \times \left(\frac{10^9 \text{ GeV}}{f_A} \right), \quad (1.11)$$

where $m_\pi = 135 \text{ MeV}$ is the pion mass, $f_\pi \simeq 92 \text{ MeV}$ is the pion decay constant, $z = m_u/m_d \simeq 0.56$ denotes the ratio of up and down-quark masses, and f_A is the axion decay constant. For sufficiently large f_A , the axion is an ultralight particle with extremely weak interactions with the Standard Model [51, 52, 49, 53]. Axions possess a universal and phenomenologically most relevant interaction with photons [54, 55, 56, 57],

$$\mathcal{L} \supset -\frac{1}{4} g_{A\gamma} A F_{\mu\nu} \tilde{F}^{\mu\nu}, \quad (1.12)$$

where the effective axion–photon coupling is given by

$$g_{A\gamma} \sim \frac{\alpha}{2\pi f_A} \sim \frac{\alpha}{2\pi} \frac{m_A}{m_\pi f_\pi} \sim 10^{-12} \text{ GeV}^{-1} \left(\frac{10^8 \text{ GeV}}{f_A} \right) \sim 10^{-12} \text{ GeV}^{-1} \left(\frac{m_A}{6 \text{ meV}} \right). \quad (1.13)$$

This effective coupling is appropriate for axions/ALPs with sufficiently light masses and is applicable when considering photons at relatively low energies. Its form follows from the full effective Lagrangian, which is discussed in Appendix A.

The smallness of this coupling follows from astrophysical constraints, which—most notably those inferred from the observed duration of the neutrino signal from SN 1987A—require a large axion decay constant f_A , or equivalently a small axion mass m_A [58],

$$f_A \gtrsim 4 \times 10^8 \text{ GeV} \implies m_A \lesssim 16 \text{ meV}. \quad (1.14)$$

Measurements of the CMB polarization by the BICEP/Keck experiments place an upper limit on the tensor-to-scalar ratio, thereby constraining the inflationary Hubble scale and, indirectly, the axion decay constant [59]. A major motivation

^bFor the QCD axion, the nonperturbative QCD effects generate a periodic potential of the form $V(A) = \Lambda_{\text{QCD}}^4 \left[1 - \cos \left(\bar{\theta} + \frac{A}{f_A} \right) \right]$. Here, Λ_{QCD} denotes the characteristic QCD confinement scale, typically of order $\sim 200\text{--}300 \text{ MeV}$. Minimization of the potential yields the vacuum expectation value $\langle A \rangle = -f_A \bar{\theta}$, so that, after shifting the axion field about the minimum, the effective CP-violating term $\propto \left(\bar{\theta} + \frac{\langle A \rangle}{f_A} \right) G_{\mu\nu}^a \tilde{G}^{a\mu\nu}$ vanishes dynamically, thereby resolving the strong CP problem.

for the QCD axions is that it provides a compelling cold dark matter candidate, produced non-thermally through the vacuum-misalignment mechanism in the early Universe [38, 39, 40, 60, 61]. Within the standard cosmological scenario, this motivates the approximate parameter range

$$10^9 \text{ GeV} \lesssim f_A \lesssim 10^{13} \text{ GeV} \implies 10^{-7} \text{ eV} \lesssim m_A \lesssim 1 \text{ meV}. \quad (1.15)$$

More stringent cosmological constraints arise by requiring that the axion relic abundance produced through the vacuum-misalignment mechanism does not exceed the observed dark matter density, leading to [62, 63, 64]

$$f_A \lesssim (10^{11}\text{--}10^{12}) \text{ GeV}, \quad m_A \sim (6\text{--}60) \mu\text{eV}. \quad (1.16)$$

These bounds are not universal and depend on the assumed cosmological history, the dominant axion production mechanism, the initial misalignment angle, and whether the Peccei-Quinn symmetry is broken before or after inflation.

The inflationary history introduces additional constraints through primordial isocurvature fluctuations. Current CMB observations constrain the tensor-to-scalar ratio $r < 0.032$ (95% C.L.), corresponding to an upper limit on the inflationary Hubble scale, $H_I \lesssim 2.5 \times 10^{13} \text{ GeV}$ [59]. If axions constitute the dominant dark matter component and no mechanism suppresses isocurvature perturbations, these bounds constrain the pre-inflationary PQ-breaking scenario, particularly for large inflationary scales [65, 66, 67, 68, 69, 70, 71, 72]. Nevertheless, pre-inflationary PQ symmetry breaking remains a viable possibility for sufficiently low inflationary scales, a suitably small initial misalignment angle, or additional mechanisms that suppress isocurvature perturbations.

In the post-inflationary PQ-breaking scenario, the PQ symmetry is restored after inflation, requiring $f_A < \frac{H_I}{2\pi}$, which, together with the current on H_I , yields

$$f_A \lesssim 2 \times 10^{13} \text{ GeV}, \quad m_A \gtrsim 0.3 \mu\text{eV}. \quad (1.17)$$

In addition to cosmological bounds on the axion decay constant, a variety of laboratory and astrophysical experiments place direct constraints on the axion–photon coupling and axion mass. Solar axion searches with the CERN Axion Solar Telescope (CAST) exclude couplings above

$$g_{A\gamma} \gtrsim 6.6 \times 10^{-11} \text{ GeV}^{-1} \quad \text{for } m_A \lesssim 0.02 \text{ eV}, \quad (1.18)$$

as the absence of a signal in the 9T magnet data rules out stronger couplings in this mass range [73, 74, 75]. Next-generation helioscopes such as the International Axion Observatory (IAXO) are expected to improve sensitivity by roughly an order of magnitude in $g_{A\gamma}$ [76].

Haloscope experiments searching for dark matter axions through microwave cavity resonances, most notably the Axion Dark Matter eXperiment (ADMX), exclude axion models in limited mass windows (e.g. near $m_A \sim \mathcal{O}(1\text{--}10) \mu\text{eV}$) with sensitivities reaching couplings as low as $g_{A\gamma} \sim 10^{-15} \text{ GeV}^{-1}$ under the assumption that axions constitute the local dark matter halo [77, 78, 79, 80, 81].

Additional haloscope efforts such as the CAST-CAPP microwave searches have reported exclusions of axion–photon couplings down to

$$g_{A\gamma} \lesssim 8 \times 10^{-14} \text{ GeV}^{-1}, \quad (1.19)$$

in specific narrow mass ranges around 20–25 μeV [82].

1.3 Motivation for Axion Like Particles

Beyond the QCD axion, there exists strong theoretical motivation for the presence of additional axion-like particles (ALPs) (see, e.g., [83, 84, 85]). Such states arise naturally as pseudo–Nambu–Goldstone bosons associated with the spontaneous breaking of global symmetries, including lepton number [86, 87] or family symmetries [88, 89, 90]. Owing to an (approximate) shift symmetry, these particles can remain naturally light compared to the electroweak or even the QCD scale, making them compelling candidates for new physics beyond the Standard Model.

In particular, low-energy effective field theories emerging from string theory generically predict not only the QCD axion, but also a multitude of additional ALPs. Upon compactification of the extra spatial dimensions, the Kaluza–Klein zero modes of antisymmetric tensor fields—present in the massless spectrum of the bosonic string in ten dimensions—manifest themselves as axion and ALP degrees of freedom in four dimensions [91, 92, 93, 94, 95, 96], with their multiplicity determined by the topology of the compactified manifold. Furthermore, ALP candidates may also arise as pseudo–Nambu–Goldstone bosons associated with accidental global $U(1)$ symmetries that emerge at low energies from exact discrete gauge symmetries in string compactifications [97, 98, 99].

Intriguingly, very light and weakly coupled ALPs a_i , with decay constants comparable to that of the axion, share an important cosmological property with the axion itself: they can constitute cold dark matter produced via the vacuum-realignment mechanism [101, 102, 103, 104]. In the strongly favored post-inflationary symmetry-breaking scenario, their relic abundance is approximately given by

$$\Omega_{a_i} h^2 \simeq 0.053 \left(\frac{m_{a_i}}{\text{eV}} \right)^{1/2} \left(\frac{f_{a_i}}{10^{11} \text{ GeV}} \right)^2. \quad (1.20)$$

These considerations provide strong motivation to search not only for the QCD axion A , but also for additional light ALPs a_i . At low energies, their interactions with photons can be described by the effective Lagrangian

$$\mathcal{L} \supset \frac{1}{2} \partial_\mu a_i \partial^\mu a_i - \frac{1}{2} m_{a_i}^2 a_i^2 - \frac{g_{a_i\gamma\gamma}}{4} a_i F_{\mu\nu} \tilde{F}^{\mu\nu}. \quad (1.21)$$

Unlike the QCD axion—whose mass m_A and photon coupling $g_{A\gamma}$ are determined by non-perturbative QCD effects associated with chiral symmetry breaking—the masses m_{a_i} and photon couplings $g_{a_i\gamma\gamma}$ of generic ALPs are model dependent. As a result, ALPs may be hierarchically more strongly coupled to photons than axions of comparable mass, rendering them potentially easier to detect experimentally.

From a phenomenological standpoint, a broad array of probes—including low-energy precision observables, cosmological considerations, and dedicated laboratory

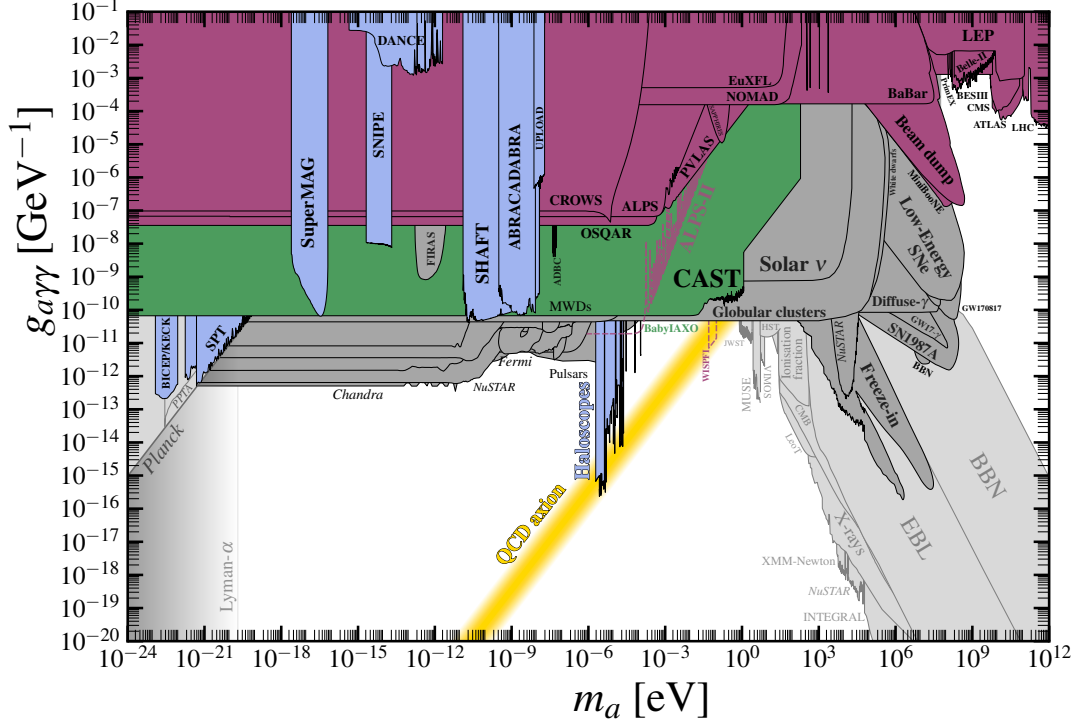


Figure 1.1: Summary of current experimental, astrophysical, and cosmological constraints on axions and axion-like particles (ALPs) in the axion mass (m_a) vs axion-photon coupling ($g_{a\gamma\gamma}$) parameter space. Shaded regions indicate parameter space excluded or probed by laboratory experiments, astrophysical observations, and cosmological considerations, including helioscopes, haloscopes, stellar cooling, supernovae, and early-Universe bounds. Also shown are well-motivated theoretical regions, such as the QCD axion band, illustrating the broad range of masses and couplings accessible to axions and ALPs. The figure is reproduced from the public website [100].

searches—already constrain substantial regions of the ALP parameter space, particularly in terms of their masses and couplings to photons and electrons [105, 106, 107, 108]. Among these, helioscope experiments play a central role in testing light and weakly coupled ALPs.

Looking ahead, next-generation helioscopes such as the International Axion Observatory (IAXO) [76] are expected to significantly extend experimental sensitivity to ALPs. This reach will be complemented by dedicated fixed-target and beam-dump searches, including the NA64 fixed-target experiment at CERN [109], collider-based ALP searches at Belle II [110] and BESIII [111], constraints on ALP couplings to electrons and photons from the ArgoNeuT and MiniBooNE beam-dump facilities [112], and searches for ALPs via ultraperipheral heavy-ion collisions [113], which together are expected to extend experimental sensitivity to ALPs. In addition, collider-based searches provide complementary coverage of heavier ALP scenarios and further broaden the reach of the ALP discovery program [114, 115, 116].

The diverse phenomenology of axions and axion-like particles is conveniently summarized in the plane spanned by their mass and effective coupling to photons, shown in Fig. [1.1]. The allowed parameter space for ALPs spans many orders of magnitude in mass and coupling strength.

Owing to their weak interactions and broad possible mass range, axions and ALPs are constrained by a wide variety of laboratory experiments, astrophysical observations, and cosmological probes. Stellar evolution, supernova cooling, and solar observations impose stringent bounds at low masses and moderate couplings, while laboratory searches such as helioscopes, haloscopes, and light-shining-through-wall experiments probe complementary regions of parameter space. Cosmological considerations further restrict both very light and very heavy axions, depending on their production mechanisms. Together, these constraints delineate the remaining viable parameter space and highlight the strong motivation for exploring new environments and observational channels, particularly those involving astrophysical sources and propagation effects. Nevertheless, large regions of parameter space remain untested, particularly in mass regimes where astrophysical processes offer competitive sensitivity.

1.4 ALP as dark matter candidate

The purpose of this section is to review the cosmological production of ALPs and to explain how they can account for the present dark matter (DM) abundance. We begin by recalling the basic results from early-Universe cosmology and thermodynamics. We then focus on the misalignment mechanism, which provides a robust and model-independent source of cold axion dark matter.

On large scales the Universe is well described as homogeneous and isotropic, and its evolution is governed by Einstein’s field equations

$$G_{\mu\nu} = 8\pi G_N T_{\mu\nu}, \quad (1.22)$$

where $G_N = 1/m_{\text{Pl}}^2$ defines the Planck mass $m_{\text{Pl}} = 1.22 \times 10^{19} \text{ GeV}$. For a spatially flat Universe, the line element is given by the Friedmann–Lemaître–Robertson–

Walker (FLRW) metric

$$ds^2 = dt^2 - R^2(t) (dx^2 + dy^2 + dz^2), \quad (1.23)$$

where $R(t)$ is the scale factor.

Assuming a perfect fluid with energy density ρ and pressure P , the Friedmann equations read

$$H^2 = \left(\frac{\dot{R}}{R} \right)^2 = \frac{8\pi}{3m_{\text{Pl}}^2} \rho, \quad (1.24)$$

$$\frac{\ddot{R}}{R} = -\frac{4\pi}{3m_{\text{Pl}}^2} (\rho + 3P), \quad (1.25)$$

where H is the Hubble expansion rate. Combining these equations yields energy conservation,

$$\dot{\rho} + 3H(\rho + P) = 0. \quad (1.26)$$

The Universe is dominated by a fluid with equation of state $P = \omega \rho$, for which $H \propto R^{-3(\omega+1)}$. During radiation domination ($\omega = 1/3$), the energy and entropy densities are

$$\rho = \frac{\pi^2}{30} g_*(T) T^4, \quad s = \frac{2\pi^2}{45} g_{*s}(T) T^3, \quad (1.27)$$

The quantities g_* and g_{*s} represent the effective numbers of relativistic degrees of freedom associated with the energy density and entropy density of the Universe, respectively, following the definitions given in Ref. [117]. In this regime the Hubble rate is

$$H(T) \simeq 1.66 g_*^{1/2}(T) \frac{T^2}{m_{\text{Pl}}}. \quad (1.28)$$

The misalignment mechanism is based on the assumption that light scalar fields in the early Universe acquire random initial values. Such initial conditions may arise, for example, from quantum fluctuations during inflation. Due to the expansion of the Universe, these initial field values become frozen as long as the Hubble expansion rate exceeds the mass scale of the field. A scalar field with mass m_a evolves on a characteristic timescale $t \sim m_a^{-1}$. Once this timescale is reached, the field begins to dynamically respond by rolling toward the minimum of its potential and performing coherent oscillations. The mechanism is termed “misalignment” because the initial value of the axion field is generically displaced, or misaligned, from the true minimum of its potential, with the degree of this misalignment setting the final dark matter abundance.

To describe this mechanism quantitatively, consider a real scalar field a governed by the Lagrangian [101, 104],

$$\mathcal{L} = \frac{1}{2} \partial_\mu a \partial^\mu a - \frac{1}{2} m_a^2 a^2 + \mathcal{L}_I, \quad (1.29)$$

where $\mathcal{L}_I$ encodes interactions with other particles in the primordial plasma. We assume the Universe underwent a period of inflation during which the Hubble expansion rate $H = d \ln R / dt$ satisfy $H \gg m_a$. As a consequence, after inflation the

scalar field is approximately homogeneous, and its initial configuration can be characterized by a single value a_i . After inflation, the Universe reheats and subsequently enters a radiation-dominated phase. The cosmological evolution of the ALP field in an expanding FLRW Universe is governed by the equation of motion

$$\ddot{a} + 3H\dot{a} + m_a^2 a = 0. \quad (1.30)$$

In general, the scalar mass may receive thermal corrections through $\mathcal{L}_I$, so $m_a = m_a(t)$ should be understood as time-dependent.

The evolution of the field can be separated into two regimes. At early times, when $3H \gg m_a$, the system is overdamped and the field remains effectively frozen, $\dot{a} \simeq 0$. At a later time t_1 , defined by

$$3H(t_1) = m_a(t_1) \equiv m_1, \quad (1.31)$$

the damping becomes undercritical and the field begins to oscillate about the minimum of its potential.

In the oscillatory regime, the mass term dominates the dynamics and the solution can be obtained in the WKB approximation as

$$a(t) \simeq a_1 \left(\frac{m_1 R_1^3}{m_a R^3} \right)^{1/2} \cos \left(\int_{t_1}^t m_a(t') dt' \right), \quad (1.32)$$

where $a_1 \simeq a_i$, since the field evolution was frozen prior to t_1 . This expression describes rapid oscillations with a slowly varying amplitude.

Defining the amplitude

$$A(t) = a_1 \left(\frac{m_1 R_1^3}{m_a R^3} \right)^{1/2}, \quad (1.33)$$

the energy density of the scalar field is given by^c

$$\rho_a = \frac{1}{2}\dot{a}^2 + \frac{1}{2}m_a^2 a^2 \simeq \frac{1}{2}m_a^2 A^2, \quad (1.34)$$

where terms involving derivatives of $A(t)$ have been neglected, since $m_a \gg H$ in this regime.

The pressure of the field is

$$P_a = \frac{1}{2}\dot{a}^2 - \frac{1}{2}m_a^2 a^2. \quad (1.35)$$

Averaging over many oscillations, whose timescale is much shorter than the cosmological timescale, yields

$$\langle P_a \rangle \simeq 0, \quad (1.36)$$

^cFor the ALP field, since $\dot{a} \sim HA \ll m_a A$, one has $\dot{a} = -Am_a \sin(\int_{t_1}^t m_a(t') dt')$. The corresponding energy density then becomes $\rho_a \simeq \frac{1}{2}A^2 m_a^2 \sin^2(\int_{t_1}^t m_a(t') dt') + \frac{1}{2}A^2 m_a^2 \cos^2(\int_{t_1}^t m_a(t') dt') \sim \frac{1}{2}m_a^2 A^2$.

corresponding to an equation of state

$$w = \frac{\langle P_a \rangle}{\langle \rho_a \rangle} \simeq 0, \quad (1.37)$$

which is identical to that of non-relativistic matter.

If the scalar mass varies with time, the comoving energy density $\rho_a R^3$ is not conserved. However, the quantity

$$N \equiv \frac{\rho_a R^3}{m_a} = \frac{1}{2} m_1 R_1^3 a_1^2, \quad (1.38)$$

remains constant and can be interpreted as the comoving number density of non-relativistic scalar quanta.

The present-day energy density of the scalar field is therefore

$$\rho_{a,0} = m_0 \frac{N}{R_0^3} \simeq \frac{1}{2} m_0 m_1 a_1^2 \left(\frac{R_1}{R_0} \right)^3. \quad (1.39)$$

The ratio of scale factors can be rewritten in terms of temperatures by using the conservation of comoving entropy,

$$s R^3 = \frac{2\pi^2}{45} g_{*S}(T) T^3 R^3 = \text{const.}, \quad (1.40)$$

which implies

$$\left(\frac{R_1}{R_0} \right)^3 = \frac{g_{*S}(T_0) T_0^3}{g_{*S}(T_1) T_1^3}. \quad (1.41)$$

The temperature T_1 at the onset of oscillations is determined by the condition $3H(T_1) = m_1$. Using the expression for the Hubble parameter during radiation domination given earlier,

$$H(T) = 1.66 \sqrt{g_*(T)} \frac{T^2}{m_{\text{Pl}}}, \quad (1.42)$$

one finds

$$T_1 = \left(\frac{m_1 m_{\text{Pl}}}{4.98 \sqrt{g_*(T_1)}} \right)^{1/2}. \quad (1.43)$$

Substituting these relations into [Eq. \(1.39\)](#) and expressing the result in conventional units yields the present-day dark matter density

$$\rho_{a,0} \simeq 0.17 \frac{\text{KeV}}{\text{cm}^3} \sqrt{\frac{m_0}{\text{eV}}} \sqrt{\frac{m_0}{m_1}} \left(\frac{a_1}{10^{11} \text{ GeV}} \right)^2 \left(\frac{g_*(T_1)}{3.36} \right)^{3/4} \left(\frac{g_{*S}(T_1)}{3.91} \right)^{-1}. \quad (1.44)$$

Comparing this result with the observed cold dark matter density,

$$\rho_{\text{CDM}} = (1.17 \pm 0.06) \frac{\text{KeV}}{\text{cm}^3}, \quad (1.45)$$

shows that large initial field amplitudes a_1 are required unless the mass at the onset of oscillations satisfies $m_1 \ll m_0$.

To ensure that the condensate behaves as standard cold dark matter, the field must begin oscillating no later than matter–radiation equality, at temperature $T_{\text{eq}} \simeq 1.3 \text{ eV}$. This implies the bound

$$m_1 > 3H(T_{\text{eq}}) \simeq 1.8 \times 10^{-27} \text{ eV}. \quad (1.46)$$

The ALP-photon coupling can be written as

$$g_{a\gamma\gamma} = \frac{\alpha}{2\pi} \frac{C_{a\gamma\gamma}}{f_a}, \quad (1.47)$$

where $C_{a\gamma\gamma}$ denotes the model-dependent Wilson coefficient. Using $a_1 = \theta_1 f_a$, the allowed ALP dark matter regions in the m_a – $g_{a\gamma\gamma}$ plane can be expressed in terms of the initial misalignment angle $\theta_1 \in [-\pi, \pi]$.

Further constraints arise from inflationary isocurvature perturbations, the formation of topological defects such as strings and domain walls, and possible enhancements from string decay. These effects must be treated on a model-by-model basis and determine the viability of ALPs as dark matter candidates across the parameter space.

As discussed above, a general constraint on ALP dark matter arises from the requirement that the ALP zero mode^d both accounts for the observed dark matter abundance and begins oscillating sufficiently early, such that its mass at matter–radiation equality exceeds the Hubble expansion rate. This condition is expressed by the bound in Eq. (1.46). For pNGB ALPs, the restriction applies: the field excursion is bounded by $a_1 \leq \pi f_a$, which in turn is related to the ALP–photon coupling.

Combining these requirements yields a constraint on the viable ALP parameter space. The dashed purple line in Fig. [1.2], labeled “ $m_a(T_{\text{eq}}) > 3H(T_{\text{eq}})$ ”, corresponds to this general bound evaluated for $a_1 \leq f_a$ and $C_{a\gamma\gamma} = 1$. Any viable ALP model must satisfy this condition in order for its zero mode to behave as cold dark matter prior to matter–radiation equality. Models that attempt to saturate this bound typically encounter difficulties, either in fitting cosmic microwave background (CMB) observations or in maintaining the validity of the WKB approximation employed in the analysis. Consequently, this bound should be regarded as conservative.

Notably, for $C_{a\gamma\gamma} \sim \mathcal{O}(1)$, the bound appears to exclude ultralight ALP scenarios proposed to explain the anomalously low opacity of the Universe to $\mathcal{O}(\text{TeV})$ gamma rays through photon–ALP oscillations in astrophysical magnetic fields. Such scenarios generally require couplings $g \sim 10^{-11} \text{ GeV}^{-1}$ and masses $m_\phi \lesssim \text{neV}$ (see Ref. [118] and references therein). Reconciling these models with the ALP simultaneously constituting the dark matter would necessitate either large values of $C_{a\gamma\gamma} \gg 1$, which remain conceivable, or a significant fine tuning of the initial misalignment angle θ_1 .

^dThe axion zero mode refers to the spatially homogeneous component of the ALP field through misalignment mechanism. As the $H \lesssim m_a$, the coherent classical field begins oscillating around the minimum of its potential and behaves effectively as cold dark matter.

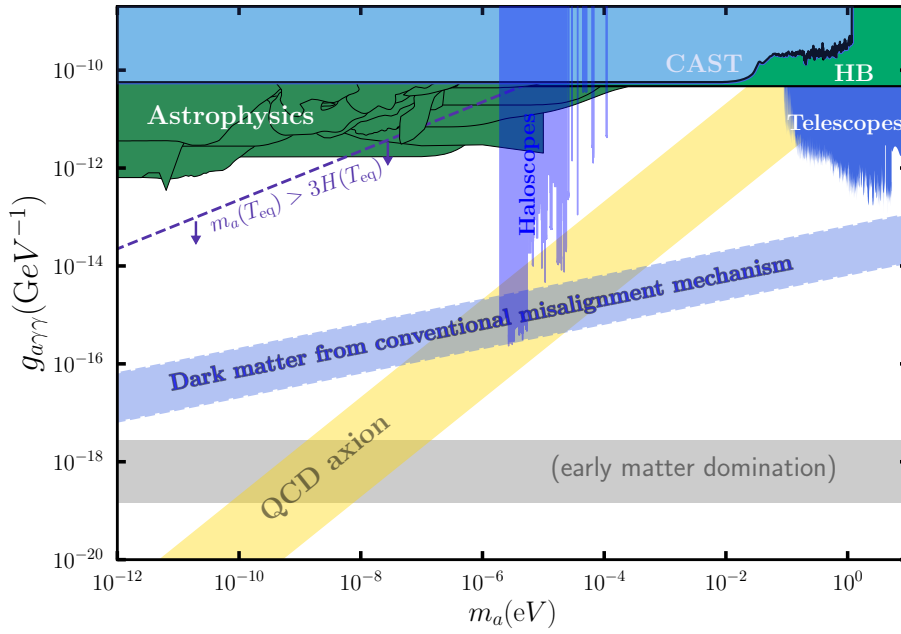


Figure 1.2: Constraints and viable regions in the axion-like particle (ALP) mass-coupling parameter space. Shown are the general consistency bound $m_a(T_{\text{eq}}) > 3H(T_{\text{eq}})$ (purple), the standard misalignment dark matter region (blue), the QCD axion band (yellow), and the early matter-dominated dark matter region (gray). Shaded areas indicate existing experimental, astrophysical, and observational constraints, including those from telescope-based measurements relevant to this thesis. The figure is redrawn using data from the `AxionLimits` website[100].

The blue band represents the standard prediction for ALP dark matter produced via the conventional misalignment mechanism labeled “Dark matter from conventional misalignment mechanism”, as given in Eq. (1.44) and Eq. (1.47) with an initial misalignment angle $\theta_i = 1$. The finite width of this band reflects uncertainties associated with model-dependent coefficients, in particular the Wilson coefficient $C_{a\gamma\gamma}$, which is varied within the range 1 to 10. The yellow band corresponds to the QCD axion ($C_{a\gamma\gamma} = E/N - 1.92$) and is defined here by allowing the ratio of electromagnetic to color anomaly coefficients, E/N , to vary over the range of viable models [119], namely $5/3 \leq E/N \leq 44/3$.

A period of early matter domination (EMD) alters the standard calculation of the ALP relic abundance by modifying the cosmological expansion history. In this scenario, the total energy density is dominated by a long-lived massive field, prior to reheating. The decay of this field injects entropy into the thermal bath, diluting the ALP energy density and thereby relaxing the tuning required in the initial misalignment angle for large decay constants f_a . As a result, EMD opens up regions of the ALP parameter space that are not accessible in the standard radiation-dominated case.

During the EMD epoch, the ratio $\rho_a/(3M_{\text{Pl}}^2 H^2)$ remains constant, allowing the ALP to constitute the observed dark matter abundance for significantly larger values of f_a . For reheating temperatures $T_{\text{RH}} \gtrsim 10$ MeV, the present-day ALP relic density is approximately

$$\Omega_a h^2 \simeq 0.12 \left(\frac{f_a \theta_1}{9 \times 10^{14} \text{ GeV}} \right)^2 \left(\frac{T_{\text{RH}}}{10 \text{ MeV}} \right), \quad (1.48)$$

valid provided that oscillations occur prior to reheating.

The corresponding region of parameter space consistent with the observed dark matter abundance in the EMD scenario is shown as the gray band in the mass-coupling plane.

In Fig. [1.2], we also display the parameter regions excluded by existing experiments and astrophysical observations. Constraints from the CAST experiment [74, 75] are shown as navy shaded regions, while limits derived from horizontal branch (HB) stars [120] are indicated in green shading. The bounds coming from study of different astrophysical systems are presented in green. Laboratory bounds from Haloscopes are presented in dark blue [100]. We also indicate constraints derived from astronomical observations using telescopes such as JWST and MUSE, shown in blue, as this parameter region is particularly relevant for the analyses presented in third chapter of this thesis.

1.5 Astrophysical Systems as Laboratories for ALPs

1.5.1 Observational Signatures of ALPs through Oscillations and Decays

This thesis contributes to the investigation of ALPs as candidates for physics beyond the SM and as viable constituents of dark matter. The search is pursued through a

range of astrophysical environments, including black holes, galactic halos, and the stochastic gravitational-wave background (SGWB), where extreme conditions offer enhanced sensitivity to weakly interacting light degrees of freedom.

ALPs can possess effective couplings to photons, fermions, gluons, and massive gauge bosons. The complete effective Lagrangian is presented in Appendix A. In the present thesis, the focus is restricted to the mass and energy regimes in which the phenomenology of ALPs is predominantly governed by the axion–photon interaction,

$$\mathcal{L} \supset -\frac{1}{4}g_{a\gamma\gamma} a F_{\mu\nu} \tilde{F}^{\mu\nu}. \quad (1.49)$$

This interaction gives rise to two distinct physical phenomena through the same $a\gamma\gamma$ vertex: (i) photon–ALP oscillations in the presence of an external electromagnetic field, where the external field effectively provides one of the two photon legs of the vertex, and (ii) radiative ALP decay, $a \rightarrow \gamma\gamma$, in vacuum, where both photon legs are real (on-shell). Conversely, in the presence of a strong external electromagnetic field, photons can convert into ALPs through the same interaction vertex, as illustrated in Fig. [1.3].

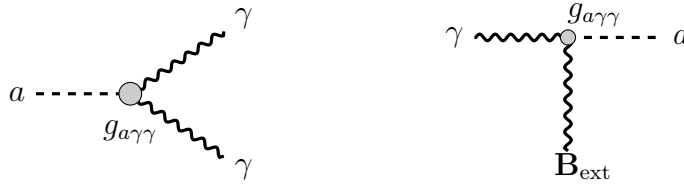


Figure 1.3: Axion–photon interaction diagrams. Left: effective $a\gamma\gamma$ vertex. Right: Conversion of a photon into an axion in an external magnetic field.

Since their theoretical conception in the twentieth century, black holes have developed into exceptional astrophysical laboratories for probing fundamental physics in the strong-gravity regime. A major observational breakthrough was achieved in 2019 with the Event Horizon Telescope’s first image of a black hole shadow, firmly establishing black holes as precision tools for testing strong-field gravity. In parallel, rapid progress in relativistic magnetohydrodynamic simulations and multi-messenger observations has significantly strengthened our ability to interpret near-horizon phenomena and their potential sensitivity to new particles.

Strong gravitational fields, such as those present in the immediate vicinity of a black hole horizon, are expected to encode detailed information about the underlying spacetime geometry. Until recently, direct observational access to this regime was unavailable. This situation changed dramatically with the imaging of the shadow of the supermassive black hole at the center of the M87 galaxy, followed by polarization-resolved observations at 230 GHz that revealed synchrotron emission on event-horizon scales [121, 122, 123, 124, 125, 126, 127]. These developments provide unprecedented opportunities to study both gravitational and plasma effects in extreme environments.

In black hole environments, photons emitted by surrounding galactic or accretion-flow sources are gravitationally drawn toward the photon sphere, where they may

orbit the black hole one or multiple times before either escaping to infinity or plunging into the event horizon. For a distant observer, these strongly lensed trajectories produce a bright ring encircling the black hole shadow. When magnetic fields are present, photon–ALP conversion can occur along these paths, potentially reducing the photon luminosity. Such conversions may manifest observationally as a measurable dimming of the photon ring [128], offering a novel avenue to probe ALPs using high-resolution observations of black hole shadows. Luminosity dimming induced by photon–ALP conversion has similarly been explored in other astrophysical contexts, including the proposed dimming of distant Type Ia supernovae by photon–axion oscillations in intergalactic magnetic fields [129, 130, 131, 132], lending further motivation to the black-hole photon-ring scenario considered here. This study was generalized to arbitrary spherically symmetric spacetimes describing supermassive black holes and demonstrated an observational avenue to distinguish between specific scenarios—such as whether M87* is described by a Schwarzschild, Reissner–Nordström, or higher-dimensional black hole (e.g., braneworld models)—through characteristic dimming signatures in the photon rings.

The signatures of dark-matter ALPs can be probed through observations of dark-matter–dominated astrophysical environments, particularly galactic halos [133, 134, 135, 136, 137]. Owing to their coupling to photons, ALPs can decay into two photons via the axion–photon interaction, provided that the decay rate is sufficiently suppressed to ensure stability on cosmological time scales, as required for dark matter. While such decays are extremely rare, the enormous dark matter content of galactic halos renders them promising targets for indirect searches.

Astronomical observations offer a powerful avenue for probing this scenario. In regions of high dark matter density, the decay of ALP dark matter would give rise to a narrow, monochromatic emission line at an energy corresponding to half the particle mass. Such spectral features can be distinguished from conventional astrophysical backgrounds using high-resolution spectroscopy. Over the past decade, dedicated searches for these decay signatures have been carried out across a wide range of frequencies, extending from radio wavelengths to X-ray energies, thereby placing constraints on ALP parameter space over many orders of magnitude [138, 139, 140, 141, 142, 143, 144, 145].

For ALP masses in the electronvolt range, the resulting decay photons fall within the infrared, optical, and ultraviolet bands. This has motivated extensive spectroscopic studies in the optical and near-infrared regimes, which have yielded competitive limits on the ALP–photon coupling as a function of the particle mass [146, 147, 148, 149, 150, 151, 152, 153, 154, 155]. Only recently have similar searches been extended into the ultraviolet domain, where observational coverage had previously been limited [156, 157].

Potential line emission from the decay of eV-scale ALP dark matter has been investigated using optical and ultraviolet spectroscopic data, drawing on observations from multiple facilities, including the Neil Gehrels Swift Observatory, whose Ultraviolet/Optical Telescope provides long-term coverage of optical and UV emissions, as well as complementary data from ground-based and space-based telescopes operating in the near-infrared, optical, and ultraviolet bands [158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168]. This analysis focused on the central region of the

Messier 87 galaxy, a dark-matter-rich environment that has been the subject of extensive multi-wavelength observational campaigns [169, 170]. The absence of narrow emission features in the relevant spectral bands allows constraints to be placed on the ALP-photon coupling by limiting the contribution from two-photon-ALP decay in this region.

1.5.2 Signatures of Fuzzy ALP Dark Matter in the Gravitational Wave Background

Compelling observational evidence from galaxy rotation curves, gravitational lensing, and measurements of the CMB provides strong support for the existence of DM as a dominant component of the Universe’s matter content [16, 171, 172, 173, 174, 137]. These observations collectively indicate that cosmic structures are embedded in a non-relativistic, non-baryonic matter component that interacts primarily through gravity and does not emit electromagnetic radiation [17]. A widely adopted framework assumes that dark matter consists of collisionless, non-relativistic particles—bosonic or fermionic—collectively referred to as Cold Dark Matter (CDM) [17]. This paradigm successfully reproduces observations on large cosmological scales, of order hundreds of megaparsecs. However, discrepancies arise on galactic and sub-galactic scales, where certain observations appear to deviate from CDM predictions.

A prominent example is the core-cusp problem, wherein collisionless CDM simulations predict steep, cuspy density profiles at the centers of galaxies, while observations favor shallow, cored distributions [175, 176, 177, 178]. The absence of unambiguous detection signals, together with these small-scale tensions, has motivated the exploration of alternative dark matter models that retain the successes of CDM on large scales while addressing its limitations at smaller scales.

Among the alternatives to Λ CDM, such as warm dark matter (WDM) [179, 180] and self-interacting dark matter (SIDM) [181, 182, 183], ultralight fuzzy dark matter (FDM) has attracted particular attention due to the emergence of quantum-mechanical effects on astrophysical scales arising from the particle’s extremely small mass [184, 185, 186, 187, 188, 189]. These effects are characterized by a macroscopic de Broglie wavelength,

$$\lambda_{\text{dB}} = \frac{h}{mv}, \quad (1.50)$$

where m denotes the particle mass and v its characteristic velocity. For $m_a \sim 10^{-22}$ eV and $v \sim 10^{-3}c$, the de Broglie wavelength can reach kiloparsec scales, allowing FDM particles to exhibit wave-like behavior across galactic distances [190].

Ultralight FDM with mass $m_a \sim 10^{-22} - 10^{-21}$ eV, which exhibits wave nature on galactic scales, has been examined as the third research direction of this thesis. This wave behavior introduces an effective quantum pressure that counteracts gravitational collapse, naturally suppressing the formation of cuspy density profiles [184, 191, 192, 193]. As a result, ULDM halos can develop stable, dense solitonic cores at their centers [134, 194]. These solitons are less centrally concentrated than CDM cusps, offering a compelling explanation for observed cored density profiles while remaining consistent with large-scale structure constraints.

A natural candidate for FDM is an ALP with mass $m_a \sim 10^{-22}$ eV. The low-energy effective action can be written as, $S = \int d^4x \sqrt{-g} \left[\frac{1}{2} (\partial_\mu a)^2 - \mu^4 \left(1 - \cos\left(\frac{a}{F}\right) \right) \right]$, where F is the symmetry breaking scale and μ is generated by nonperturbative instanton effects [134]. The ALP mass is then, $m_a = \frac{\mu^2}{F}$. The constant F is constrained by particle-physics considerations to lie within [134],

$$10^{16} \text{ GeV} \lesssim F \lesssim M_{\text{Pl}} \simeq 2.4 \times 10^{18} \text{ GeV}. \quad (1.51)$$

This range is satisfied in many string-theoretic constructions, including heterotic and type-II compactifications, and is well suited for FDM phenomenology.

In the early Universe, the ALP field is initially frozen by Hubble friction. When the Hubble parameter drops below the mass scale, $H \sim m_a$, the field begins coherent oscillations. These oscillations correspond to a Bose–Einstein condensate of ultra-light particles with negligible momentum, whose energy density redshifts as R^{-3} , exactly like cold dark matter. Assuming an initial misalignment angle $\theta_i = \mathcal{O}(1)$, the present-day ALP dark matter abundance is [134]

$$\Omega_a \sim 0.1 \left(\frac{F}{10^{17} \text{ GeV}} \right)^2 \left(\frac{m_a}{10^{-22} \text{ eV}} \right)^{1/2}, \quad (1.52)$$

demonstrating that ALPs in the above parameter range naturally reproduce the observed dark matter density without fine-tuning. Consequently, ALP FDM behaves as an exceptionally cold form of dark matter while exhibiting wave-like effects on galactic scales.

Given that galactic centers host supermassive black holes (SMBHs), their gravitational influence can significantly impact the formation and stability of ULDM solitonic cores. The dynamics of ULDM in such environments are described by a scalar field minimally coupled to gravity, governed by the coupled Gross–Pitaevskii–Poisson equations [195, 196, 197, 198]. These equations, formulated within a weak-field, slowly varying, non-relativistic framework, provide the theoretical basis for modeling solitonic dark matter halos in the presence of black holes.

The stochastic gravitational wave background (SGWB) is an incoherent superposition of gravitational wave signals from countless unresolved astrophysical and cosmological sources, forming a diffuse gravitational wave foreground that encodes information about the population and environment of its emitters. Its detection and characterization therefore offers a unique observational window into both the large-scale structure of the Universe and the physical processes operating in the vicinity of compact objects such as supermassive black hole binaries. To investigate the properties of FDM in the vicinity of SMBHs, we utilize measurements of the SGWB obtained from pulsar timing array (PTA) experiments. These observations provide a powerful probe of the population and dynamical evolution of inspiraling SMBH binaries, and are therefore well suited for constraining the influence of FDM in such environments. Several major PTA collaborations—including EPTA [199, 200], PPTA [201, 202, 203], NANOGrav [204, 205], InPTA [199], CPTA [206], and the combined IPTA dataset [207]—are actively searching for such signals. PTAs exploit the extraordinary timing stability of millisecond pulsars to detect correlated variations in pulse arrival times induced by passing gravitational waves.

The SGWB observed by PTAs is broadly consistent with expectations from inspiraling SMBH binaries [208, 209]. If the binary evolution is governed purely by gravitational-wave emission, the resulting characteristic strain follows a power-law spectrum $h_c(f) \propto f^{-2/3}$ [210]. However, realistic astrophysical environments can modify this behavior [211, 212]. In particular, the presence of a dense ULDM environment introduces an additional energy-loss channel through dynamical friction [213], altering the SGWB spectrum.

The presence of a central supermassive black hole modifies the structure of these solitons, which are obtained by solving the coupled Gross–Pitaevskii–Poisson or Schrödinger–Poisson equations, including a possible quartic self-interaction term $\lambda\phi^4$. Using the derived soliton profiles, one can compute the resulting drag forces and thereby derive constraints on the ULDM mass m , core size, and self-coupling strength λ from the current PTA observations.

1.6 Thesis Outline

The three directions above probe ALPs across a remarkably broad range of masses and energy scales:

- Photon–ALP mixing near black holes: $m_a \sim 10^{-9} - 10^{-7}$ eV
- ALP dark-matter decay in galaxies: $m_a \sim 1 - 20$ eV
- ULDM solitons and GW background: $m_a \sim 10^{-22} - 10^{-21}$ eV

In our work [214], we investigated photon–ALP conversion near the photon sphere of a black hole and its impact on the observed photon ring, showing that the resulting dimming signatures can distinguish between different black hole spacetimes such as Schwarzschild, Reissner–Nordström, and higher-dimensional (braneworld) scenarios for M87*.

In our work [215], we investigated potential line emission from the decay of eV-scale ALP dark matter using optical and ultraviolet spectroscopic data, focusing on the central region of the Messier 87 galaxy, and used the absence of narrow emission features to constrain the ALP–photon coupling.

The third research direction, based on Ref. [216], concerns ultralight FDM with mass $m_a \sim 10^{-22} - 10^{-21}$ eV, which exhibits wave nature on galactic scales and forms stable solitonic cores at galactic centers, whose interaction with SMBH binaries leaves an imprint on the stochastic gravitational-wave background.

Taken together, these studies form a multi-messenger approach to probing new physics using black-hole imaging, galaxy spectroscopy, and gravitational-wave astronomy. Each method exploits different physical conditions—strong gravity, large dark matter halos, or SMBH binary inspirals—and together they offer a coherent and complementary strategy for exploring dark-sector fields.

The remainder of the thesis is organized as follows:

- **Chapter 2** develops the theory of photon–ALP mixing near black holes and applies it to modifications of the photon ring brightness.

- **Chapter 3** studies ALP dark-matter decay signatures in galaxies through detailed spectral modeling.
- **Chapter 4** analyzes solitonic ULDM cores, self-interactions, and their imprint on the stochastic gravitational-wave background.

Notation and conventions

This thesis is based on three studies that employ slightly different conventions, reflecting the standard practices of their respective subfields. For clarity, we summarize the unit systems and sign conventions used throughout the thesis.

In the first study, we work primarily in natural units where $\hbar = c = k_B = 1$. For gravitational systems involving black holes, we additionally adopt geometrized units by setting $G = 1$. Here $\hbar$ denotes the reduced Planck constant, c the speed of light, G the gravitational constant, and k_B the Boltzmann constant. We adopt the “mostly plus” metric signature $(-, +, +, +)$, which is commonly used in gravitational physics.

In the second study, we again employ natural units with $\hbar = c = k_B = 1$; however, we use the field-theory metric signature $(+, -, -, -)$, which is standard in high-energy particle physics.

In the third study, natural units with $c = \hbar = 1$ are used throughout, although c and $\hbar$ are occasionally reinstated explicitly for clarity and dimensional transparency. The reduced Planck mass is denoted by M_{Pl} and is defined as

$$M_{\text{Pl}} = \sqrt{\frac{c\hbar}{8\pi G_N}} \simeq 2.4 \times 10^{18} \text{ GeV}, \quad (1.53)$$

where G_N is Newton’s gravitational constant. The symbol $M_\odot$ denotes the solar mass, with $M_\odot \simeq 2 \times 10^{30} \text{ kg}$.

Unless stated otherwise, these conventions are used consistently within each chapter, and all physical results are independent of the specific choice of units or metric signature.

Chapter 2

Photon-ALP oscillations in Black Hole atmosphere

The only elements in the construction of black holes are our basic concepts of space and time. They are, thus, almost by definition, the most perfect macroscopic objects there in the universe.

Subrahmanyan Chandrasekhar, Noble Lecture, 1983

Strong field gravitational interaction, such as that occurring near a black hole's horizon, is anticipated to reveal a multitude of details about the basic properties of background geometry. Direct observations exploring the near horizon zone of a black hole geometry were not available until recently. The chances of understanding the nature of strong gravity have significantly improved with the two subsequent ground-breaking discoveries, namely the gravitational wave measurements from the collision of binary black holes and neutron stars[217, 218] and imaging the shadow of the supermassive black holes at the centre of the M87 galaxy[121, 122, 123, 124, 125, 126, 127].

Recent observations of the black hole in the center of the M87 galaxy (M87*) through the Event Horizon Telescope imaged polarized synchrotron emission at 230 GHz on event horizon scales[219]. This polarized synchrotron radiation provides insights into the structure of magnetic fields and the properties of the plasma near the event horizon. Based on these observations, the EHT collaboration estimated the magnetic field strength to be around 1-30 Gauss, the average electron density (n_e) to be approximately 10^{4-7} cm^{-3} , and the electron temperature (T_e) of the radiating plasma to be about 10^{11} K [220]. Further, one can find the techniques of probing superradiant axion using photon ring birefringence with EHT data in [221, 222, 223, 224].

So it is interesting to investigate the photon-ALP conversion around black holes. ALPs interact with photons through the coupling term $\mathcal{L}^{int} = -(\frac{g_{\Phi\gamma}}{4})\Phi F_{\mu\nu}\tilde{F}^{\mu\nu}$, where $g_{\Phi\gamma}$ represents the ALP-photon coupling constant, Φ denotes the ALP field, $F_{\mu\nu}$ is the electromagnetic field strength tensor, and $\tilde{F}^{\mu\nu}$ is its dual. This interaction makes it possible for ALPs to become photons and vice versa when there is

an external magnetic field present[225, 226]. The wavelengths of both photons and ALPs are such that the magnetic field persists over a larger radial distance, facilitating a smooth conversion process. When photons pass near a black hole, they experience gravitational bending caused by the strong gravity. Consequently, these photons begin to orbit the black hole due to its powerful gravitational pull. Photons with an impact parameter below a critical value (b_c) will enter the black hole's event horizon, while those with exactly the critical impact parameter will move in an unstable circular orbit around the black hole which eventually results in a bright ring. The conversion of photons to ALPs reduces the number of photons escaping the photon sphere, resulting in a dimming effect on the bright ring. Thus, the presence of this conversion leads to darkening of the photon ring in the observed image. The nature of the gravitational theory can also be better understood by looking at the spectrum of photons, originating from the near horizon region. Dimming of the photon ring has been extensively studied for M87* black hole assuming it to be a Schwarzschild black hole[128]. In this chapter, we have provided the technique to predict the dimming rate in various alternative theories of gravity for non-rotating black holes. We have used our method to find the signature of extra-dimension by observing the dimmed photon ring.

Since this chapter focuses on axion-like particles (ALPs), we will frequently use the term “axion” for brevity. This should not be interpreted as excluding ALPs; rather, given the mass range considered in this present chapter, ALPs are the primary particles of interest, and “axion” is used as a shorthand notation throughout this chapter.

This chapter is structured as follows: In [Section 2.1](#), the mechanism of photon-axion conversion has been described. Moving on to [Section 2.2](#), we explore the time it takes for photons to orbit around the photon sphere of the black hole. The source of the photons near the galactic black holes and the impact parameter of these photons in terms of their emission angle has been reviewed in [Section 2.3](#). The next section i.e. [Section 2.4](#) has been dedicated to the discussion on the axion-photon conversion probability in the spherically symmetric spacetime along with the effect of extra dimension on the dimming of the photon ring of the black hole. Concluding remarks with a discussion on major findings have been provided in [Section 2.5](#).

2.1 Photon-Axion Conversion Mechanism

In this section, we study the conversion between photons and axions in a constant external magnetic field in a flat space. We consider the following photon-axion system[226, 227, 228, 229]

$$\mathcal{S} = \int d^4x \left[-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2} \partial_\mu \Phi \partial^\mu \Phi - \frac{1}{2} m_\Phi^2 \Phi^2 - \frac{1}{4} g_{\Phi\gamma} \Phi F_{\mu\nu} \tilde{F}^{\mu\nu} \right], \quad (2.1)$$

where Φ is the pseudoscalar field (axion) with mass m_Φ , $g_{\Phi\gamma}$ is the axion-photon coupling constant, $F_{\mu\nu} \equiv (\partial_\mu A_\nu - \partial_\nu A_\mu)$ is the electromagnetic field strength tensor for electromagnetic gauge field A_μ . The dual of $F_{\mu\nu}$ is represented by $\tilde{F}_{\mu\nu} \equiv \frac{1}{2} \epsilon_{\mu\nu\rho\sigma} F^{\rho\sigma}$ with $\epsilon_{\mu\nu\rho\sigma}$ being completely anti-symmetric tensor in its indices. The fourth term in the action represents the CP -conserving interaction between the axion and the

electromagnetic field. Such interactions may arise in the context of one of the solutions of strong CP problem in QCD as well as the requirement for canceling gauge anomaly in a string inspired model through the resulting electromagnetic Chern-Simons term[230, 231, 232, 233]. From the action given in Eq. (2.1), we have the Klein-Gordon equation for the axion field as

$$(\square - m_\Phi^2)\Phi = \frac{1}{4}g_{\Phi\gamma}F_{\mu\nu}\tilde{F}^{\mu\nu}, \quad (2.2)$$

and the Maxwell's equations are given by

$$\partial_\mu F^{\mu\nu} = -g_{\Phi\gamma}\tilde{F}^{\rho\nu}\partial_\rho\Phi, \quad (2.3)$$

where we have used the Bianchi identity $\partial_\mu\tilde{F}^{\mu\nu} = 0$ to derive the equations. Needless to say, although the Bianchi identity remains unchanged, Maxwell's equations get modified due to the presence of the photon-axion coupling term.

We analyse a scenario, in which electromagnetic waves spread against a constant magnetic field $\mathbf{B}$. This is essential because this magnetic field will induce the photon-axion conversion phenomenon, as we will show in this section. The magnetic field and electromagnetic waves combine to form the electromagnetic field

$$F_{\mu\nu} = \bar{F}_{\mu\nu} + \partial_\mu A_\nu - \partial_\nu A_\mu. \quad (2.4)$$

In the simplest scenario, a constant magnetic field has been considered as our background electromagnetic field such that the background field $\bar{F}_{\mu\nu}$ is represented as

$$\bar{F}_{0i} = \bar{E}_i = 0 \quad (2.5a)$$

$$\tilde{\bar{F}}_{0i} = \frac{1}{2}\epsilon_{0ijk}\bar{F}_{jk} = B_i \quad (2.5b)$$

We select Coulomb gauge for the propagating photons A_μ as $\nabla \cdot \mathbf{A} = 0$ so that the third component of $\mathbf{A}$ i.e A_z vanishes. In the leading approximation with the dispersion relation $w \simeq k$, we write the 3-vector $\mathbf{A}$ as a plane-wave solution:

$$\mathbf{A}(z, t) = i \begin{pmatrix} A_x(z) \\ A_y(z) \\ 0 \end{pmatrix} e^{-i\omega t}. \quad (2.6)$$

assuming that the spatial extent of magnetic field is substantially larger than that of photon or axion wavelength. If one further assumes the relativistic axion with momentum $k \gg m_\Phi$, the solution of Φ field can be written as

$$\Phi(z, t) = \Phi(z)e^{-i\omega t}. \quad (2.7)$$

We consider a monochromatic light beam traveling along z direction. Magnetic field $\mathbf{B}$ lies in the $x - z$ plane as shown in Fig. [2.1], where Θ is the angle between the direction of $\mathbf{B}$ and z axis. A_x and A_y denote components of the vector potential parallel and perpendicular to $\mathbf{B}_T$ respectively, where $\mathbf{B}_T$ denotes the projection of $\mathbf{B}$ on the x axis.

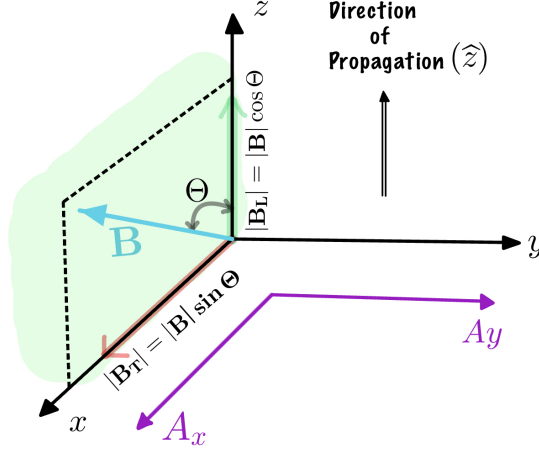


Figure 2.1: Direction of magnetic field has been shown in the cartesian coordinate system. The magnetic field $\mathbf{B}$ lies in the x - z plane making an angle Θ with the propagation vector of the photon. The transverse component of the magnetic field is along the x -axis and is denoted by $\mathbf{B}_T$. This component of the magnetic field is parallel to the component A_x , whereas A_y is perpendicular to the transverse part of the magnetic field.

The spatial components of Eq. (2.3), written under Coulomb gauge, are given by

$$\square \mathbf{A} - \nabla \dot{A}^0 = g_{\Phi\gamma} \mathbf{B} \dot{\Phi}. \quad (2.8)$$

Here, the dot denotes the time derivative. Time component of gauge field satisfies the following equation

$$\nabla^2 A^0 = -g_{\Phi\gamma} \mathbf{B} \cdot \nabla \Phi. \quad (2.9)$$

The equation of motion of the axion can be rewritten as

$$(\square - m_\Phi^2) \Phi = -g_{\Phi\gamma} \mathbf{B} \cdot (\dot{\mathbf{A}} + \nabla A^0). \quad (2.10)$$

It is clear from Eq. (2.8) and Eq. (2.10) that only the component of $\mathbf{A}$ parallel to $\mathbf{B}$ mixes with the axion. We have assumed $\mathbf{B}$ to be in the $x - z$ plane without losing generality. Now we have the following components for the magnetic field and propagating photons in (x, y, z) coordinate system:

$$\mathbf{B} = (|\mathbf{B}| \sin \Theta, 0, |\mathbf{B}| \cos \Theta), \quad (2.11)$$

$$\mathbf{A} = (iA_x(z), iA_y(z), 0) e^{-i\omega t}, \quad (2.12)$$

where Θ is the angle between the direction of $\mathbf{B}$ and z axis. A_x and A_y denote components of the photon field along the x axis and y axis respectively. The vacuum polarizability effect due to photon-photon interaction, in the limit where photon frequencies are small in comparison to electron mass and magnetic field strengths are weak in comparison to critical field strengths, in the form of Euler-Heisenberg effective Lagrangian is given by[234]

$$\mathcal{L}_{EH} = \frac{\alpha^2}{90m_e^4} \left[(F_{\mu\nu} F^{\mu\nu})^2 + \frac{7}{4} (\tilde{F}_{\mu\nu} F^{\mu\nu})^2 \right], \quad (2.13)$$

where $\alpha = \left(\frac{1}{137}\right)$ is the fine structure constant and $m_e \equiv 511 \text{ KeV}$ is the electron mass. This Lagrangian, arises due to the quantum field theoretic effect describing the one-loop corrections to classical electrodynamics, contributing the following term in the right-hand side of Eq. (2.3)

$$\frac{4\alpha^2}{45m_e^4} \partial_\mu \left(F_{\alpha\beta} F^{\alpha\beta} F^{\mu\nu} + \frac{7}{4} F_{\alpha\beta} \tilde{F}^{\alpha\beta} \tilde{F}^{\mu\nu} \right). \quad (2.14)$$

Using Eq. (2.12) and assuming plane-wave solution, we have the following equations in the linear order of $\mathbf{A}$

$$F_{\alpha\beta} F^{\alpha\beta} = 2|\mathbf{B}|^2 + 4\omega|\mathbf{B}| \sin \Theta A_y. \quad (2.15)$$

$$F_{\alpha\beta} \tilde{F}^{\alpha\beta} = -4\omega|\mathbf{B}| \sin \Theta A_x. \quad (2.16)$$

The Euler-Heisenberg Lagrangian modifies Maxwell's equation by the term

$$\frac{4\alpha^2}{45m_e^4} \partial_\mu \left[(2|\mathbf{B}|^2 + 4\omega|\mathbf{B}| \sin \Theta A_y) F^{\mu\nu} + \frac{7}{4} (-4\omega|\mathbf{B}| \sin \Theta A_x) \tilde{F}^{\mu\nu} \right]. \quad (2.17)$$

With this term added to the right side of Eq. (2.8) we have

$$\square A_x + 7\omega^2 \xi \sin^2 \Theta A_x + \omega g_{\Phi\gamma} |\mathbf{B}| \sin \Theta \Phi = 0, \quad (2.18)$$

$$\square A_y + 4\omega^2 \xi \sin^2 \Theta A_y = 0, \quad (2.19)$$

where,

$$\xi = \frac{\alpha}{45\pi} \left(\frac{|\mathbf{B}|}{|\mathbf{B}|_{\text{crit}}} \right)^2, \quad |\mathbf{B}|_{\text{crit}} \equiv \frac{m_e^2}{e}. \quad (2.20)$$

The interaction of the electromagnetic field with the background plasma current J_μ is described by the Lagrangian term $-J_\mu A^\mu$. In a plasma, this interaction modifies the photon dispersion relation and gives rise to an effective in-medium photon mass characterized by the plasma frequency. Consequently, this effect is introduced by adding a term $-\omega_{\text{pl}}^2 A_x$ to the equations of motion, where ω_{pl} is plasma frequency,

$$\omega_{\text{pl}} \equiv \sqrt{\frac{4\pi\alpha n_e}{m_e}} = 3.7 \times 10^{-11} \text{ eV} \sqrt{\frac{n_e}{\text{cm}^{-3}}}. \quad (2.21)$$

Here n_e is the number density of the electron in the medium in which the photon propagates. Here m_e is the mass of the electron and α is the fine-structure constant. The A_x component only mixes with the axion, so the relevant equations are

$$\square A_x - \omega_{\text{pl}}^2 A_x + 7\omega^2 \xi \sin^2 \Theta A_x + \omega g_{\Phi\gamma} |\mathbf{B}| \sin \Theta \Phi = 0. \quad (2.22a)$$

$$(\square - m_\Phi^2) \Phi + \omega g_{\Phi\gamma} |\mathbf{B}| \sin \Theta A_x = 0. \quad (2.22b)$$

The solutions of the fields as given in Eq. (2.6) and Eq. (2.7) are now expressed as

$$A_x(t, z) = \tilde{A}(z) e^{-i(\omega t - kz)} + \text{h.c.} \quad (2.23a)$$

$$\Phi(t, z) = \tilde{\Phi}(z) e^{-i(\omega t - kz)} + \text{h.c.} \quad (2.23b)$$

The photons propagating over a distance in the z -direction convert into axions and their amplitudes also vary with the distance z . It is anticipated that these amplitudes would vary slowly in the sense that the ratio of the second-order partial derivative and the first-order partial derivative of the amplitudes with respect to the propagation direction z is much less than the momentum k associated with the fields i.e $|\partial_z^2 \tilde{A}(z)| \ll k|\partial_z \tilde{A}(z)|$ and $|\partial_z^2 \tilde{\Phi}(z)| \ll k|\partial_z \tilde{\Phi}(z)|$.

Assuming the ultra-relativistic limit $|m_\phi|^2 \ll w^2$, and $\omega \sim k$, the box operator becomes,

$$\square = (\omega + i\partial_z)(\omega - i\partial_z) = (\omega + i\partial_z)(\omega + k) \approx 2\omega(\omega + i\partial_z). \quad (2.24)$$

With this lowest-order approximation, we have

$$\square A_x(t, z) \simeq 2i\omega\partial_z \tilde{A}(z)e^{-i(\omega t - kz)} + \text{h.c.} \quad (2.25a)$$

$$\square \Phi(t, z) \simeq 2i\omega\partial_z \tilde{\Phi}(z)e^{-i(\omega t - kz)} + \text{h.c.} \quad (2.25b)$$

The equations of motion reduce to

$$i\frac{d}{dz}\tilde{A}(z) = \left(\frac{\omega_{\text{pl}}^2}{2\omega} - \frac{28\alpha^2\omega}{90m_e^4}(|\mathbf{B}|\sin\Theta)^2\right)\tilde{A}(z) - \frac{1}{2}g_{\Phi\gamma}|\mathbf{B}|\sin\Theta\tilde{\Phi}(z). \quad (2.26a)$$

$$i\frac{d}{dz}\tilde{\Phi}(z) = \left(-\frac{1}{2}g_{\Phi\gamma}|\mathbf{B}|\sin\Theta\right)\tilde{A}(z) + \frac{m_\Phi^2}{2\omega}\tilde{\Phi}(z). \quad (2.26b)$$

It is convenient to rewrite the equations as

$$i\frac{d}{dz}\Psi(z) = \mathbf{M}\Psi(z), \quad (2.27)$$

where we have used the notations

$$\Psi(z) = \begin{bmatrix} \tilde{A}(z) \\ \tilde{\Phi}(z) \end{bmatrix} \quad (2.28)$$

$$\mathbf{M} = \begin{bmatrix} \Delta_{\parallel} & -\Delta_{\text{M}} \\ -\Delta_{\text{M}} & \Delta_{\Phi} \end{bmatrix} \quad (2.29)$$

with

$$\Delta_{\parallel} = \Delta_{\text{pl}} - \Delta_{\text{vac}}, \quad (2.30a)$$

$$\Delta_{\text{pl}} \equiv \frac{\omega_{\text{pl}}^2}{2\omega} = 6.9 \times 10^{-25} \text{eV} \left(\frac{n_e}{\text{cm}^{-3}}\right) \left(\frac{\text{KeV}}{\omega}\right), \quad (2.30b)$$

$$\Delta_{\text{vac}} \equiv \frac{28\alpha^2\omega}{90m_e^4}(|\mathbf{B}|\sin\Theta)^2 = 9.3 \times 10^{-29} \text{eV} \left(\frac{\omega}{\text{KeV}}\right) \left(\frac{|\mathbf{B}|}{\text{Gauss}}\right)^2 \sin^2\Theta, \quad (2.30c)$$

$$\Delta_{\text{M}} \equiv \frac{1}{2}g_{\Phi\gamma}|\mathbf{B}|\sin\Theta = 9.8 \times 10^{-23} \text{eV} \left(\frac{g_{\Phi\gamma}}{10^{-11}\text{GeV}^{-1}}\right) \left(\frac{|\mathbf{B}|}{\text{Gauss}}\right) \sin\Theta, \quad (2.30d)$$

$$\Delta_{\Phi} \equiv \frac{m_\Phi^2}{2\omega} = 5 \times 10^{-22} \text{eV} \left(\frac{m_\Phi}{\text{neV}}\right)^2 \left(\frac{\text{KeV}}{\omega}\right). \quad (2.30e)$$

These Δ_{is} ($i=[\text{pl}, \text{vac}, \text{M}, \Phi]$) have been estimated for our study of M87* and are all defined positive in contrast with the definition in [226]. The matrix $\mathbf{M}$ has eigenvalues that are

$$\lambda_{\pm} = \frac{(\Delta_{\parallel} + \Delta_{\Phi}) \pm \sqrt{(\Delta_{\parallel} - \Delta_{\Phi})^2 + (2\Delta_{\text{M}})^2}}{2}. \quad (2.31)$$

Now we introduce an orthogonal matrix $\mathbf{O}$ to diagonalize $\mathbf{M}$ such that

$$\mathbf{O}^T \mathbf{M} \mathbf{O} = \begin{bmatrix} \lambda_+ & 0 \\ 0 & \lambda_- \end{bmatrix}, \quad \mathbf{O} \equiv \begin{bmatrix} \cos \vartheta & \sin \vartheta \\ -\sin \vartheta & \cos \vartheta \end{bmatrix} \quad (2.32)$$

where ϑ is the *mixing angle* related as

$$\vartheta = \frac{1}{2} \arctan \left(\frac{2\Delta_{\text{M}}}{\Delta_{\Phi} - \Delta_{\parallel}} \right). \quad (2.33)$$

Using the matrix $\mathbf{O}$, Eq. (2.27) reduces to

$$i \frac{d}{dz} (\mathbf{O}^T \Psi(z)) = \begin{bmatrix} \lambda_+ & 0 \\ 0 & \lambda_- \end{bmatrix} (\mathbf{O}^T \Psi(z)). \quad (2.34)$$

Solving the equation, we have

$$\Psi(z) = \mathbf{O} \begin{bmatrix} e^{-i\lambda_+ z} & 0 \\ 0 & e^{-i\lambda_- z} \end{bmatrix} \mathbf{O}^T \Psi(0). \quad (2.35)$$

In the end, we have found the general solutions as

$$\tilde{A}(z) = (\cos^2 \vartheta e^{-i\lambda_+ z} + \sin^2 \vartheta e^{-i\lambda_- z}) \tilde{A}(0) + \sin \vartheta \cos \vartheta (e^{-i\lambda_+ z} - e^{-i\lambda_- z}) \tilde{\Phi}(0). \quad (2.36a)$$

$$\tilde{\Phi}(z) = \sin \vartheta \cos \vartheta (e^{-i\lambda_+ z} - e^{-i\lambda_- z}) \tilde{A}(0) + (\sin^2 \vartheta e^{-i\lambda_+ z} + \cos^2 \vartheta e^{-i\lambda_- z}) \tilde{\Phi}(0). \quad (2.36b)$$

Assuming $\tilde{\Phi}(0) = 0$ and $\tilde{A}(0) = 1$ as initial axion density is negligibly small with respect to that of photons. With this initial condition, one can obtain the probability of conversion of photon into axion as a function of distance z as

$$\begin{aligned} P_{\gamma \rightarrow \Phi}(z) &= |\tilde{\Phi}(z)|^2 \\ &= \sin^2 2\vartheta \sin^2 \left(\frac{\Delta_{\text{osc}}}{2} z \right) \\ &= \left(\frac{\Delta_{\text{M}}}{\Delta_{\text{osc}}/2} \right)^2 \sin^2 \left(\frac{\Delta_{\text{osc}}}{2} z \right). \end{aligned} \quad (2.37)$$

where we have defined the *oscillation length* Δ_{osc}^{-1} as

$$\Delta_{\text{osc}} \equiv \lambda_+ - \lambda_- = \sqrt{(\Delta_{\Phi} - \Delta_{\parallel})^2 + (2\Delta_{\text{M}})^2}. \quad (2.38)$$

2.1.1 Photon–Axion Conversion Parameters Associated with the SMBH

The key parameters for the photon-axion conversion are:

- frequency of the propagating photons, ω
- magnetic field perpendicular to the direction of photon propagation, $\mathbf{B}_T$
- axion mass, m_Φ
- axion–photon coupling, $g_{\Phi\gamma}$
- number density of electrons in the medium, n_e

The conversion of photons to axions depends critically on determining Δ_M using $g_{\Phi\gamma}$ and $|\mathbf{B}|_T$. The conversion process is suppressed by the contributions of finite axion mass (Δ_Φ), plasma contribution (Δ_{pl}), and one-loop corrections of electrons (Δ_{vac}). For relativistic axions, ω is significantly larger than the axion mass (m_Φ), and for photons travelling through a medium, ω is much larger than the plasma frequency (ω_{pl}). At least, three of the following requirements must be met for the current framework to be valid:

- $\frac{\alpha}{45\pi} \left(\frac{|\mathbf{B}|}{|\mathbf{B}|_{\text{crit}}} \right)^2 \ll 1$ where critical magnetic field is given by $|\mathbf{B}|_{\text{crit}} \equiv \frac{m_e^2}{\sqrt{4\pi\alpha}} = 4 \times 10^{13} \text{ Gauss}$,
- $\omega \gg m_\Phi$ for validating the assumption of relativistic axions,
- $\omega \gg \omega_{\text{pl}}$ in order for photons to move in the surrounding plasma.

Further, from Eq. (2.37) and Eq. (2.38), it can be concluded that the conversion probability becomes more effective when

$$(\Delta_{\text{pl}} - \Delta_{\text{vac}} - \Delta_\Phi)^2 \ll 4\Delta_M^2. \quad (2.39)$$

For such effective conversion, the characteristic length scale of the conversion can be expressed as

$$\Delta_{\text{osc}}^{-1} \simeq (2\Delta_M)^{-1} \quad (2.40)$$

$$= 3.4 \times 10^2 \times (2 \times 10^9 M_\odot) \times \left(\frac{\text{Gauss}}{|\mathbf{B}|_T} \right) \left(\frac{10^{-11} \text{ GeV}^{-1}}{g_{\Phi\gamma}} \right). \quad (2.41)$$

It can be observed that for $|\mathbf{B}| \sim 10^{1-2} \text{ Gauss}$, once the conversion length is compared to the Schwarzschild radius of a supermassive black hole with mass $\sim 10^{9-10} M_\odot$, the two become comparable if $g_{\Phi\gamma} \sim 10^{-11} \text{ GeV}^{-1}$. It is worth mentioning here that the supermassive black hole M87* with mass $6.2_{-0.5}^{+1.1} \times 10^9 M_\odot$ [235] possesses the magnetic field of strength $\sim (1 - 30) \text{ Gauss}$ [220] in its vicinity. Since photons can remain in the photon sphere of black holes for a while, we can anticipate that the conversion to axions takes place efficiently. From Eq. (2.39), one can notice that the efficient conversion can occur if these trivial conditions hold: $\Delta_{\text{pl}} \ll \Delta_M$, $\Delta_{\text{vac}} \ll \Delta_M$, and $\Delta_\Phi \ll \Delta_M$. These conditions can be realised better in terms of the photon-axion coupling, magnetic field associated with the black hole

and mass of axion as

$$7 \times 10^{-3} \left(\frac{n_e}{\text{cm}^{-3}} \right) \ll \left(\frac{g_{\Phi\gamma}}{10^{-11} \text{GeV}^{-1}} \right) \left(\frac{\omega}{\text{KeV}} \right), \quad (2.42a)$$

$$\left(\frac{\omega}{\text{KeV}} \right) \left(\frac{|\mathbf{B}|}{\text{Gauss}} \right) \ll 1.1 \times 10^6 \left(\frac{g_{\Phi\gamma}}{10^{-11} \text{GeV}^{-1}} \right), \quad (2.42b)$$

$$5.1 \left(\frac{m_\Phi}{\text{neV}} \right) \ll \left(\frac{g_{\Phi\gamma}}{10^{-11} \text{GeV}^{-1}} \right) \left(\frac{\omega}{\text{KeV}} \right) \left(\frac{|\mathbf{B}|}{\text{Gauss}} \right). \quad (2.42c)$$

Eq. (2.42a) and Eq. (2.42c) set the the lower bound on the frequency ω for efficient conversion. On the other hand, Eq. (2.42b) can be used to determine the upper bound of the frequency. Even when Δ_{pl} is comparable to Δ_M , under certain scenarios we can have the efficient conversion. The conditions are

- The first case is $\Delta_{\text{pl}} \simeq \Delta_\Phi$ and $\Delta_{\text{vac}} \ll \Delta_\Phi$, which in turn can be translated as

$$\left(\frac{m_\Phi}{\text{neV}} \right)^2 \simeq 1.4 \times 10^{-3} \left(\frac{n_e}{\text{cm}^{-3}} \right). \quad (2.43)$$

$$\left(\frac{\omega}{\text{KeV}} \right)^2 \left(\frac{|\mathbf{B}|}{\text{Gauss}} \right)^2 \ll 5.4 \times 10^6 \left(\frac{m_\Phi}{\text{neV}} \right)^2. \quad (2.44)$$

- The second case is $\Delta_{\text{pl}} \simeq \Delta_{\text{vac}}$ and $\Delta_\Phi \ll \Delta_{\text{vac}}$, which can be re-written as

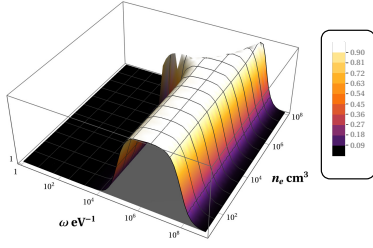
$$\left(\frac{\omega}{\text{KeV}} \right)^2 \left(\frac{|\mathbf{B}|}{\text{Gauss}} \right)^2 \simeq 7.4 \times 10^3 \left(\frac{n_e}{\text{cm}^{-3}} \right). \quad (2.45)$$

$$5.4 \times 10^6 \left(\frac{m_\Phi}{\text{neV}} \right)^2 \ll \left(\frac{\omega}{\text{KeV}} \right)^2 \left(\frac{|\mathbf{B}|}{\text{Gauss}} \right)^2. \quad (2.46)$$

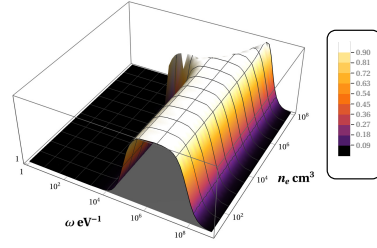
We aim to visualize the conversion probability in the parameter space, considering the observations from the Event Horizon Telescope. Based on the observations, it is known that the electron number density ranges from approximately 10^4 to 10^7 cm^{-3} and the magnetic field near M87* is estimated to be around 1-30 Gauss[220]. We fix the axion-photon coupling to be 10^{-11}GeV^{-1} . The conversion probability involves two factors: a space-independent factor, $[\Delta_M/(\Delta_{\text{osc}}/2)]^2$, which is unaffected by the spacetime geometry, and a space-dependent oscillation factor that depends on the geometry of the spacetime.

Only with the space independent factor $[\Delta_M/(\Delta_{\text{osc}}/2)]^2$ in ω - n_e plane and with axion masses $m_\Phi = 1 \text{ neV}$, 10 neV and 100 neV , there exists a region where the conversion probability can reach its maximum value as shown in Fig. [2.2]. In the case of $m_\Phi = 1 \text{ neV}$, the plasma effect Δ_{pl} dominates over the axion mass effect Δ_Φ throughout the entire space when $n_e \gtrsim 10^3 \text{ cm}^{-3}$. This has been shown in Fig. [2.2c]. From Fig. [2.2a], Fig. [2.2b] and Fig. [2.2c], one can infer that the frequency zone narrows at higher axion masses, such as $m_\Phi = 10 \text{ neV}$ or 100 neV , setting an upper bound on ω for effective conversion.

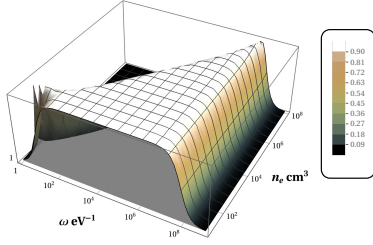
In Fig. [2.2d], the contour plots are arranged vertically for fixed values of m_Φ . The horizontal white regions extending toward the low- ω regime (most prominently in the upper panel) correspond to resonances where Δ_{pl} becomes comparable to Δ_Φ . Furthermore, for a fixed axion mass, these white bands gradually narrow toward



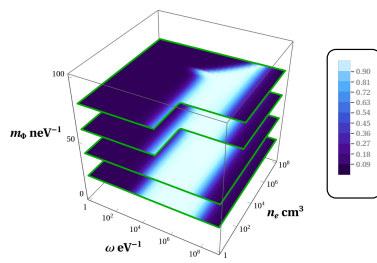
(a) Variation of the conversion probability (without the sinusoidal part) in the ω - n_e plane assuming $m_\Phi = 100$ neV, $g_{\Phi\gamma} = 10^{-11}$ GeV $^{-1}$ and $|\mathbf{B}| = 30$ Gauss.



(b) Same as (a) but for $m_\Phi = 10$ neV.



(c) Same as (a) but for $m_\Phi = 1$ neV.



(d) Variation of the conversion probability (without the sinusoidal part) in the ω - n_e plane for $m_\Phi = (1-100)$ neV.

Figure 2.2: Dependence of the conversion probability (without the sinusoidal part) on various parameters. The photon frequency and electron density are varied in the ranges $(1-10^9)$ eV and $(1-10^8)$ cm $^{-3}$, respectively. The magnetic field is fixed at $|\mathbf{B}| = 30$ Gauss and the axion-photon coupling at $g_{\Phi\gamma} = 10^{-11}$ GeV $^{-1}$.

the upper-right region of the plots, eventually approaching the resonance condition $\Delta_{\text{pl}} \sim \Delta_{\text{vac}}$. Considering that the probability of conversion from photon to axion is influenced by the magnetic field and the electron number density (n_e), we will adopt reference values of $|\mathbf{B}| = 30$ Gauss for the magnetic field, $n_e \approx 10^4$ cm $^{-3}$ for the number density, and an axion mass on the order of 10^{-9} eV. At frequencies greater than or equal to 10^6 eV, the creation of electron-positron ($e^- - e^+$) pairs becomes relevant. However, in the present work, the photon-axion conversion takes place at relatively lower frequencies.

2.2 Photon orbiting-time around the photon sphere

In this section, we discuss the orbiting path length of the photon-axion beam, or equivalently, the time spent orbiting the photon sphere before escaping to infinity and reaching the observer. We take into account the geodesic motion of a photon in a static, spherically symmetric spacetime. The magnetic field strength in this context is on the order of Gauss, and it is not substantial enough to exert any discernible back reaction on the spherically symmetric geometry. The general static, spherically

symmetric metric that satisfies Einstein's gravitational field equations is given by

$$ds^2 = -f(r)dt^2 + \frac{1}{g(r)}dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2). \quad (2.47)$$

The geodesic of a photon is represented by $x^\mu(\lambda) = (t(\lambda), r(\lambda), \theta(\lambda), \phi(\lambda))$ in these coordinates, where λ is affine parameter. Due to the symmetry of the spacetime, without loss of generality, we consider the geodesics in the equatorial plane, i.e. $\theta = \pi/2$. The metric given in Eq. (2.47) is invariant under time translation and possesses a time-like Killing vector given by $t^\mu = (1, 0, 0, 0)$. Similarly, due to the invariance of metric on azimuthal angle gives rise to the space-like Killing vector $\phi^\mu = (0, 0, 0, 1)$. The presence of these Killing vectors inspires one to introduce two conserved quantities along the geodesics of both massless and massive particles. These conserved quantities are given by

$$E = -(\partial_t)_\mu \frac{dx^\mu}{d\lambda} = f(r) \frac{dt}{d\lambda}, \quad (2.48)$$

$$L = (\partial_\phi)_\mu \frac{dx^\mu}{d\lambda} = r^2 \frac{d\phi}{d\lambda}, \quad (2.49)$$

such that E and L represent the energy and angular momentum for massless particles and energy per unit mass and angular momentum per unit mass for massive particles, respectively. In our analysis, we will focus on photon particles i.e. particles with zero rest mass. Hence, from now on, we will use E and L as energy and angular momentum of the photon particle, respectively. The impact parameter b for a photon that approaches a black hole can be expressed in terms of these conserved quantities as

$$b = \frac{L}{E}. \quad (2.50)$$

Now, let us consider a dimensionful parameter s such that the four-momentum is given by $p^\mu = E \frac{dx^\mu}{ds}$, where E is the conserved energy. With this definition, the null geodesic satisfies

$$\frac{dr}{ds} = \pm \sqrt{\mathbf{R}(r)}, \quad \mathbf{R}(r) = h(r) (1 - V_{eff}), \quad (2.51a)$$

$$\frac{d\phi}{ds} = \frac{b}{r^2}, \quad (2.51b)$$

$$\frac{dt}{ds} = \frac{1}{f(r)}, \quad (2.51c)$$

where we have defined

$$h(r) = \frac{g(r)}{f(r)}. \quad (2.52a)$$

$$V_{eff} = \frac{b^2}{r^2} f(r). \quad (2.52b)$$

One can note that for Schwarzschild and Reissner-Nordström spacetime, we have $h(r) = 1$.

The photon sphere is the radius at which photons have zero radial velocity, allowing them to revolve around a central gravitating object with a fixed radial distance. This radius of photon sphere can be determined with the following conditions: $\mathbf{R}(r) = 0 = \frac{d\mathbf{R}(r)}{dr}$. Let us assume that the photon orbits have a photon sphere at $r = r_{ph}$, which can be determined by the following algebraic equation

$$r_{ph}f'(r_{ph}) - 2f(r_{ph}) = 0. \quad (2.53)$$

The critical impact parameter for the photons of zero radial velocity at the photon sphere is given by

$$b_c = \frac{r_{ph}}{\sqrt{f(r_{ph})}}. \quad (2.54)$$

Now, we will study the behaviour of photon geodesics near the photon sphere in perturbative approach[236]. We define dimensionless fractional deviations around the radius of the photon sphere and the critical impact parameter as follows

$$r = r_{ph}(1 + \delta r) \quad (2.55a)$$

$$b = b_c(1 + \delta b) \quad (2.55b)$$

Expanding the impact parameter around its critical value we have

$$b = b_c \left[1 + \left(\frac{1}{2} - \frac{1}{4}b_c^2 f''(r_{ph}) \right) \delta r^2 + \left(-\frac{1}{12}b_c^2 r_{ph} f'''(r_{ph}) + \frac{1}{2}b_c^2 f''(r_{ph}) - 1 \right) \delta r^3 + \mathcal{O}(\delta r^4) \right]. \quad (2.56)$$

For our purposes, i.e., orbits close to the photon sphere, δr and δb must be brought simultaneously to zero at a rate $\delta b \propto \delta r^2$. This approximation, which keeps the leading term, offers the form of $\mathbf{R}(r)$ around the photon sphere as

$$\mathbf{R}(r) \approx h(r_{ph}) \left\{ \left[2r_{ph} \frac{f'(r_{ph})}{f(r_{ph})} - \frac{1}{2}r_{ph}^2 \frac{f''(r_{ph})}{f(r_{ph})} - 3 \right] \delta r^2 - 2\delta b \right\}. \quad (2.57)$$

From now we define

$$2r_{ph} \frac{f'(r_{ph})}{f(r_{ph})} - \frac{1}{2}r_{ph}^2 \frac{f''(r_{ph})}{f(r_{ph})} - 3 = 1 - \frac{1}{2}r_{ph}^2 \frac{f''(r_{ph})}{f(r_{ph})} \equiv \mathbf{p}, \quad (2.58)$$

and

$$\left[\frac{1}{2} - \frac{1}{4}b_c^2 f''(r_{ph}) \right] \left(\frac{M}{r_{ph}} \right)^2 \equiv \mathbf{a}. \quad (2.59)$$

The turning point is determined by setting the potential to zero. With the leading order approximation near the photon sphere, we have

$$\delta r_{turn} = \sqrt{\frac{2\delta b}{\mathbf{p}}}. \quad (2.60)$$

Now, we inquire how much affine parameter gathers in an area close to the photon orbit specified by

$$-\delta R < \delta r < \delta R \quad (2.61)$$

with the assumption $0 < \delta R \ll 1$.

Let us first consider the case when $\delta b < 0$. Eq. (2.57) suggests that for $\delta b < 0$ there exists no turning point. In this particular case, from Eq. (2.51a) we get

$$\Delta s = \int_{-\delta R}^{\delta R} \frac{r_{ph} d\delta r}{\sqrt{h(r_{ph})} \sqrt{\mathfrak{p}\delta r^2 - 2\delta b}} \quad (2.62)$$

$$= \frac{r_{ph}}{\sqrt{\mathfrak{p}} \sqrt{h(r_{ph})}} \ln \left[\frac{\mathfrak{p}\delta R^2}{-2\delta b} \left(1 + \sqrt{1 - \frac{2\delta b}{\mathfrak{p}\delta R^2}} \right)^2 \right]. \quad (2.63)$$

From Eq. (2.63) it is evidently distinct that there exists a logarithmic divergence as δb approaches zero.

Now let us consider the case $\delta b > 0$. In this scenario we have a single turning point. In such a scenario, we have to take $\delta R > \delta r_{turn}$ for the photon to enter the region of interest. When $\delta b > 0$, the trajectories can reach infinity and involve outer turning points and for that we have

$$\Delta s = 2 \times \frac{r_{ph}}{\sqrt{h(r_{ph})}} \int_{\sqrt{\frac{2}{\mathfrak{p}}\delta b}}^{\delta R} \frac{d\delta r}{\sqrt{\mathfrak{p}\delta r^2 - 2\delta b}} \quad (2.64)$$

$$= \frac{r_{ph}}{\sqrt{\mathfrak{p}} \sqrt{h(r_{ph})}} \ln \left[\frac{\mathfrak{p}\delta R^2}{2\delta b} \left(1 + \sqrt{1 - \frac{2\delta b}{\mathfrak{p}\delta R^2}} \right)^2 \right] \quad (2.65)$$

$$= \frac{r_{ph}}{\sqrt{\mathfrak{p}} \sqrt{h(r_{ph})}} \ln \left[\frac{\mathfrak{p}\delta R^2}{2|\delta b|} \left(1 + \sqrt{1 - \frac{2\delta b}{\mathfrak{p}\delta R^2}} \right)^2 \right] \Theta(\delta R^2 - (2/\mathfrak{p})\delta b). \quad (2.66)$$

When $\delta b \ll (\mathfrak{p}/2)\delta R^2$, we have

$$\Delta s \approx \frac{r_{ph}}{\sqrt{\mathfrak{p}} \sqrt{h(r_{ph})}} \ln \left[2\mathfrak{p} \frac{\delta R^2}{\delta b} \right]. \quad (2.67)$$

Using Eq. (2.67) one can find the lapse in t near the photon sphere, which is given by

$$\Delta t \simeq \frac{1}{f(r_{ph})} \frac{r_{ph}}{\sqrt{\mathfrak{p}} \sqrt{h(r_{ph})}} \ln \left[2\mathfrak{p} \frac{\delta R^2}{\delta b} \right] \quad (2.68)$$

$$= -\frac{1}{f(r_{ph})} \frac{r_{ph}}{\sqrt{\mathfrak{p}} \sqrt{h(r_{ph})}} \ln \left[\frac{b - b_c}{2\mathfrak{p}b_c\delta R^2} \right] \quad (2.69)$$

$$= -\frac{1}{f(r_{ph})} \frac{r_{ph}}{\sqrt{\mathfrak{p}} \sqrt{h(r_{ph})}} \ln \left[\frac{2(b - b_c)}{\mathfrak{p}b_c} \times \frac{r_{ph}^2}{\epsilon^2 M^2} \right] \equiv \mathcal{T}(b). \quad (2.70)$$

where we have redefined the small parameter as

$$\epsilon = \left(\frac{r_{ph}}{M} \right) \delta r. \quad (2.71)$$

Here M is the mass parameter of the given spacetime.

2.3 Photons approaching the photon sphere

To calculate the photon flux entering the photon sphere or escaping from the unstable orbit to infinity, we must first understand the geometry surrounding the SMBH. We therefore begin by introducing the coordinate system required to describe the relevant parameters.

2.3.1 Relation between emission angle and impact parameter

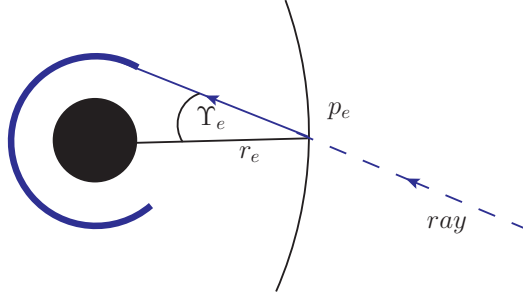


Figure 2.3: The blue curve represents a light ray emitting from a point p_e toward the photon sphere. The point p_e is positioned on a sphere with a radius labeled as r_e and it is centred around the black hole within a spherically symmetric coordinate system. The angle Υ_e is the zenith angle.

Consider a light beam that is directed towards a black hole photon sphere with an impact parameter of b from a point p_e that is situated at r_e in spherically symmetric spacetime coordinates. As illustrated in Fig. [2.3], we introduce an angle Υ_e between the initial direction of the incident photon and the direction to the black hole's centre, in order to characterise the track of the photon. For spherically symmetric spacetime as given in Eq. (2.47) and assuming $g(r) = f(r)$ the tetrads are given by

$$e^{(0)} = \left(\sqrt{f(r)}, 0, 0, 0 \right) \quad (2.72a)$$

$$e^{(1)} = \left(0, \frac{1}{\sqrt{f(r)}}, 0, 0 \right) \quad (2.72b)$$

$$e^{(2)} = (0, 0, r, 0) \quad (2.72c)$$

$$e^{(3)} = (0, 0, 0, r \sin \theta) \quad (2.72d)$$

The tetrads $e^{(1)}$ and $e^{(3)}$ are the representative of orthonormal bases parallel and normal to the direction to the center of the black hole. Hence, the angle Υ_e is given by

$$\tan \Upsilon_e = \left| \frac{k^\mu e_\mu^{(3)}}{k^\mu e_\mu^{(1)}} \right|_{p_e} = \left| r \sqrt{f(r)} \sin \theta \frac{d\phi}{dr} \right|_{p_e}, \quad (2.73)$$

where $k^\mu = \frac{dx^\mu}{d\lambda}$ is the tangent vector to the geodesic of the photon with the affine parameter λ . For simplicity, we assume that the plane of the geodesic is at $\theta = \pi/2$

plane. Using Eq. (2.51), one can find

$$b = \frac{r_e}{\sqrt{f(r_e)}} \sin \Upsilon_e. \quad (2.74)$$

Eq. (2.74) enables us to relate the emission angle Υ_e , defined at a point p_e , with the impact parameter.

Having established the geometric framework, we can now estimate the number of photons that may enter the photon sphere; this will be discussed in the following section.

2.3.2 Flow into photon sphere from spherical region

We picture the black hole as being located in the centre of a sphere, where photons are isotropically emitted from each point with a specific emissivity. Here, we provide a ballpark estimate of the number of photons that approach a photon sphere. For this purpose, we will also consider a spherically symmetric spacetime as our model geometry. Let the number of photons with a frequency width of $d\omega_e$, emitted from dV_e , and traveling through an infinitesimal solid angle of $d\Omega_e$, per unit time τ_e as seen from the emission point p_e be expressed as

$$d^6 \left(\frac{dN}{d\tau_e} \right) = J_e(\omega_e, r_e) d\Omega_e dV_e d\omega_e \quad (2.75)$$

Here, the volume V_e , frequency ω_e , time τ_e , solid angle Ω_e are all measured in a local inertial frame at p_e . Choosing the point p_e as the origin, let us denote Ψ_e as the azimuthal angle in the plane normal to the direction of the black hole from point p_e and Υ_e as the zenithal angle measured from that direction. With more simplification, we assume that the emission from the point p_e is isotropic. With this assumption, we can safely choose J_e to be independent of Υ_e and Ψ_e .

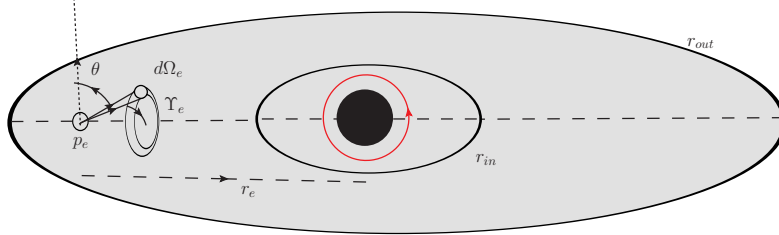


Figure 2.4: Photons coming from a spherical region in between r_{in} and r_{out} towards the black hole photon sphere which is shown in red circle.

We express equation Eq. (2.75) in terms of the impact parameter b of a photon. We use Eq. (2.74) and $db = \frac{r_e}{\sqrt{f(r_e)}} \cos \Upsilon_e d\Upsilon_e$ to find the desired result, for a fixed r_e . Integrating Eq. (2.75) over the angle Ψ_e , we have the number of photons, emitted in the direction of photon sphere, per unit of time as

$$d^5 \left(\frac{dN}{d\tau_e} \right) = \frac{1}{2} \times 2\pi J_e(\omega_e, r_e) \frac{\sqrt{f(r_e)}}{r_e} \frac{b}{\sqrt{\frac{r_e^2}{f(r_e)} - b^2}} \times db dV_e d\omega_e. \quad (2.76)$$

Due to the fact that only photons with $0 \leq \Upsilon_e \leq \pi/2$ can approach the photon sphere, the $1/2$ factor is used in Eq. (2.76). In a local inertial frame at p_e , the time element $d\tau_e$ and volume element dV_e are given by $d\tau_e = \sqrt{f(r_e)}dt$ and $dV_e = \frac{r_e^2}{\sqrt{f(r_e)}} \sin\theta dr_e d\theta d\phi$ in spherically symmetric spacetime (SSS) coordinate.

The number of photons with impact parameter lying in the range $(b, b + db)$, released from a spherical shell with a width of dr_e and a unit frequency of ω_e is obtained by integrating Eq. (2.76) over θ and ϕ , which is given by

$$\left(\frac{d^4 N}{dt d\omega_e db dr_e} \right) = 4\pi^2 J_e(\omega_e, r_e) \frac{br_e \sqrt{f(r_e)}}{\sqrt{(r_e^2/f(r_e)) - b^2}}. \quad (2.77)$$

It should be noted that ω_e is the frequency in a local inertial frame at the emission point p_e , which is given by

$$\omega_e = k^\mu e_\mu^{(0)} \Big|_{p_e} = \frac{dt}{d\lambda} \sqrt{f(r_e)} \Big|_{p_e} = \frac{E}{\sqrt{f(r_e)}}, \quad (2.78)$$

where $k^\mu = \frac{dx^\mu}{d\lambda}$ is the tangent vector to the geodesic, λ is the affine parameter and $e_\mu^{(\alpha)}$ is the local tetrad at p_e . The frequency measured in a local inertial frame at the photon sphere located at $r = r_{ph}$ is $\omega_c = E/\sqrt{f(r_{ph})}$. As a result, the number of photons reaching the photon sphere per unit time t and unit frequency ω_c with impact parameter $(b, b + db)$ is given by

$$\left(\frac{d^3 N}{dt d\omega_c db} \right) = 4\pi^2 \int_{r_{in}}^{r_{out}} dr_e J_e \left(\frac{\sqrt{f(r_{ph})}\omega_c}{\sqrt{f(r_e)}}, r_e \right) \times \frac{br_e \sqrt{f(r_e)}}{\sqrt{(r_e^2/f(r_e)) - b^2}}, \quad (2.79)$$

where the emission region is in between a spherical region with inner diameter r_{in} and outer diameter r_{out} . Since we are concerned about the region near the photon sphere, integrating Eq. (2.79) in the range $(b_c, b_c(1 + \alpha\epsilon^2))$, we find

$$\frac{d^2 N}{dt d\omega_c} \simeq 4\pi^2 \alpha\epsilon^2 b_c^2 \int_{r_{in}}^{r_{out}} dr_e J_e \left(\frac{\sqrt{f(r_{ph})}\omega_c}{\sqrt{f(r_e)}}, r_e \right) \times \frac{r_e \sqrt{f(r_e)}}{\sqrt{(r_e^2/f(r_e)) - b_c^2}}. \quad (2.80)$$

Here, it has been assumed that r_e is sufficiently larger than the radius of photon sphere such that $r_{in}^2/f(r_{in}) > b^2$ holds for $b \in (b_c, b_c(1 + \alpha\epsilon^2))$.

2.3.3 Sources of photons near galactic black holes

We now discuss the sources of photons that may enter the photon sphere and their distribution. From recent observations, it is known that in the vicinity of a super-massive black hole electromagnetic plasma exists[126]. Photons with frequencies much larger than the plasma frequency ($\omega \gg \omega_{pl}$) can travel through this plasma. The presence of such high-frequency radiation is a result of a charged particle being accelerated in the Coulomb field of another charge. This phenomenon is known as free-free emission or bremsstrahlung. To fully comprehend this process, a quantum

treatment is necessary, as it allows for the production of photons with energies similar to that of the emitting particle. The radiated energy by this process per unit time, per unit frequency, and unit volume is given by[237]

$$\left(\frac{dW}{d\tau_e d\omega_e dV_e}\right) = \frac{2^4 \alpha^3}{3m_e} \left(\frac{2\pi}{3m_e}\right)^{1/2} T_e^{-1/2} n_e^2 e^{-\omega_e/T_e} \bar{g}_{ff}, \quad (2.81)$$

where T_e is the electron temperature, n_e is the electron number density in the plasma and $\bar{g}_{ff}$ is velocity averaged Gaunt factor. We express the emission rate as an approximate classical result times the free emission Gaunt factor $\bar{g}_{ff}$ accounting for quantum-mechanical Born approximation. $\bar{g}_{ff}$ is a certain function of energy of the electron and of the frequency of emission. But for order magnitude estimation, this factor can be considered to be unity approximately. We have assumed that ion density n_i is equal to n_e . For isotropic radiation $J_e^{(N)}$ as defined in Eq. (2.75) leads to

$$J_e^{(N)}(\omega_e, r_e) = \frac{1}{4\pi\omega_e} \left(\frac{2^4 \alpha^3}{3m_e}\right) \left(\frac{2\pi}{3m_e}\right)^{1/2} T_e^{-1/2} n_e^2 e^{-\omega_e/T_e} \bar{g}_{ff}. \quad (2.82)$$

In the case of M87*, observations from the Event Horizon Telescope (EHT) have provided evidence for the presence of a hot accretion disk. The EHT images of M87* show a bright ring-like structure, corresponding to the emission from the inner part of the accretion disk. The observed properties of the emission, such as its spectrum and variability, are consistent with expectations from a hot accretion flow. Hot accretion refers to the accretion of gas with high temperatures, typically in the range of millions to billions of Kelvin. The temperature of hot accretion is almost virial, $T \simeq GMm_p/6k_B r \approx 10^{12} K (r/M)^{-1}$ with m_p being proton mass[238, 239]. The study of accretion processes around supermassive black holes like M87* often involves the use of theoretical models to understand the properties and behaviour of the accreting material. One commonly used model is the spherical accretion model, which assumes that the accretion flow onto the black hole is spherically symmetric. It is important to note that while the spherical accretion model provides a useful starting point for understanding the overall behaviour of the accretion flow around M87*, it is a simplification of the actual complex processes that may occur. Then the spherical mass accretion rate can be written as $\dot{M} = 4\pi r^2 \rho v_r$ with mass density ρ and radial velocity v_r . Assuming constant mass accretion and free falling gas $v_r \propto r^{-1/2}$, we have $\rho \propto (r/M)^{-3/2}$. Then we can assume that the electron temperature and number density of electrons obey the power law:

$$T_e = T_{e,c} \left(\frac{r_e}{r_{ph}}\right)^{-1}, \quad (2.83a)$$

$$n_e = n_{e,c} \left(\frac{r_e}{r_{ph}}\right)^{-3/2}, \quad (2.83b)$$

where $T_{e,c}$ and $n_{e,c}$ are the values at the photon sphere.

2.4 Conversion of photons into axions and relative luminosity

2.4.1 Dependence of the conversion probability and conversion factor on spacetime metric

In [Section 2.2](#), we have made a discussion on the existence of the photon sphere in a general spherically symmetric spacetime and also derived the photon time-lapse in such spacetimes. As a further simplification, we choose the general spherically symmetric metric in the form

$$ds^2 = -f(r)dt^2 + \frac{1}{f(r)}dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2), \quad (2.84)$$

where $f(r)$ corresponds to any spherically symmetric black hole geometry. The time that photons stay close to the photon sphere while their impact parameter (b) is close to (b_c) around the photon sphere ($r_{ph} < r < r_{ph}(1 + \delta r)$) is given by

$$\mathcal{T}(b) = -\frac{1}{f(r_{ph})} \frac{r_{ph}}{\sqrt{\mathfrak{p}}} \ln \left[\frac{2(b - b_c)}{\mathfrak{p}b_c} \frac{r_{ph}^2}{\epsilon^2 M^2} \right]. \quad (2.85)$$

To calculate the number of photons that transform to axions, near the photon sphere, per unit time t and unit frequency ω_c , the following equation can be used

$$\frac{d^2 N_{\gamma \rightarrow \Phi}}{dt d\omega_c} = \int_{b_c}^{b_c(1+\mathfrak{a}\epsilon^2)} db \frac{1}{2} \left(\frac{d^3 N}{dt d\omega_c db} \right) P_{\gamma \rightarrow \Phi} \left(\sqrt{f(r_{ph})} \mathcal{T}(b) \right), \quad (2.86)$$

where

$$P_{\gamma \rightarrow \Phi} \left(\sqrt{f(r_{ph})} \mathcal{T}(b) \right) = \left(\frac{2\Delta_M}{\Delta_{osc}} \right)^2 \times \sin^2 \left(-\frac{\Delta_{osc}}{2} \frac{r_{ph}}{\sqrt{\mathfrak{p}} \sqrt{f(r_{ph})}} \ln \left[\frac{2(b - b_c)}{\mathfrak{p}b_c} \frac{r_{ph}^2}{\epsilon^2 M^2} \right] \right). \quad (2.87)$$

The factor of $1/2$ introduced in [Eq. \(2.86\)](#) accounts for the conversion of photons only with polarization parallel to the external magnetic field into axions. By integrating over the interval $(b_c, b_c(1 + \mathfrak{a}\epsilon^2))$, the number of photons that enter the region $r_{ph} < r < r_{ph}(1 + \epsilon M/r_{ph})$ and escape to infinity can be determined. As no significant change in the number of photons occurs with the change in impact parameter in this scenario, it can be approximated by taking its value at $b = b_c$ out of the integral. Then the [Eq. \(2.86\)](#) can be written as

$$\frac{d^2 N_{\gamma \rightarrow \phi}}{dt d\omega_c} \simeq \frac{1}{2} \left(\frac{d^3 N}{dt d\omega_c db} \right) \Big|_{b=b_c} \times \int_{b_c}^{b_c(1+\mathfrak{a}\epsilon^2)} db P_{\gamma \rightarrow \Phi} \left(\sqrt{f(r_{ph})} \mathcal{T}(b) \right) \quad (2.88)$$

$$= \frac{1}{2} \left(\frac{d^3 N}{dt d\omega_c db} \right) \Big|_{b=b_c} \times \left(\frac{2\Delta_M}{\Delta_{osc}} \right)^2 \times \int_{b_c}^{b_c(1+\mathfrak{a}\epsilon^2)} \sin^2 \left(\frac{\Delta_{osc}}{2} \mathcal{T}(b) \sqrt{f(r_{ph})} \right) db. \quad (2.89)$$

The associated integral can be simplified as

$$I = \int_{b_c}^{b_c(1+\mathfrak{a}\epsilon^2)} \sin^2 \left(\frac{\Delta_{\text{osc}}}{2} \mathcal{T}(b) \sqrt{f(r_{ph})} \right) db \quad (2.90)$$

$$\begin{aligned} &= \frac{1}{2} \mathfrak{a} b_c \epsilon^2 \left[1 - \frac{\sqrt{\mathfrak{p}} \sqrt{f(r_{ph})}}{\Delta_{\text{osc}} r_{ph}} \int_0^\infty \cos x \exp \left(-\frac{\sqrt{\mathfrak{p}} \sqrt{f(r_{ph})}}{\Delta_{\text{osc}} r_{ph}} x \right) dx \right] \\ &= \frac{1}{2} \left(\frac{1}{1 + \mathcal{X}^2} \right) \mathfrak{a} b_c \epsilon^2. \end{aligned} \quad (2.91)$$

where we have defined the dimensionless quantity $\mathcal{X}$ as

$$\mathcal{X} = \frac{\sqrt{\mathfrak{p}} \sqrt{f(r_{ph})}}{\Delta_{\text{osc}} r_{ph}}, \quad (2.92)$$

and we have used the relation $\frac{2\mathfrak{a}r_{ph}^2}{\mathfrak{p}M^2} = 1$.

The fraction of photons entering the region near the photon sphere that are converted into axions is given as

$$\frac{d^2 N_{\gamma \rightarrow \phi}}{dt d\omega_c} \bigg/ \frac{d^2 N}{dt d\omega_c} \simeq \frac{1}{4} \left(\frac{2\Delta_{\text{M}}}{\Delta_{\text{osc}}} \right)^2 \left(\frac{1}{1 + \mathcal{X}^2} \right). \quad (2.93)$$

For the efficient conversion, we have $\Delta_{\text{osc}} \simeq 2\Delta_{\text{M}}$. For such a scenario, the conversion factor (CF) can be given by

$$CF = \frac{dN_{\gamma \rightarrow \Phi}}{dt d\omega_c} \bigg/ \frac{d^2 N}{dt d\omega_c} \bigg|_{\text{efficient conversion}} \simeq \frac{1}{4} \left(\frac{1}{1 + \mathcal{X}^2} \right) \quad (2.94)$$

$$= \frac{1}{4} \left[\frac{\left(\frac{r_{ph}}{M} \right)^2 (2M\Delta_{\text{M}})^2}{\left(\frac{r_{ph}}{M} \right)^2 (2M\Delta_{\text{M}})^2 + \mathfrak{p}f(r_{ph})} \right]. \quad (2.95)$$

Now, inserting [Eq. \(2.80\)](#) into the [Eq. \(2.89\)](#), we have

$$\begin{aligned} \frac{d^2 N_{\gamma \rightarrow \Phi}}{dt d\omega_c} &\simeq \mathfrak{a} \pi^2 \epsilon^2 b_c \left(\frac{2\Delta_{\text{M}}}{\Delta_{\text{osc}}} \right)^2 \left[\frac{(\Delta_{\text{osc}} r_{ph})^2}{(\Delta_{\text{osc}} r_{ph})^2 + \mathfrak{p}f(r_{ph})} \right] \\ &\quad \times \int_{r_{\text{in}}}^{r_{\text{out}}} dr_e J_e \left(\frac{\sqrt{f(r_{ph})} \omega_c}{\sqrt{f(r_e)}}, r_e \right) \left[\frac{br_e \sqrt{f(r_e)}}{\sqrt{(r_e^2/f(r_e)) - b^2}} \right]. \end{aligned} \quad (2.96)$$

The above equation i.e [Eq. \(2.96\)](#) enables us to find the number of axions created via the conversion of the photons to axions.

2.4.2 Spherically symmetric spacetime in brane-world scenario

Here, we will discuss the spherically symmetric spacetime in the presence of extra dimension i.e in the brane-world scenario. The effective gravitational field equations

on the brane can be solved for a spherically symmetric scenario in the vacuum by projecting the bulk gravitational field equations onto the brane hypersurface. This method of getting the gravitational field equations on the brane is feasible because it avoids the need for extensive knowledge of the bulk geometry, which is presumptively lacking for any observer of the brane. If the bulk gravitational field equations are the bulk Einstein equations, then the effective gravitational field equations on the brane can be obtained by projecting the bulk Einstein tensor onto the brane hypersurface using the Gauss, Codazzi, and Mainardi equations[240, 241, 242]. The effective field equations of vacuum brane consist of the following terms

- The four dimensional Einstein tensor $G_{\mu\nu}$, not relating to the additional spatial dimension.
- Projection of the bulk Weyl tensor W_{ABCD} , onto the brane hypersurface, which yield $E_{\mu\nu} = W_{ABCD}e_\mu^A e_\nu^C n^B n^D$, where e_μ^A are the projectors and n_A are the normals to the brane hypersurface.

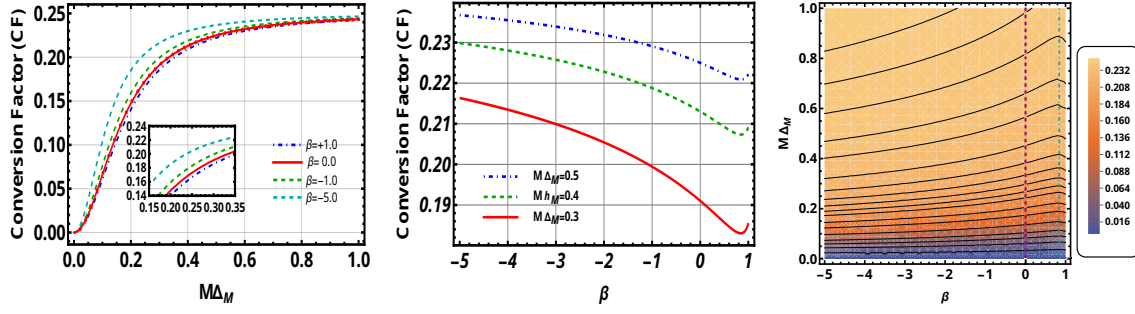
We would not address the additional contributions from the matter sector in the case of non-vacuum brane at this time. In the effective gravitational field equations, the term $E_{\mu\nu}$, which is derived from the bulk Weyl tensor, incorporates the fact that vacuum branes have an additional spatial dimension. Due to its relationship to the Weyl tensor and the symmetries of that tensor, it follows that the tensor $E_{\mu\nu}$ is traceless and so resembles the energy-momentum tensor of a Maxwell field with a crucially important overall negative value[243]. As a result of the effective gravitational field equations on the brane, the static and spherically symmetric vacuum brane spacetime has already been studied in [244]. Due to the fact that the tensor $E_{\mu\nu}$ shares the same characteristics as the energy-momentum tensor of the Maxwell field, the resulting static spherically symmetric solution will resemble Reissner-Nordström spacetime, with the accompanying line element having the following form

$$ds^2 = -f(r) dt^2 + \frac{1}{f(r)} dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2), \quad (2.97)$$

with

$$f(r) = \left(1 - \frac{2M}{r} + \frac{\beta M^2}{r^2}\right). \quad (2.98)$$

Here M is the mass parameter of the black hole. The dimensionless parameter β , which appears in Eq. (2.98), is inherited from the presence of higher dimensions and therefore can take on positive and negative values. This is in stark contrast to the Maxwell field, where the corresponding contribution to the metric components is made through Q^2 , which is strictly positive regardless of the sign of the electric charge Q . Unlike black holes with an electromagnetic field, where Eq. (2.97) resembles a Reissner-Nordström black hole with an event horizon and a Cauchy horizon for positive values of β , however, the scenario for negative values of β has no parallel in general relativity and so offers a real indicative of the extra spatial dimensions[243]. We will look for schwarzschild metric when β is zero as well as for the black holes with positive β and negative β values as special cases.



(a) Variation of the conversion factor (CF) with $M\Delta_M$ for different values of the extra-dimensional parameter β . The conversion factor reaches its maximum value as $M\Delta_M \rightarrow 1$, independent of β . (b) For fixed values of $M\Delta_M$, the variation of the conversion factor (CF) with the extra-dimensional parameter β is shown. The conversion factor reaches its minimum at $\beta = 27/32$ for any fixed $M\Delta_M$. (c) Contour plot showing that larger values of $M\Delta_M$ and more negative values of the extra-dimensional parameter β enhance the conversion efficiency.

Figure 2.5: Variation of the conversion factor (CF) with $M\Delta_M$ and the extra-dimensional parameter β for efficient conversion, corresponding to the condition $2\Delta_M = \Delta_{osc}$.

In Fig. [2.5] we have displayed the variation of the conversion factor, in the effective conversion regime, with the variation of $M\Delta_M$ and the extra-dimensional parameter β . In such regime, the conversion factor(CF) increases with the increase in axion-photon couplings for obvious reasons. If one choose for efficient conversion, the conversion factor(CF) reaches its lowest value for $\beta = \frac{27}{32}$ as shown in Fig. [2.5b]. On the other hand, when β takes on negative values, the dimming intensifies, revealing a distinct indication of the presence of an additional dimension. For a specific black hole which has a specific mass and magnetic field associated with it, the variation of conversion factor with the variation of dimensionless quantity $M\Delta_M$ is of immense importance. This variation, as shown in Fig. [2.5a], suggests that the increasing value of $M\Delta_M$ or equivalently the increasing value of axion-photon coupling constant leads to higher conversion rate of photons into axions. Such kind of variation of conversion factor with change in coupling constant holds for any value of extra-dimensional parameter β .

2.4.3 Dimming of the photon ring luminosity and observation of the axions

Dimming of the photon sphere

Once the conversion factor is known to us, we only need a count of the photon numbers that approached the photon sphere and escape to infinity to find the number of converted photons and newly generated axions. Using Eq. (2.80) and Eq. (2.82),

we have the photon count as

$$\frac{d^2 N}{dt d\omega_c} = L_\omega^0 \omega_{\text{obs}}^{-1} 2\alpha \left(\frac{b_c}{M}\right)^2 \left(\frac{r_{ph}}{M}\right) \times \int_{x_{\text{in}}}^{x_{\text{out}}} dx_e \left[\frac{x_e^{-3/2} f(x_e)}{\sqrt{x_e^2/f(x_e) - \left(\frac{b_c}{r_{ph}}\right)^2}} \right] e^{-\frac{\omega_{\text{obs}}}{T_{e,c}} \frac{x_e}{\sqrt{f(x_e)}}}, \quad (2.99)$$

where we have defined the variable x as $x = r/r_{ph}$ and also L_ω^0 is defined as

$$\begin{aligned} L_\omega^0 &= 2\pi^2 \epsilon^2 M^3 \left(\frac{4\alpha^3}{3\pi m_e}\right) \left(\frac{2\pi}{3m_e}\right)^{1/2} T_{e,c}^{-1/2} n_{e,c}^2 \bar{g}_{ff} \\ &= 6.48 \times 10^{27} \text{erg} \cdot \text{sec}^{-1} \cdot \text{KeV}^{-1} \epsilon^2 \left(\frac{M}{6.2 \times 10^9 M_\odot}\right)^3 \left(\frac{T_{e,c}}{10^{11} \text{K}}\right)^{-1/2} \left(\frac{n_{e,c}}{10^4 \text{cm}^{-3}}\right)^2 \bar{g}_{ff}. \end{aligned} \quad (2.100)$$

The observed frequency ω_{obs} is related to that frequency ω_c at the photon sphere as $\omega_{\text{obs}} = \sqrt{f(r_{ph})} \omega_c$ due to gravitational redshift. The distance to M87*, denoted as D , is approximately 16.8 Mpc. The cosmological red-shift of M87* is estimated to be around $z \approx \frac{H_0 D}{c} \approx 0.004$, where H_0 represents the present value of the Hubble parameter (taken to be 71 km/s/Mpc) and c is the speed of light. In the local universe, galaxies, including Messier galaxy, typically exhibit peculiar velocities on the order of a few hundred kilometers per second. However, when considering the dominant gravitational redshift, we neglect small effects such as peculiar velocities and cosmic expansion.

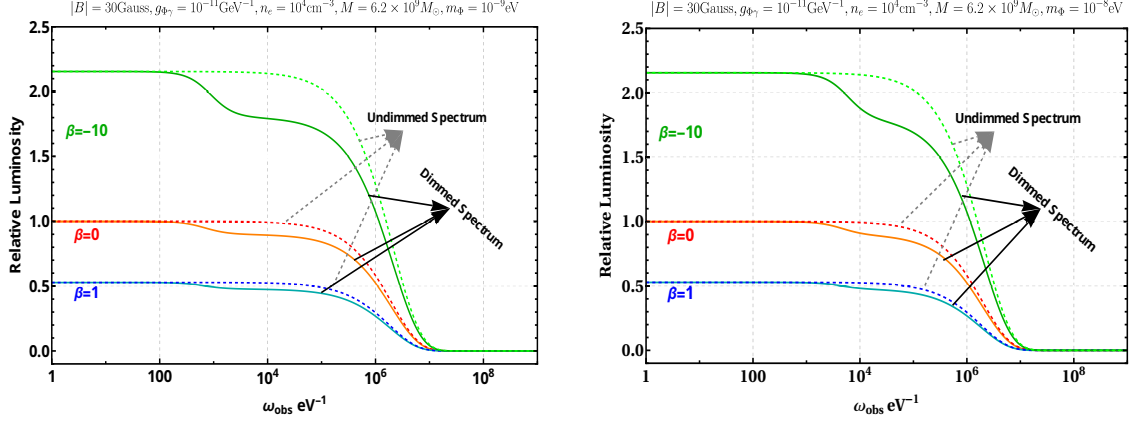
We define the following quantity to be the *relative luminosity* (RL) of photons before any conversion as

$$\begin{aligned} \frac{L_\omega^\gamma}{L_\omega^0} &\equiv \frac{\omega_{\text{obs}}}{L_\omega^0} \left(\frac{d^2 N}{dt d\omega_c} \right) \\ &= 2\alpha \left(\frac{b_c}{M}\right)^2 \left(\frac{r_{ph}}{M}\right) \int_{x_{\text{in}}}^{x_{\text{out}}} dx_e \left[\frac{x_e^{-3/2} f(x_e)}{\sqrt{x_e^2/f(x_e) - \left(\frac{b_c}{r_{ph}}\right)^2}} \right] \exp \left[-\frac{\omega_{\text{obs}}}{T_{e,c}} \frac{x_e}{\sqrt{f(x_e)}} \right]. \end{aligned} \quad (2.101)$$

The relative luminosity for the axions as produced during photon-axion conversion is given by

$$\frac{L_\omega^{\gamma \rightarrow \Phi}}{L_\omega^0} = \frac{L_\omega^\gamma}{L_\omega^0} \frac{1}{4} \left(\frac{2\Delta_M}{\Delta_{\text{osc}}} \right)^2 \left(\frac{1}{1 + \mathcal{X}^2} \right). \quad (2.102)$$

The spectral variation of the relative luminosity(RL) has been depicted in [Fig. \[2.6\]](#) for various values of the extra-dimensional parameter β . The spectrum has been normalised with respect to the undistorted relative luminosity for Schwarzschild black hole in very low frequency. The dashed lines indicate the RL if there is no conversion of the photons, approaching the photon sphere. While the solid lines show that the spectrum is attenuated as a result of the fraction of photons becoming axions. The effect of the attenuation of the spectrum from its undistorted part



(a) Dimmed and undimmed relative luminosity spectrum of photons for axion mass $m_\Phi = 1 \text{ neV}$, shown for different values of the extra-dimensional parameter β .

(b) Same as (a), but for axion mass $m_\Phi = 10 \text{ neV}$.

Figure 2.6: Relative luminosity of photons before (dashed curves) and after (solid curves) photon–axion conversion. The magnetic field associated with the black hole is taken to be $|\mathbf{B}| = 30 \text{ Gauss}$, the axion–photon coupling $g_{\Phi\gamma} = 10^{-11} \text{ GeV}^{-1}$, the electron number density $n_e = 10^4 \text{ cm}^{-3}$, and the black hole mass $M = 6.2 \times 10^9 M_\odot$. The plots are shown for two different axion mass values.

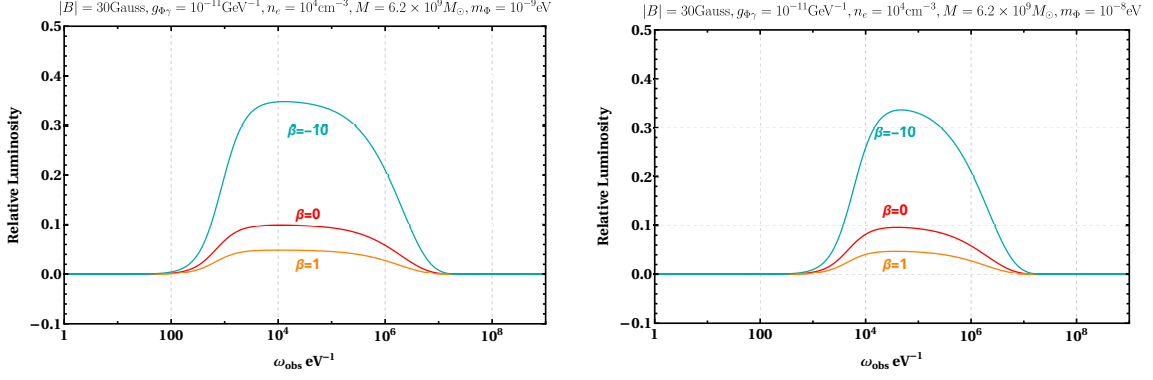
becomes more visible when the parameter β is increased in its negative direction. Moreover, by comparing Fig. [2.6a] and Fig. [2.6b], one can infer that the attenuation point of the spectrum also depends on the mass value of the axion. The rapid downtrend of the spectrum occurs due to the presence of the exponential factor in Eq. (2.99) which in turn appeared through the Eq. (2.82). Because there are so few high-temperature electrons that release photons with such high frequencies in the electromagnetic plasma, there is a suppression of these photon luminosities. The exponential factor in Eq. (2.82) can be put to unity if the following condition holds

$$\frac{\omega_e}{T_e} = \frac{\omega_e}{T_{e,c}} \left(\frac{r_e}{r_{ph}} \right) = \frac{\omega_c}{T_{e,c}} \sqrt{\frac{f(r_c)}{f(r_e)}} \left(\frac{r_e}{r_{ph}} \right) \ll 1. \quad (2.103)$$

Under such conditions, the RL reaches its maximum value and an almost flat part of the spectrum is obtained.

In Fig. [2.7], the relative luminosity spectrum of axion has been depicted. The contributing axions are created from the photons near the photon sphere. From the spectrum, it can again be inferred that the more negative value of the extra-dimensional parameter β induces the higher production rate of the axions. Moreover, one can also notice the broadening of the axion spectrum as the mass of the axion decreases. This has explicitly also been shown in Fig. [2.8] and discussed in corresponding subsection.

2.4. Conversion of photons into axions and relative luminosity



(a) Relative luminosity spectrum of axions for axion mass $m_\Phi = 1 \text{ neV}$ with different values of the extra-dimensional parameter.

(b) Same as (a), but for axion mass $m_\Phi = 10 \text{ neV}$.

Figure 2.7: Relative luminosity spectra of axions. The magnetic field strength is taken to be $|\mathbf{B}| = 30 \text{ Gauss}$, the axion-photon coupling $g_{\Phi\gamma} = 10^{-11} \text{ GeV}^{-1}$, the electron number density $n_e = 10^4 \text{ cm}^{-3}$, and the black-hole mass $M = 6.2 \times 10^9 M_\odot$. Results are shown for two different axion masses.

Spectral flux of photons

The spectral flux coming from the nearby region of the photon sphere incorporating the photon-to-axion conversion is approximately estimated as

$$F_{\mathcal{D},\omega} \simeq \frac{1}{4\pi D^2} \left(\frac{d^2 N}{dt d\omega_c} - \frac{d^2 N_{\gamma \rightarrow \Phi}}{dt d\omega_0} \right) \quad (2.104)$$

$$\begin{aligned} &= 1.2 \times 10^{-15} \text{ cm}^{-2} \cdot \text{sec}^{-1} \cdot \text{KeV}^{-1} \left(1 - \frac{1}{4} \left(\frac{\Delta_M}{\Delta_{\text{osc}}/2} \right)^2 \left[\frac{(\Delta_{\text{osc}} r_{ph})^2}{(\Delta_{\text{osc}} r_{ph})^2 + \mathbf{p}f(r_{ph})} \right] \right) \\ &\times 2\mathbf{a} \left(\frac{b_c}{M} \right)^2 \left(\frac{r_{ph}}{M} \right) \times \int_{x_{\text{in}}}^{x_{\text{out}}} dx_e \left[\frac{x_e^{-3/2} f(x_e)}{\sqrt{x_e^2/f(x_e) - (b_c/r_{ph})^2}} \right] e^{-\frac{\omega_{\text{obs}}}{T_{e,c}} \frac{x_e}{\sqrt{f(x_e)}}} \\ &\times \left(\frac{16.8 \text{ Mpc}}{D} \right)^2 \left(\frac{M}{6.2 \times 10^9 M_\odot} \right)^3 \left(\frac{T_{e,c}}{10^{11} \text{ K}} \right)^{-1/2} \left(\frac{n_{e,c}}{10^4 \text{ cm}^{-3}} \right)^2 \left(\frac{\text{KeV}}{\omega_{\text{obs}}} \right) \epsilon^2 \bar{g}_{ff}. \end{aligned} \quad (2.105)$$

For the supermassive black hole M87*, located at a distance of $D = 16.8 \text{ Mpc}$, the conversion reaches its maximum in the frequency range from nearly few eV to MeV, depending upon the axion mass for a fixed value of the parameter β . The luminosity and fluxes of photons in this range vary depending on the extra-dimensional parameter β of the black hole as shown in Fig. [2.6]. From the plot, in the case of the Schwarzschild blackhole (i.e. for $\beta = 0$), the dimming fraction is about 10%. With this fraction and setting the parameters to approximate values, such as $\epsilon \sim \bar{g}_{ff} \sim 1$, the flux is estimated to be $F_{\mathcal{D},\omega} \sim 6.28 \times 10^{-15} \times (1 - \text{conversion factor}) \text{ cm}^{-2} \cdot \text{sec}^{-1} \cdot \text{KeV}^{-1}$ for ω_{obs} around the KeV range. In comparison, for the black hole Sgr A*, although it is closer at a distance of $D = 8 \text{ Kpc}$, its mass is three orders of magnitude smaller (approximately $4 \times 10^6 M_\odot$). Since the flux depends on the mass and distance according to M^3/D^2 , the flux for Sgr A* is much

smaller compared to that of M87* and it is difficult to get a flux from the black hole centred in the Milky Way.

Observation limit on axion mass

Referring to Fig. [2.8], a narrow range of converted axion masses can be demonstrated. When the mass of m_ϕ exceeds 100 neV, the dimmed spectra align with the spectra of source photons, resulting in minimal observable impact. Consequently, an upper bound on the mass is imposed, with m_ϕ being less than or approximately 100 neV, for effective photon-axion conversion near the photon sphere. Furthermore, if the axion mass decreases below $\sim$ neV, the spectra merge, indicating saturation in mass. In the figure, we observe an intriguing reversal of spectral behavior between 3.7 neV and 1 neV. This is due to specific parameter values where a resonance occurs in photon-axion conversion at an axion mass of 3.7 neV. So the parameter space with finite photon-axion conversion which can give maximum observable opportunity falls in the axion mass range $\approx (1 - 100)$ neV. The plot as shown in Fig. [2.9]

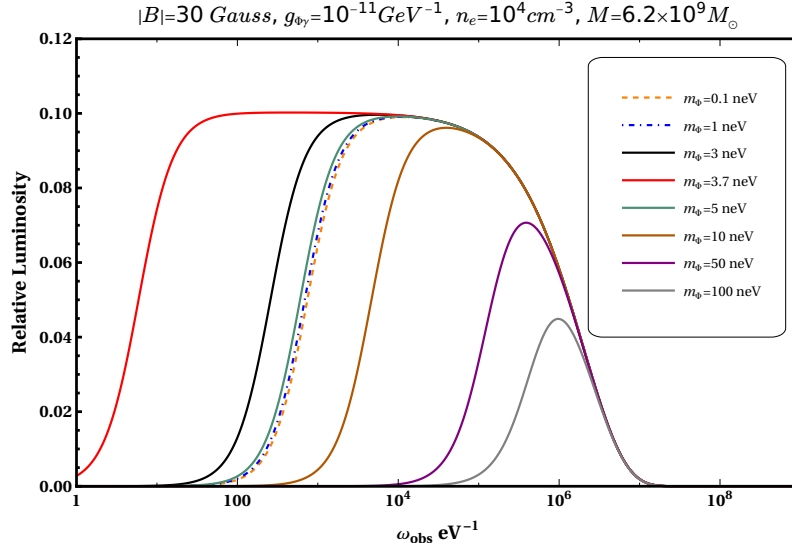


Figure 2.8: The plot describes the expected spectral luminosity of axions arising from photons in proximity to the photon sphere, considering different axion masses within the framework of a Schwarzschild black hole.

gives the bounds on coupling and mass of axions from different astrophysical observations. In the present study we have found that there is upper bound on axion mass at $m_\phi = 10^{-7}$ eV as demonstrated from the spectra of Fig. [2.8]. For axion mass lower than 1 neV, the spectra merges as seen from Fig. [2.8]. This is the reason why the likelihood of conversion remains unaffected by mass throughout the entire region, which is denoted as sky blue in the figure for $m_\phi \lesssim 1$ neV. The displayed window having axion mass in the range 10^{-9} eV $- 10^{-7}$ eV, showcases a valuable parameter space on coupling and mass for the M87* supermassive black hole. In our calculations, we have utilized the coupling constant of $g_{\phi\gamma} = 10^{-11}$ GeV $^{-1}$, which nearly falls within the permissible region on the $g_{\phi\gamma} - m_\phi$ plane. To be precise, in the plot, we have highlighted a specific data point with a red dot, where the coupling constant and axion mass are approximately 5×10^{-12} GeV $^{-1}$ and 4 neV, respectively. At this point, the photon-axion conversion probability reaches approximately 3.6%

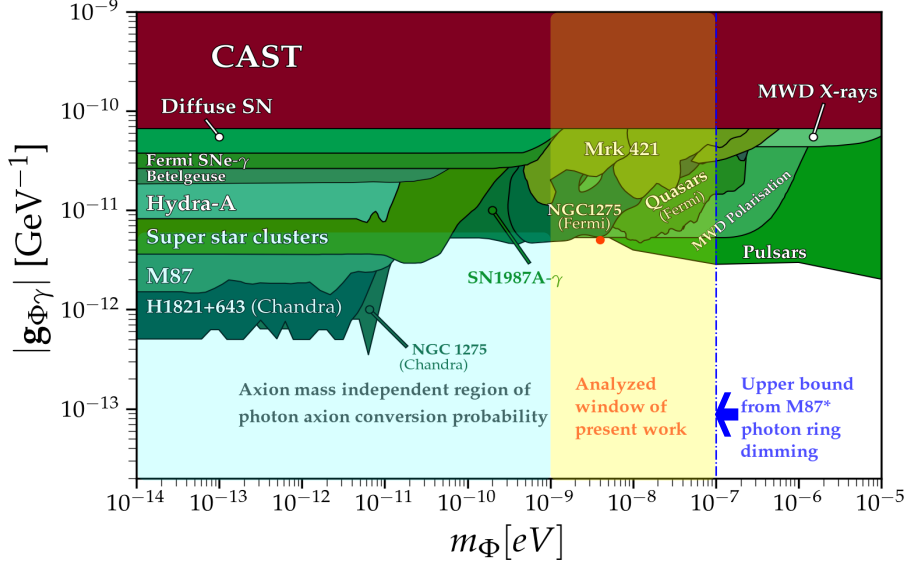


Figure 2.9: The plot describes the low mass astrophysical bounds on the axion-photon coupling from different observations[100]. The region in light-yellow demonstrates the window for which the dimming of the photon ring can be expected to be observed. For axion mass $m_{\Phi} \lesssim 10^{-9} \text{eV}$ there is no significant change in photon-axion conversion probability due to a change in axion mass. This mass range has been shown in sky blue region of $g_{\Phi\gamma} - m_{\Phi}$ plane.

when considering a Schwarzschild black hole ($\beta = 0$).

Required resolution of the black hole image

The Chandra X Observatory has achieved an angular resolution of 1 arc-sec for observing objects beyond the event horizon scale[245] which is more specifically the photons that originated through bremsstrahlung in nearby plasma well outside the photon sphere. The total luminosity of photons emitting from a region of space in between $r_{\text{ISCO}} < r_e < \mathfrak{R}$ with energies in KeV – MeV band is approximately given by,

$$L_{\text{total}} = 4\pi \int_{r_{\text{ISCO}}}^{\mathfrak{R}} dr_e r_e^2 \frac{dW}{d\tau_e d\omega_e dV_e} \quad (2.106)$$

$$\sim \frac{16\pi}{\epsilon^2} \left(\frac{r_{ph}}{M} \right)^3 \left[\left(\frac{\mathfrak{R}}{r_{ph}} \right)^{1/2} - \left(\frac{r_{\text{ISCO}}}{r_{ph}} \right)^{1/2} \right] L_{\omega}^0, \quad (2.107)$$

where r_{ISCO} stands for the radius of the Innermost Stable Circular Orbit. So when observing objects at a distant scale where the radius is much larger than the innermost stable circular orbit ($\mathfrak{R} \gg r_{\text{ISCO}}$), the majority of the total luminosity is accounted for, resulting in an insignificant dimming effect caused by the photon-axion conversion. However, when the distance scale is smaller than approximately 3 times the mass of the object ($\mathfrak{R} \lesssim 3M$) i.e. size of photon sphere of a Schwarzschild black hole, this scenario changes. In such cases, the resolution will be

$$\theta = \frac{\mathfrak{R}}{D} \bigg|_{\beta=0} \lesssim 1.09 \times 10^{-5} \text{arc-sec} \left(\frac{M}{6.2 \times 10^9 M_{\odot}} \right) \left(\frac{16.8 \text{Mpc}}{D} \right). \quad (2.108)$$

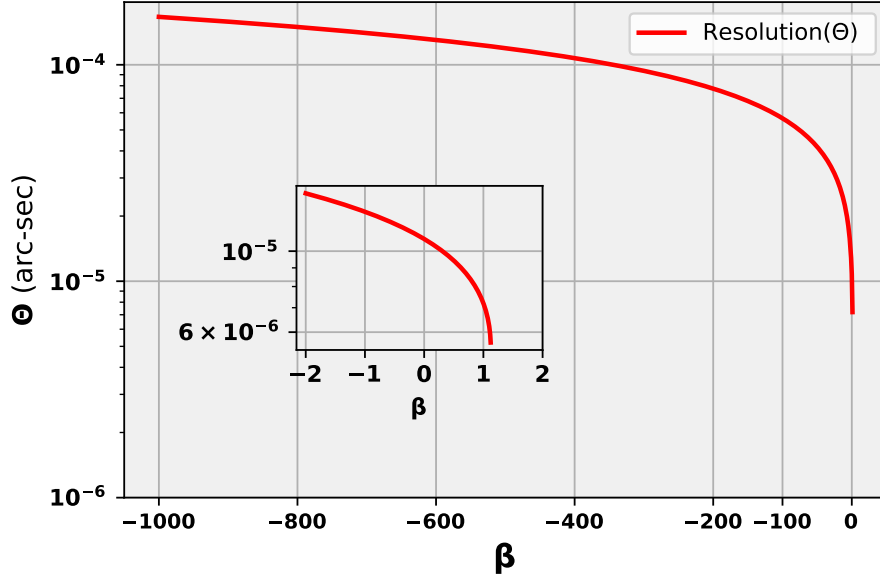


Figure 2.10: The red plot illustrates the relationship between the charge parameter and the required resolution for observing the dimming effect in a spherically symmetric black hole. Specifically, it demonstrates how the necessary resolution changes as the charge parameter varies. In the context of a Schwarzschild black hole, the point where the plot intersects the $\beta = 0$ line symbolizes the minimum resolution that is required. Inset figure shows the variation of resolution for small β values.

For the black holes having spherical symmetry as mentioned in Eq. (2.98), the resolution will vary with the β parameter as indicated in Fig. [2.10]. The photons originating from a thermal bremsstrahlung process are depicted as undimmed spectra in Fig. [2.6], exhibiting a resolution akin to that of the Chandra observatory, approximately 1 arc-sec. However, to effectively study the region near the photon sphere, it becomes imperative to enhance the angular resolution by at least a factor of 10^5 compared to Chandra's resolution. Notably, the Event Horizon Telescope, boasting a resolution of $\theta = 10^{-5}$ arc-sec, has successfully imaged structures near the horizon, although its current operations are conducted in the radio band. As depicted in the aforementioned Fig. [2.10], the required resolution for observing the vicinity of the black hole's horizon decreases in tandem with the increased value of the extra-dimensional parameter(β) in negative direction. If observations at a lower resolution manage to match the predicted diminished luminosity, it would provide substantial support for the existence of extra dimensions within the theoretical framework. Therefore, it becomes imperative to explore the X-ray to gamma-ray frequency range $\approx (10^2 - 10^6 \text{ eV})$ and to achieve a resolution for observing the photon sphere in the ballpark of $\theta \lesssim 10^{-5}$ arc-sec for a Schwarzschild black hole, and the corresponding resolution values for observing brane-world black holes, as indicated in the figure above.

2.5 Conclusion

If ALPs are present in the universe, photons traveling through a magnetic field can convert into ALPs through the coupling term $g_{\Phi\gamma}\Phi F^{\mu\nu}\tilde{F}^{\mu\nu}$. This chapter focuses

on investigating the phenomenon of photon-ALP conversion around black holes. By considering a magnetic field of approximately $|B| \sim (1 - 30)$ Gauss and an ALP-photon coupling $\sim 10^{-11} \text{ GeV}^{-1}$, electron number density 10^4 cm^{-3} we find that the propagation length required for conversion is of the order of a milli-parsec, comparable to the Schwarzschild radius of a supermassive black hole with a mass of $10^9 M_\odot$.

Our studies also demonstrate that photons in the X-ray and gamma-ray bands can efficiently convert into ALPs for masses $\lesssim 100$ neV. Thus, observing the vicinity of the black hole with electromagnetic waves may reveal a dimming of the photon ring in those wavelengths, provided sufficiently high resolution is achieved in the future. Larger values of M correspond to greater dimming, making supermassive black holes like M87* excellent candidates for observing photon ring dimming. While the Event Horizon Telescope has succeeded in imaging the region in the radio band, achieving such high-resolution observations in the X-ray and gamma-ray bands remains a challenge.

In this chapter, we have obtained the path length of photons following the geodesic of a spherically symmetric black hole, enabling us to calculate the photon-ALP conversion probability in the spacetime of a black hole with spherical symmetry, given the mass of the black hole is known. The general photon-ALP conversion formula of [Section 2.1](#) is applicable to any spherically symmetric black hole, including the Schwarzschild black hole as a special case which has been derived in [Eq. \(2.96\)](#). Through our generalization, we have expanded upon the concepts introduced in [\[128\]](#), resulting in a more inclusive approach. Importantly, we have also enhanced the accuracy by reducing several approximations in our calculations. Furthermore, by numerically solving the spectra, we have bolstered the credibility of the conversion factor for the Schwarzschild black hole case.

From our analysis, we have found that in presence of extra dimension for supermassive black holes, the maximum conversion possible is 25%, and the range of ALP mass that exhibits a finite conversion factor lies approximately between (1 - 100) neV. Observations related to the dimming of the photon ring demand very high resolution, approaching $\theta \sim 10^{-5}$ arc-sec if the black hole is a Schwarzschild black hole. Similarly, for the black holes with positive β values also require resolution at that order of magnitude, as shown in [Fig. \[2.10\]](#). On the other hand, the brane world black hole (i.e for negative β values) case would require a lower resolution with a higher charge parameter. In [Section 2.4.3](#), we explored an approximate estimation of the luminosity after conversion and its related flux, considering the mass of M87*, nearby plasma density, and ALP properties such as coupling and mass. If this estimation proves to be accurate in the future, it could offer valuable insights into the coupling and mass of ALPs.

This study is based on calculations that consider a uniform magnetic field surrounding the photon sphere. However, it is possible to conduct an analysis considering an inhomogeneous magnetic field. The plasma near the photon sphere is assumed to be homogeneous, but in realistic scenarios, some level of inhomogeneity might be necessary. In [Section 2.1](#), the formula for the total probability of photon-ALP conversion was derived in Minkowskian spacetime. The rationale behind using a flat metric for the probability formula, even in the curved geometry of

a black hole, is due to our focus on the geodesic motion of photons, which can be accurately represented within a local coordinate system. However, a more comprehensive formula that considers the curved spacetime effects can also be rigorously formulated[246]. The observations give insight on possessing a spin parameter of the M87*[126, 247]. However, for simplicity, we do not take the rotation of the black hole in the present work. The calculations for a rotating case i.e. a Kerr space time[248, 249, 250] will be more realistic and robust and we will keep it for a future prospect. Even a rotating and charged black hole i.e Kerr-Sen black hole can also be pursued. The polarization of photons which are escaping the photon sphere can be affected by the conversion mechanism[251]. So analysis of these polarized emissions would give valuable insights into ALP properties. It is also worth studying the photon ALP conversion in the presence of cosmic ALP background[252, 253]. The dark matter aspects can also be explored if this ALP background is present near the photon sphere. The fading or brightening of the photon sphere produced by black hole mergers can also be modified by the presence of photon ALP conversion mechanisms[94, 254, 255]. To have a higher flux of ALPs we can explore the same mechanism for even heavier black holes or a collection of black holes. The cosmological expansion and relative velocity of the galaxy are not taken into account in the present work but for galactic black holes which are even farther away we should have to consider this. In the present work, the source photon distribution is assumed to originate from radiative bremsstrahlung processes occurring outside the photon sphere. However, alternative photon distributions arising from synchrotron emission or inverse-Compton-scattered gamma rays may also be considered within a similar framework.

Chapter 3

ALP Dark matter Decay signal from Messier 87 Galaxy

The universe is a laboratory far beyond the reach of terrestrial experiments.

Martin Rees

Numerous astrophysical and cosmological observations firmly support the existence of DM, a non-baryonic, minimally interacting, cold matter component that provides about 25% of the energy density of our universe [17]. However, the microscopic features of DM are still unknown. Extensive efforts in its search suggest that DM resides in the form of new, unknown particles.

Due to its couplings to photons, ALPs can decay into two photons, although the decay rate should be slow enough so that they are stable on the cosmological time scale to explain the observed DM. Astronomical telescopes can provide a powerful means of searching for signals from such ALP DM decay. The decay of ALP DM particle inside a DM dense region such as a galactic DM halo can produce a distinctive monochromatic emission line that can be distinguished from astrophysical backgrounds. Over the years, searches for these emission lines have spanned a broad range of frequencies, from radio waves to X-rays [138, 139, 140, 141, 142, 143, 144, 145]. In particular, if the mass m_a of the ALP particles is on the order of electronvolts (eVs) then their decay would produce photons in the infrared, optical, and ultraviolet bands. In this context, rigorous spectroscopic searches have been performed in the infrared, optical to ultraviolet frequency bands [146, 147, 148, 149, 150, 151, 152, 153, 154, 156, 155] which impose constraints on the $g_{a\gamma\gamma} - m_a$ plane for ALPs, further encouraging the exploration of ALPs at the eV scale.

In this study, we constrain the line emission from the decay of eV scale ALP DM using the optical-ultraviolet bands in spectroscopic data collected by the Neil Gehrels Swift Observatory (which has been operational for 19 years). Its Ultraviolet/Optical Telescope (UVOT) is designed to capture optical and UV emissions from astrophysical sources. Additionally, we utilized data from the Kanata Telescope at the Hiroshima Astrophysical Science Center, which operates in the near-infrared and optical bands [256]. We also explored the near-UV and visible bands using

the UV Imaging Telescope (UVIT) onboard the Astrosat mission, which features dedicated near-UV, visible, and far-UV channels [257, 168]. This work focuses on observations of the central region of Messier 87 (M87) galaxy, a DM-rich environment. In such regions, decay of ALP DM particles into two photons could lead to strong emission lines, non-observation of which by current telescopes targeting at the central part of M87 put constraint on the ALP-photon coupling. Two major Multi Wavelength EHT campaigns (MWL2017[169] and MWL2018[170]), conducted by the EHT Multi-wavelength Science Working Group (MWL WG), analyzed multi-band data from millimeter waves to TeV γ -rays for the central region of M87 galaxy. This study leverages data from the 2018 campaign (MWL2018) (with a focus on the ultraviolet and optical bands) as well as the data from previous observations [158, 159, 160, 161, 162, 163, 164, 165, 166, 167] by Spitzer and International Ultraviolet Explorer focused towards M87 center in the infrared-ultraviolet band to constrain the decay of eV scale ALP DM.

3.1 Dark Matter Profiles in M87 Galactic Halo

In this section we first discuss the DM profiles in the M87 galaxy. Since the inner DM profile of M87 is subject to significant astrophysical uncertainties, we consider several benchmark density profiles motivated by numerical simulations and dynamical studies.

Navarro–Frenk–White profile. As a reference model, we adopt the Navarro–Frenk–White (NFW) profile [258, 259],

$$\rho_{\text{NFW}}(r) = \frac{\rho_0}{(r/r_0)(1+r/r_0)^2}, \quad (3.1)$$

where ρ_0 and r_0 are the scale density and scale radius, respectively. For the M87 galaxy within the Virgo cluster, we use the parameters $\rho_0 = 6.96 \times 10^5 M_\odot \text{Kpc}^{-3}$ and $r_0 = 403.8 \text{ Kpc}$, corresponding to a virial radius of approximately 1.7 Mpc and a distance to Earth of $d = 17.2 \text{ Mpc}$ [259]. We also consider an alternative NFW parameterization adopted in Ref. [260] to assess the impact of halo parameter uncertainties.

Burkert (cored) profile. As an alternative to the cuspy NFW profile, we consider a cored DM distribution described by the Burkert profile [261, 262],

$$\rho_{\text{B}}(r) = \frac{\rho_0}{(1+r/r_0)[1+(r/r_0)^2]}, \quad (3.2)$$

which approaches a constant density in the central region. For M87, the best-fit parameters are $\rho_0 = 6.94 \times 10^6 M_\odot \text{Kpc}^{-3}$ and $r_0 = 91.2 \text{ Kpc}$, with a virial radius $R_{200} = 1.3 \text{ Mpc}$ and a total halo mass of $1.3 \times 10^{14} M_\odot$ [261].

Central density spike. The presence of a SMBH at the center of M87, with mass $M_\bullet \simeq 6.5 \times 10^9 M_\odot$, can induce a dark matter density spike within its radius of influence r_{sp} . Inside this region, the DM density is modeled as a power law,

$$\rho_{\text{sp}}(r) = \rho(r_{\text{sp}}) \left(\frac{r}{r_{\text{sp}}} \right)^{-\gamma_{\text{sp}}}, \quad r < r_{\text{sp}}. \quad (3.3)$$

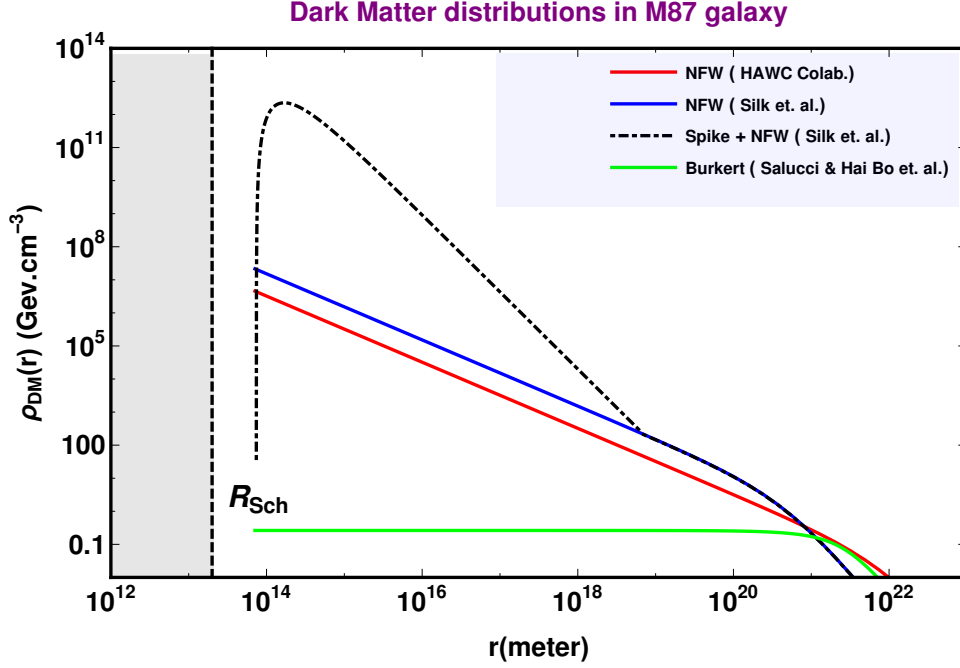


Figure 3.1: Dark matter density profiles of the M87 galaxy adopted in this chapter. The red solid curve shows the NFW profile with parameters taken from Ref. [259], while the blue solid curve corresponds to the NFW profile from Ref. [260]. The black dot-dashed curve represents the NFW profile including a central black-hole-induced density spike following Ref. [260]. The green solid curve shows the cored Burkert profile from Ref. [261, 262].

For self-interacting dark matter with velocity-dependent scattering cross section $\sigma \propto v^{-a}$, the spike slope is given by

$$\gamma_{\text{sp}} = \frac{9 - 2\gamma}{4 - \gamma}. \quad (3.4)$$

For example, we show the spike for $\gamma_{\text{sp}} = 7/4$ [260] and that transit to a NFW profile at larger radii. At radii smaller than the marginally bound orbit of the SMBH, $r_m \simeq 1.24 \times 10^{-3} \text{ pc} (\sim 4R_s)$, the spike density falls sharply, following the standard treatment adopted from the literature.

Contracted NFW (cNFW) profile. In addition to the standard NFW profile, we also consider a contracted NFW (cNFW) profile with a steeper inner logarithmic slope. Following Ref. [260], we adopt an inner slope $\gamma = 1.25$, which leads to a more centrally concentrated dark matter distribution compared to the canonical NFW case with $\gamma = 1$. The density profile is described by the generalized NFW form,

$$\rho_{\text{cNFW}}(r) = \frac{\rho_0}{(r/r_0)^\gamma (1 + r/r_0)^{3-\gamma}}, \quad (3.5)$$

where the scale radius is fixed to $r_0 = 20 \text{ Kpc}$, identical to the value used for the standard NFW profile. The normalization ρ_0 is chosen such that the total dark matter mass enclosed within 50 Kpc remains the same as in the $\gamma = 1$ case. This prescription allows for a direct comparison between the NFW and cNFW profiles while isolating the impact of the steeper inner slope.

Fig. [3.1] shows the dark matter density profiles of M87 corresponding to the NFW and Burkert parameterizations used in this chapter. The figure illustrates the qualitative differences between cuspy and cored halo models, as well as the enhancement of the central density due to the presence of a black-hole-induced spike. These profiles bracket the astrophysical uncertainties associated with the dark matter distribution in the inner region of M87 and are used as benchmarks in the computation of the ALP-induced photon signal.

3.2 ALP DM decay signal

In this chapter, we look for infrared-optical-ultraviolet emission lines from ALP DM decay ^a. The ALP can decay to two photons via the interaction[151, 144]

$$\mathcal{L} = -\frac{1}{4}g_{a\gamma\gamma}aF_{\mu\nu}\tilde{F}^{\mu\nu}, \quad (3.6)$$

where a represents the ALP field, $F_{\mu\nu}$ and $\tilde{F}^{\mu\nu}$ denote the electromagnetic field strength tensor and its dual, respectively, and $g_{a\gamma\gamma}$ is the ALP-photon coupling. The ALP decay rate is given by:

$$\Gamma_a = \frac{g_{a\gamma\gamma}^2 m_a^3}{64\pi}, \quad (3.7)$$

where m_a is the ALP mass. Note that the lifetime ($1/\Gamma_a$) should be larger than the age of the universe so that the ALP can explain the observed DM density of the Universe. The ALP decay rate into two photon has been discussed in Section A.2.

The photon flux density, i.e., the radiated power per unit area per unit frequency, from the decay of ALP DM is given by [148]:

$$S_{decay} = \frac{\Gamma_a}{4\pi} \int d\theta \, 2\pi \sin \theta \int_{l.o.s} dl \, \frac{\rho_a(r(l, \theta))}{\Delta\nu(r(l, \theta))}. \quad (3.8)$$

Here $\Delta\nu$ is the width of the produced photon line, given by $\Delta\nu = \nu_c \sigma_d/c$, where ν_c is the central frequency ($= m_a/4\pi$) of the photon line, σ_d is the velocity dispersion of DM particles within the DM halo and c is the speed of light. The decay signal will follow a Gaussian distribution, centered around the flux S_{decay} as the peak value, with $\Delta\nu$ representing the spread of the signal. We study ALP DM decay signal from the central region of the M87 galaxy which hosts a supermassive black hole (SMBH). Around the SMBH the velocity dispersion of DM is given by $\sqrt{\frac{2G_N M_{BH}}{r}}$ [263], where G_N is the gravitational constant and M_{BH} is the black hole mass which for the central SMBH in M87 is $M_{BH} \simeq 6.5 \times 10^9 M_\odot$ [170]. Additionally, there is also an outer halo component of the DM velocity dispersion which is of the order of $10^{-3}c$. The integral in Eq. (3.8) is performed over the line of sight (l.o.s) l between the emitting object and the location of Earth and the angle θ covered by the observing telescope. The radial distance from the center of M87 galaxy $r(l, \theta)$ is given by $r = \sqrt{l^2 + d^2 - 2ld \cos \theta}$, where d is the distance of the M87 center from Earth. In order to take into account the effect of absorption of the DM induced

^aOur study, however is sensitive to any DM candidate which produces narrow emission signals.

photons, we remove the contribution from the central region $r \lesssim 100R_s$ (with R_s being the Schwarzschild radius of the SMBH) from Eq. (3.8), where the absorption can be stronger [170]. In Eq. (3.8), $\rho_a(r)$ describes the DM density in the inner region of the M87 galaxy, as discussed in Section 3.1.

In the bottom panel of Fig. [3.2], the blue line shows an example of the ALP DM signal from the central region of M87. Such a signal is estimated with the DM profile parameters mentioned above. The ALP mass is considered to be $m_a = 9.51$ eV, with the value of $g_{a\gamma\gamma}$ set to $1.52 \times 10^{-11} \text{ GeV}^{-1}$ which corresponds to the existing bound on the coupling at this mass. This m_a produces an observed frequency of $\nu = 1.15 \times 10^{15} \text{ Hz}$. The angular width of observation is taken to be the one that corresponds to the resolution used at that frequency.

In the literature, the DM density near the center of galaxies (such as M87) is often described by a spike profile (due to the presence of a SMBH at the center) [264, 260, 265, 262]. However, for this study, we only adopt the canonical NFW profile. We checked that our DM decay signal remains almost unaffected even when a spike profile is superimposed on the NFW profile.

3.3 Observations and Data

ν (Hz)	Angular Scale [$''$]	νS_ν (erg cm $^{-2}$ s $^{-1}$)	Telescope/Observatory
2.4×10^{14}	10.0	$< 5.18 \times 10^{-12}$	Kanata
4.7×10^{14}	10.0	$< 4.96 \times 10^{-12}$	Kanata
5.48×10^{14}	3.0	$1.05 \pm 0.41 \times 10^{-11}$	Swift-UVOT
6.83×10^{14}	3.0	$(1.14 \pm 0.18) \times 10^{-11}$	Swift-UVOT
8.65×10^{14}	3.0	$(5.22 \pm 0.82) \times 10^{-12}$	Swift-UVOT
1.15×10^{15}	3.0	$(1.69 \pm 0.79) \times 10^{-12}$	Swift-UVOT
1.27×10^{15}	0.1	$(1.92 \pm 0.96) \times 10^{-12}$	HST
1.33×10^{15}	3.0	$(4.45 \pm 0.54) \times 10^{-12}$	Swift-UVOT
1.55×10^{15}	3.0	$(2.48 \pm 0.76) \times 10^{-12}$	Swift-UVOT
1.95×10^{15}	300.0	$< 5.13 \times 10^{-12}$	AstroSat-UVIT

Table 3.1: This table summarizes the observed data for infrared, optical and ultraviolet emissions from the central region of the M87 galaxy. These data were gathered during 2018 campaign (by MWL WG) by various telescopes such as Swift-UVOT, Kanata, Astrosat-UVIT, HST at the frequencies listed in the first column. The angular resolutions used by different telescopes are provided in arc-secs($''$) corresponding to their respective working frequencies. The data collected by the Swift-UVOT and HST include the experimental errors in flux measurements, while the data collected by Kanata and Astrosat-UVIT only provide upper limits on the flux. Data in this table are reproduced from [170] under the Creative Commons Attribution 4.0 International license.

Our analysis emphasizes the infrared, optical, and UV observations within M87's broad spectrum, from which we can constrain the ALP-photon coupling.

The dataset utilized in this chapter, is drawn from the campaign paper [170] which details the second multi-wavelength observational campaign on M87 by the

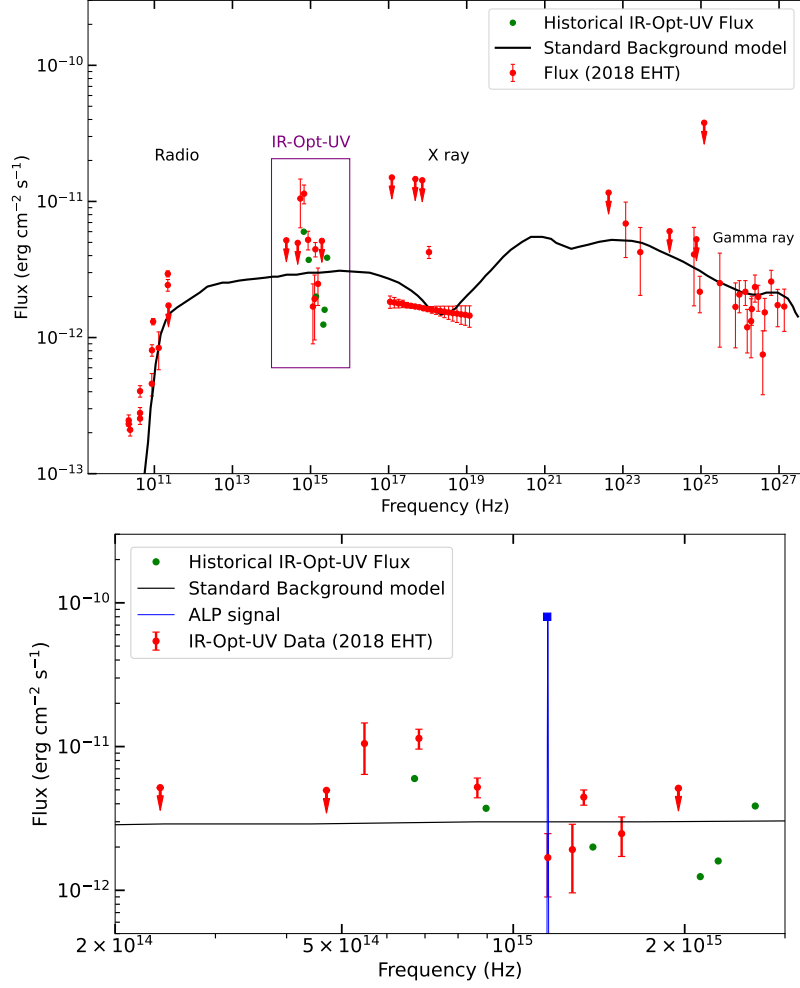


Figure 3.2: The observed broadband SED of M87, measured during the April 2018 Event Horizon Telescope (EHT) campaign using various instruments, are shown in red points. The measurements, represented by red points with error bars, indicate observed flux with experimental uncertainties, while red downward arrows denote upper limits on flux estimates. The black line depicts the best-fit model (modified model A) applied to the entire dataset, as mentioned in [170]. Additionally, the green dots represent the historical data in the IR-optical-UV frequency range from earlier observations of M87, compiled from studies such as [158, 159, 160, 164, 165, 163, 167, 161, 162, 166]. *Bottom panel:* The IR-optical-UV frequency range which is highlighted by a box in the top panel, is shown in the bottom panel as a zoomed-in view. Here the ALP DM decay signal from M87 is shown by the blue line. Such a signal is estimated for $m_a = 9.51$ eV with the value of $g_{a\gamma\gamma}$ set to the corresponding existing bound, i.e., $g_{a\gamma\gamma} = 1.52 \times 10^{-11} \text{ GeV}^{-1}$.

MWL WG and is presented in Table 3.1. The measured spectral energy distribution (SED) data for M87 can be accessed via the data archive at [266]. The MWL WG collaboration used state-of-the-art data reduction techniques to publish this dataset, covering an electromagnetic flux range from 10^{10} to 10^{27} Hz. These 2018 flux measurements are represented as red points in Fig. [3.2].

During this observational campaign, in the IR-Optical-UV region, four telescopes have been utilized for observations. The Swift UV-Optical Telescope (UVOT) ob-

served the M87 galaxy with an angular resolution of $3''$, collecting data over a wavelength range of approximately 192–547 nm. The Kanata Telescope contributed data up to an upper flux limit at 634.5 nm and 1250 nm within a $10'' \times 10''$ field of view. The UV Imaging Telescope (UVIT), part of the AstroSat mission, provided observations with two telescopes: one focused on the near-UV (2000–3000 Å) and visible (3200–5500 Å) bands, and the other on the far-UV (1300–1800 Å), offering upper flux limits with a large $300''$ field of view. We will be using single data point obtained in the near-UV channel. The Hubble Space Telescope (HST), operating in the UV-optical range, measured the flux at a central wavelength of 236 nm but with a relatively small resolution of $0.1''$. In total, there are 10 data points in the IR-Optical-UV bands, with seven having associated error bars for flux estimation.

The M87 SED are also informed by a series of earlier datasets. These older datasets include core emission measurements (ranging from millimeter to X-ray frequencies) [158, 159, 160, 164, 165, 163, 167, 161, 162, 166], MOJAVE VLBA observations at radio frequencies [267], Chandra X-ray data from 2009 [268], Fermi-LAT data [269], HESS data [270], VERITAS data [271], and MAGIC data [272]. The infrared-optical-UV frequency range of this older dataset, referred to in this chapter as “Historical data”, is displayed as green points in Fig. [3.2].

3.4 Methodology & Results

The contributions of the ALP DM signal at the observed frequencies (i.e., in IR, Optical and UV) are estimated within the angular resolution of the telescopes, focused towards the center of M87. The angular resolution is determined by the specifications of each telescope for specific frequency windows across the electromagnetic spectrum. Comparing the ALP signal with the observed data, we derive bounds on the ALP-two-photon coupling $g_{a\gamma\gamma}$. This section of this chapter presents two separate approaches for deriving the bounds on $g_{a\gamma\gamma}$. Below, we describe each approach and the corresponding constraints obtained in each case ^b.

3.4.1 Conservative approach

This approach, serving as a conservative estimate, asserts that the DM signal must not exceed the observed data in any of the frequency bins that we are considering. This approach does not involve any modeling of the background, which makes it conservative in that regard. There are total ten data points (including observed data with error bars as well as upper-limits) from the EHT2018 in the IR-Optical-UV range. In addition, the historical dataset has six observed data points. For details, see Fig. [3.2]. For a given ALP mass m_a that corresponds to any of these observed frequencies, we obtained bound on $g_{a\gamma\gamma}$ requiring that the photon signal arising from the ALP decay (given in Eq. (3.8)) does not exceed the observed data. The bounds obtained with this approach are shown in the $m_a - g_{a\gamma\gamma}$ plane in Fig. [3.3].

As can be seen from Fig. [3.3], we are able to put constraints in the mass window (2–16) eV for the EHT(2018) dataset (shown by red solid and dashed curves) and

^bA discussion on the possible impact of systematic uncertainties on the results is provided in Section 3.6.

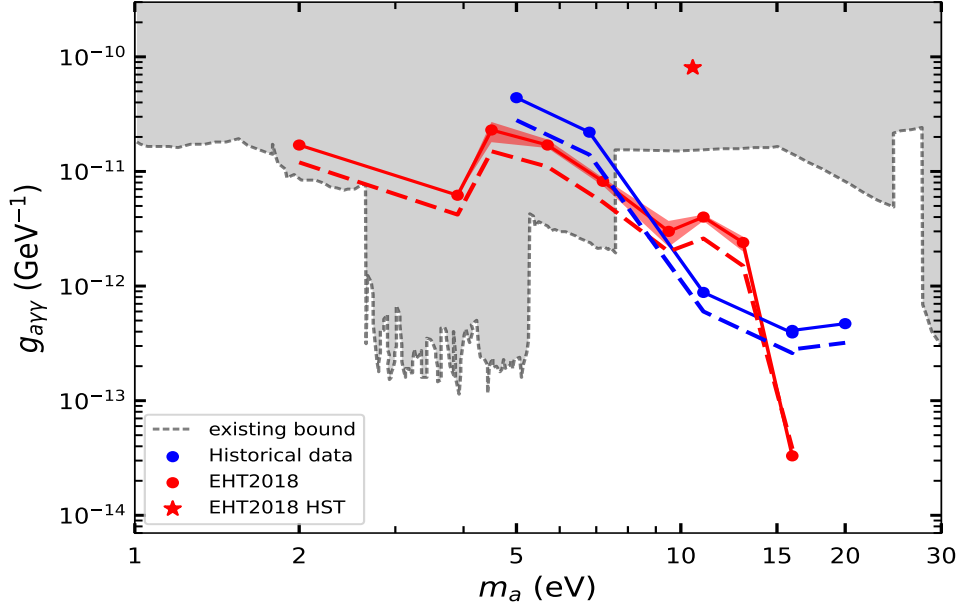


Figure 3.3: Constraints on the axion photon coupling $g_{a\gamma\gamma}$ considering the conservative approach (described in Section 3.4.1), is shown in the figure. The red and blue lines correspond to the 2018 observational campaign and the historical observational dataset of M87, respectively. The band around the constraint shown by the red solid line demonstrates the observational uncertainty in the data at relevant frequencies (see Table 3.1). The single red star corresponds to the EHT(2018) HST observed data point. The dashed red and blue lines represent the constraints derived for an alternative NFW DM profile discussed in the text. The gray shaded region indicates the strongest existing bound in the $m_a - g_{a\gamma\gamma}$ plane [151, 152, 147, 146, 144, 139, 149].

in the mass window (5-20) eV for the historical dataset (blue solid and dashed curves). The constraints shown by the solid red and blue lines correspond to the NFW DM profile with the parameters obtained from [259] as mentioned in Section 3.2. In addition, we also show by the dashed red and blue lines the constraints that correspond to a NFW DM profile, but with different parameters: $\rho_0 = 6.5 \times 10^7 M_\odot/\text{Kpc}^3$, $r_0 = 20 \text{ Kpc}$ (obtained from [260]). Apart from these, we also checked that considering a Burkert profile as reported in [261], with their best-fit parameters: $\rho_0 = 4.7 \times 10^{-25} \text{ g/cm}^3 = 6.94 \times 10^6 M_\odot/\text{Kpc}^3$ and $r_0 = 91.2 \text{ Kpc}$, the constraints derived are weakened by a factor of ~ 1.8 compared to that obtained using the former NFW profile. The rest of our discussion will be based on the former NFW profile parameters. In Section 3.5 we present a comparative analysis highlighting how the constraints vary with different choices of the dark matter profile for M87 galaxy.

We compare our bounds with the existing constraints (obtained from [151, 152, 147, 146, 144, 139, 149])^c that are shown as the gray-shaded region in the $m_a - g_{a\gamma\gamma}$ plane. We find that, our constraints can surpass the existing bounds on $g_{a\gamma\gamma}$ by an

^cThe existing constraints are estimated from the data files corresponding to the relevant mass range, available on the website [100]. The strongest limit on the coupling was identified and used as the upper boundary of the allowed parameter space.

order of magnitude for the ALP mass range $8\text{ eV} \lesssim m_a \lesssim 20\text{ eV}$. Only for the single data from the HST observation, the constraint on the coupling (shown as red star) falls inside the existing bound due to the weak sensitivity of HST towards DM decay signal, owing to its much smaller angular resolution (see Table 3.1).

Our constraints on $g_{a\gamma\gamma}$ are comparatively stronger in the higher ALP mass range ($8\text{ eV} \lesssim m_a \lesssim 20\text{ eV}$) than those at the lower ALP mass range ($2\text{ eV} \lesssim m_a \lesssim 8\text{ eV}$). The photon flux collected by telescopes such as Kanata, Swift-UVOT, and IUE in the lower frequency range (corresponding to $2\text{ eV} \lesssim m_a \lesssim 8\text{ eV}$) are relatively higher than those in the higher frequency band (as evident from the bottom panel of Fig. [3.2]). Since the constraints on $g_{a\gamma}$ from ALP decay are derived considering these observed flux as the upper limits (under conservative assumptions), a larger observed flux in the low frequency range results in comparatively weaker bounds in the low ALP mass range. Also, independent of the observational data, the flux from ALP decay scales as $\propto g_{a\gamma}^2 m_a^3$. As a consequence, in the low-frequency regime (see again the bottom panel of Fig. [3.2]), a smaller axion mass leads to a suppressed signal (for a fixed coupling), and hence, a stronger coupling is required to yield a comparable flux, indicating that constraints on $g_{a\gamma}$ become progressively weaker for lower ALP masses. All these reasons inherently make the bounds on $g_{a\gamma}$ weaker at lower masses (i.e. $2\text{ eV} \lesssim m_a \lesssim 8\text{ eV}$) in comparison to the higher masses.

3.4.2 χ^2 fit constraints

In the second approach, the bounds on the coupling are derived considering a background fit model. We refer to this approach here as the χ^2 fit constraints. Based on the astrophysical background model that describes the observed data over the whole frequency range, we will incorporate photon emissions resulting from ALP decay to constrain the parameter space.

Constraints from standard background

Given that the observed data are unevenly distributed across a wide frequency range, it is practical to fit a single model that accounts for the entire spectrum uniformly. A comprehensive review of astrophysical models, including their parameters, is provided in MWL2018 [170]. In this analysis, we adopt the modified Model A from [170] as the standard astrophysical background.

Now to estimate the coupling $g_{a\gamma\gamma}$, we perform a χ^2 analysis where χ^2 is defined as:

$$\chi^2(m_a, g_{a\gamma\gamma}) = \sum_{i=1}^N \left(\frac{S_{\text{th}}^i - S_{\text{obs}}^i}{\sigma_{\text{obs}}^i} \right)^2. \quad (3.9)$$

Here, S_{th}^i is the theoretical flux in the i -th frequency bin, computed as $S_{\text{th}}^i = S_{\text{signal}}^i + S_{\text{background}}^i$. The S_{signal}^i is the ALP DM signal computed using Eq. (3.8), and $S_{\text{background}}^i$ is derived from the modified Model A. S_{obs}^i is the observed flux, and σ_{obs}^i represents the associated observational uncertainty.

Here the summation runs over all frequency bins. For a fixed DM mass m_a , the function χ^2 depends only on $g_{a\gamma\gamma}$. The significance of a signal from ALP decay can then be computed using:

$$\Delta\chi^2 = \chi^2(g_{a\gamma\gamma}) - \chi_{\text{min}}^2. \quad (3.10)$$

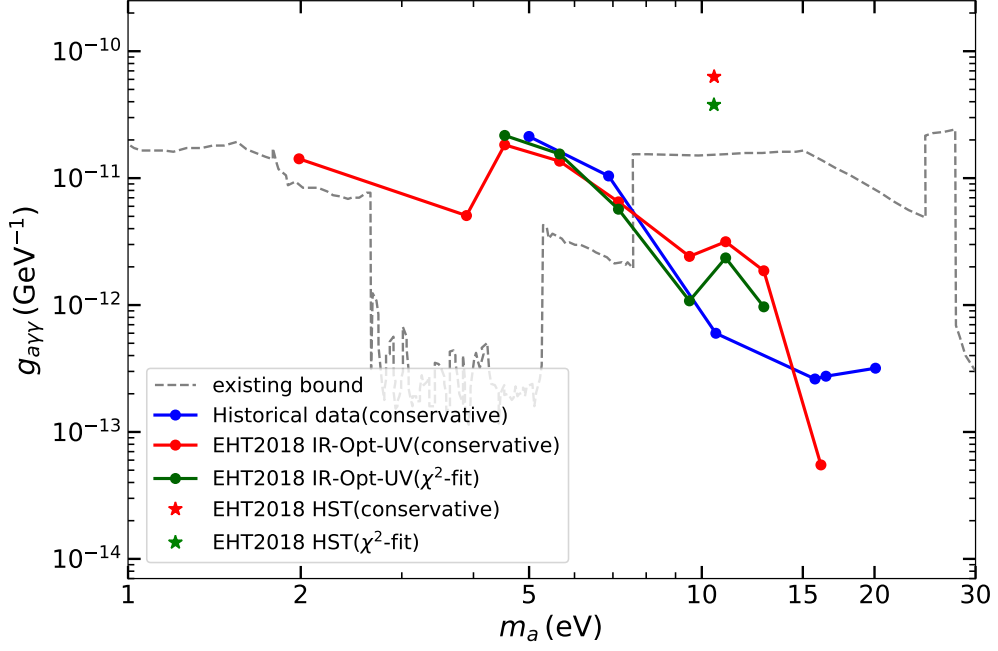


Figure 3.4: Upper-limit on $g_{a\gamma\gamma}$ obtained (at 95% C.L.) considering a background model (modified model A from [170]) is shown by the green line (see Section 3.4.2). The result corresponds to the HST data is shown separately by red and green star. For comparison, constraints derived using the conservative approach are also presented: the red line corresponds to the EHT (2018) dataset, and the blue line corresponds to the historical dataset.

For a 95% confidence level (C.L.) and one degree of freedom, $\Delta\chi^2 = 2.71$.

The quantity $\chi^2_{\min}$, the minimum value of χ^2 , corresponds to the best-fitted background model [170] without any signal (that is, for a vanishing axion-photon coupling). Taking into account the EHT(2018) whole dataset we first obtain $\chi^2_{\min}$, then we add the DM signal in the frequency bin corresponding to m_a with varying $g_{a\gamma\gamma}$ and estimate the constraint on it within 95% C.L.

The constraint on $g_{a\gamma\gamma}$ obtained in this method is shown in Fig. [3.4] by the green solid line, along with the conservative limit discussed in Section 3.4.1 (shown as red solid line) for comparison. The χ^2 -analysis of the entire dataset gives constraints in the mass range $4.3\text{ eV} \lesssim m_a \lesssim 12.8\text{ eV}$, of which the mass range $8\text{ eV} \lesssim m_a \lesssim 12.8\text{ eV}$ have almost an order of magnitude stronger constraints compared to the existing bounds. The green star demonstrates the limit on the coupling for the frequency bin corresponding to HST data.

Constraints from fitted background

We also performed a fit to the infrared-optical-UV data points using a higher-degree polynomial model, as shown on the top panel of Fig. [3.5]. Due to the irregular distribution of these data points relative to the standard background, a conventional power-law model does not provide a good fit in this frequency range. Instead, we identified the best-fit curve for these 7 data points using a polynomial model. Based on this fitted model, the constraints on the $m_a - g_{a\gamma\gamma}$ parameter space are obtained.

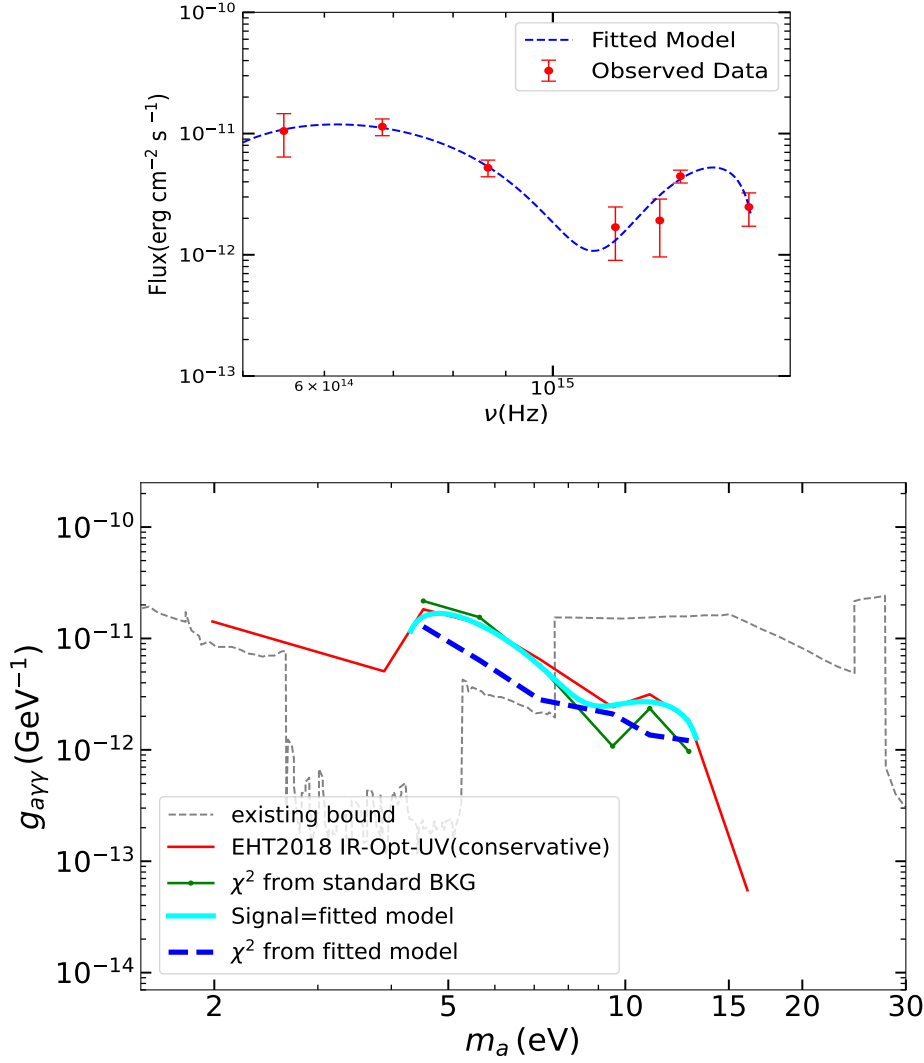


Figure 3.5: *Top panel:* The blue dashed line represents the fit to the 2018 (EHT) data in the IR-Opt-UV frequency range (see [Section 3.4.2](#) for details). *Bottom panel:* The dashed blue line shows the constraint on $g_{a\gamma\gamma}$ derived (at 95% C.L.) using the fitted background model shown in the top panel. The solid cyan line corresponds to the constraint obtained assuming the fitted background model as the upper limit for the ALP DM decay signal. Constraints obtained using the standard background (BKG) model and the conservative approach are also presented for comparison.

The fitted model is expressed as:

$$F(\nu) = a_0 + a_1(\nu/\nu_0) + a_2(\nu/\nu_0)^2 + a_3(\nu/\nu_0)^3 + a_4(\nu/\nu_0)^4, \quad (3.11)$$

where ν_0 is a reference frequency chosen to be 10^{15} Hz. The fit yielded a $\chi^2/d.o.f = 0.98$ with the best-fit parameters: $a_0 = -2.05 \times 10^{-10} \text{ erg cm}^{-2} \text{ s}^{-1}$, $a_1 = 9.92 \times 10^{-10} \text{ erg cm}^{-2} \text{ s}^{-1}$, $a_2 = -1.605 \times 10^{-9} \text{ erg cm}^{-2} \text{ s}^{-1}$, $a_3 = 1.08 \times 10^{-9} \text{ erg cm}^{-2} \text{ s}^{-1}$ and $a_4 = -2.56 \times 10^{-10} \text{ erg cm}^{-2} \text{ s}^{-1}$. This fitted model was then treated as the background for further analysis.

Using this background model, the same χ^2 -fit method (discussed in [Section 3.4.2](#) with a focus on the infrared-optical-UV part of the data) was applied to constrain the

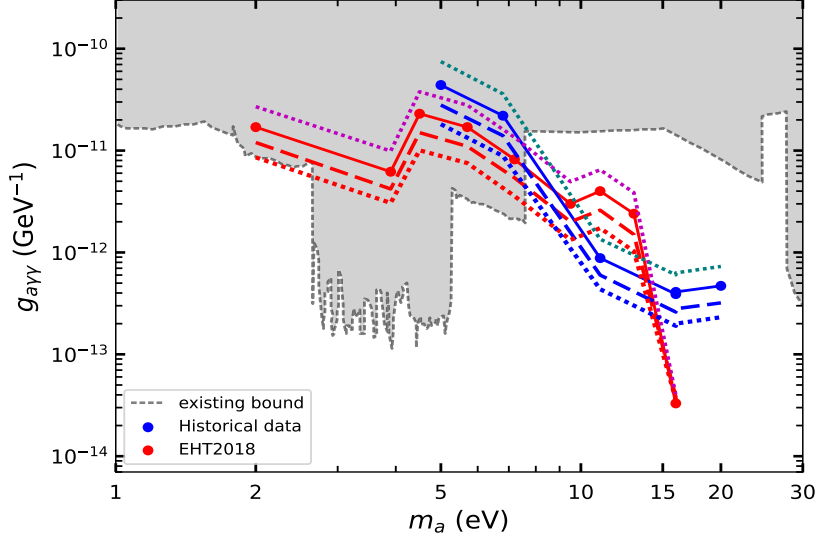


Figure 3.6: The plot illustrates the variation of the constraints with different assumptions for the DM halo profile of M87, using both the new and historical dataset. Solid lines represent an NFW profile as adopted in [259], while dashed lines correspond to an alternative NFW profile from [260]. They are same as the ones shown in Fig. [3.3]. Constraints derived using a contracted NFW (cNFW) profile with $\gamma = 1.25$ are depicted with dotted red and blue lines. Bounds corresponding to a cored Burkert profile are shown with dotted purple and green lines.

coupling $g_{a\gamma\gamma}$ for a given m_a . In the bottom panel of Fig. [3.5], the blue dashed line represents the 95% C.L. upper limit in the $g_{a\gamma\gamma} - m_a$ plane for the fitted background model. Additionally, the solid cyan line shows the constraints if the fitted model flux is directly interpreted as the upper bound of the ALP signal (i.e., without performing any χ^2 analysis with the signal). For comparison, the conservative limit (discussed in Section 3.4.1) as well as constraints obtained using the modified model A as background (discussed in Section 3.4.2) are also shown in the figure.

As shown in the figure, the fitted model provides slightly stronger constraints at the lower end of the mass range, whereas the standard background (the modified model A from [170]) yields tighter constraints at the higher mass range.

3.5 Robustness on the results due to different DM halo profile

Here we elaborate the dependence of our results on the choices of DM halo profile considered for the M87 galaxy. We explore several DM profiles. As discussed in the main text, we consider the NFW profile from the recent study [259] (Eq. (3.1)), and an alternative NFW profile from [260], both assuming a typical inner slope parameter of $\gamma = 1$. The solid and dashed (red and blue) lines, respectively, depict the limits derived using these two NFW profiles and are same as the ones shown in Fig. [3.3]. Note that the profile from [259] is used to obtain our main results.

In addition, we also adopt the NFW profile from [260] with a steeper inner slope $\gamma = 1.25$ [273], leading to a contracted NFW (cNFW) profile. The corresponding constraints obtained using this cNFW profile are represented in Fig. [3.6] by dotted red and blue lines.

Incorporating a central DM spike on top of the NFW profile from [260] (with $\gamma = 1$) leads to only a marginal change in the results ($\lesssim 1\%$) compared to that obtained for the NFW profile. This is mainly due to the fact that the DM decay signal is mostly sensitive to the overall spread of the DM distribution. As this impact is negligible, the corresponding result has not been displayed in the figure.

Additionally, we include the constraints based on a cored Burkert profile [261], shown as dotted purple and green lines in Fig. [3.6]. The specific values of the profile parameters such as ρ_0 and r_0 correspond to their best-fit values from [261].

3.6 Systematic Uncertainty

The dataset published by the MWL WG, which is utilized in this study, does not explicitly detail the systematic uncertainties arising from detector calibration. However, several systematic factors could influence the results. In this section, we will discuss a few of them.

3.6.1 Background modeling

In our χ^2 analysis in Section 3.4.2, we employed the modified Model A as the background model, as presented in [170]. If we instead consider alternative background models, such as Model A or Model B from that reference, which provide the best fit to the overall spectrum, the resulting constraints on the coupling will exhibit only minimal variations. The Table 3.2 below illustrates the changes in the results (i.e. upper limit on $g_{a\gamma\gamma}$) when using Model A and Model B in comparison to the modified Model A.

Mass Bin $m_a(\text{eV})$	Modified Model A $g_{a\gamma\gamma} (\times 10^{-12} \text{ GeV}^{-1})$	Model A $g_{a\gamma\gamma} (\times 10^{-12} \text{ GeV}^{-1})$	Model B $g_{a\gamma\gamma} (\times 10^{-12} \text{ GeV}^{-1})$
4.53	21.7	22.8	23.2
5.65	15.5	16.4	16.6
7.16	5.69	6.95	7.14
9.51	1.07	1.73	1.80
11.00	2.35	3.20	3.20
12.82	0.97	1.58	1.57

Table 3.2: The table presents the upper bounds on the photon-ALP coupling $g_{a\gamma\gamma}$ across various mass bins, derived using three distinct background models, taken from [170].

As discussed in Section 3.4.2, we have also modeled the background by fitting

the data with a fourth-order polynomial, as it provides the best fit with the lowest $\chi^2/\text{d.o.f.}$ compared to polynomials of other orders. For example, considering a third-order polynomial, $F(\nu) = b_0 + b_1(\nu/\nu_0) + b_2(\nu/\nu_0)^2 + b_3(\nu/\nu_0)^3$, where ν_0 is a reference frequency chosen to be 10^{15} Hz with fitting parameters: $b_0 = 8.99 \times 10^{-11} \text{ erg cm}^{-2} \text{ s}^{-1}$, $b_1 = -2.14 \times 10^{-10} \text{ erg cm}^{-2} \text{ s}^{-1}$, $b_2 = 1.73 \times 10^{-10} \text{ erg cm}^{-2} \text{ s}^{-1}$ and $b_3 = -4.64 \times 10^{-11} \text{ erg cm}^{-2} \text{ s}^{-1}$, results in $\chi^2/\text{d.o.f.}$ of 4.01. Using this third-order polynomial would alter the constraints by no more than 5%, but it would increase $\chi^2/\text{d.o.f.}$, leading to a poorer fit to the data.

3.6.2 Detector calibration

Systematic errors in detector calibration arise from factors such as inaccuracies in converting count rates to flux, changing sensitivity over time (aging effects), and the Point Spread Function (PSF). In observations of M87, the PSF influences how light from its extended structure is spread and blurred by the telescope, affecting measurements of surface brightness, flux accuracy, and the ability to resolve fine details in both the central and outer regions.

The choice of background regions can also contribute to systematic effects, especially in cases where diffuse emissions from the galaxy or variations in the sky background are present. Furthermore, in observations of bright sources, coincidence loss occurs when multiple photons arrive in the same detector region within a short time, causing non-linear count rates.

For Swift-UVOT calibration, a systematic error of approximately 2.3% is assumed for individual UVOT photometric measurements, as reported in calibration studies by [274]. The analysis by [275] indicate a $\sim 1\%$ per year decline in UVOT sensitivity across all filters. More recently, [276] claimed that for the wavelength range $2800 \text{ \AA} < \lambda < 4000 \text{ \AA}$, the systematic uncertainty in the extraction of the source data is about 11.2%.

AstroSat-UVIT, a space-based telescope operating in the near-UV region, maintains systematic uncertainties below $\lesssim 15\%$ [257, 277]. Meanwhile, the ground-based Kanata telescope, subject to atmospheric and instrumental systematics, typically experiences uncertainties in the range of $\sim (3 - 5)\%$ [278, 279, 280].

Old infrared observations (for the historical data) of M87 conducted with the Spitzer telescope had a photometric uncertainty of approximately $\sim 3\%$ [281, 282]. Similarly, ultraviolet observations by the IUE exhibited an uncertainty in the range of $\sim (3 - 5)\%$ [283, 284]. For a more detailed discussion and rigorous analysis of these uncertainties, we refer the reader to the cited references.

As long as these uncertainties stay below the level of the statistical uncertainties, their influence on the results appears to be limited, with only minor variations observed in the constraints across different mass bins.

3.7 Conclusion

Due to its coupling with the SM photon, an axion-like particle (ALP) can decay into two photons. Assuming that ALPs account for the entire observed cold DM

abundance of the Universe, their decay near the central region of galaxies, which is expected to be DM-rich, can give rise to observable line emission signals.

Based on this, in this work we have constrained the ALP-two-photon coupling $g_{a\gamma\gamma}$ of the ALP DM with mass in the range $2\text{ eV} \lesssim m_a \lesssim 20\text{ eV}$, using the observation of M87 galaxy in infrared, optical and ultraviolet emission. For this we have used the dataset of the central region of M87, obtained by the EHT Multi-wavelength Science Working Group (MWL WG) during their 2018 observational campaign, as well as the old dataset for the same region collected over the years. The recent data were collected by Swift-UVOT, HST, Astrosat-UVIT and Kanata telescope. The historical data were collected by the spitzer space telescope and International Ultraviolet Explorer. Based on such data we exclude $g_{a\gamma\gamma} \gtrsim 10^{-11}\text{ GeV}^{-1} - 10^{-13}\text{ GeV}^{-1}$ in the above-mentioned mass window. Fig. [3.3] shows the results obtained in a conservative approach without assuming any background model. The χ^2 -fit analysis make the constraints stronger in the mass range $4\text{ eV} \lesssim m_a \lesssim 13\text{ eV}$ which is presented in Fig. [3.4] and Fig. [3.5]. The better constraints are obtained by Swift-UVOT, Astrosat-UVIT and Kanata telescope due to their wide field observations. Such constraints surpass the existing bounds on $g_{a\gamma\gamma}$ by an order of magnitude in the mass range $8\text{ eV} \lesssim m_a \lesssim 20\text{ eV}$. On the other hand, HST has a low sensitivity for the ALP DM decay signal due to its small angular resolution.

The constraints can be improved through increasing the sensitivity of the observing telescopes and onboard spectrometers. One can improve the bounds by enhancing the data reduction maneuvers from the Swift-UVOT telescope in the upcoming future. Future telescopes like the LUVOIR and on-board spectrograph LUMOS [285] will be able to obtain higher sensitivity for the Infrared-optical-UV emission lines, and thus will be able to further improve the constraints on the ALP DM decay. UVEX [286] can serve as a future telescope in the UV frequency range.

Chapter 4

Exploring Self-coupling of Fuzzy ALP DM via GWB

What we observe is not nature itself, but nature exposed to our method of questioning.

Werner Heisenberg, Physics and Philosophy, 1958

Fuzzy ultralight dark matter (ULDM) composed of Axion-like particles (ALPs) behave as an exceptionally cold form of dark matter while exhibiting wave-like effects on galactic scales, leading to the formation of solitonic cores in virialized dark matter halos. In realistic astrophysical environments, however, galactic centers are known to host supermassive black holes (SMBHs), whose strong gravitational fields can significantly alter the structure and dynamical role of these solitonic ULDM configurations.

The primary objective of the present chapter is to investigate the influence of ULDM solitons^a on the dynamics of supermassive black hole binaries and the resulting imprints on the stochastic gravitational-wave background (SGWB). In particular, the presence of a solitonic ULDM environment introduces an additional source of dynamical friction, which modifies the inspiral evolution of SMBH binaries beyond that driven solely by gravitational-wave emission. Such modifications can lead to departures from the canonical power-law form of the SGWB spectrum expected in vacuum.

Recent measurements of the SGWB by pulsar timing array (PTA) experiments provide a unique observational window into the environments of inspiraling SMBH binaries. While the observed signal is broadly consistent with expectations from gravitational-wave-dominated evolution, environmental effects associated with ULDM can induce measurable deviations. Consequently, PTA observations offer a sensitive probe of the properties of ULDM through its impact on binary dynamics, without requiring direct detection of the dark matter field.

^aIn the literature, solutions of the coupled Schrödinger–Poisson or Gross–Pitaevskii–Poisson equations are commonly referred to as solitons. The solitonic solutions discussed here are non-topological in nature and should not be confused with topological solitons or instantons; their stability arises from the balance between gravity and quantum pressure rather than from a conserved topological charge.

To model these effects, we describe ULDM as a non-relativistic scalar field minimally coupled to gravity and obtain solitonic dark matter profiles by solving the coupled Gross–Pitaevskii–Poisson (or equivalently Schrödinger–Poisson) equations in the presence of a central SMBH. Possible self-interaction effects are included insofar as they influence the soliton structure and constitute a central aspect of the present analysis. Using the resulting soliton profiles, we compute the dynamical friction experienced by SMBH binaries and assess its impact on the SGWB spectrum measured by PTA collaborations.

Several recent works have explored related but distinct aspects of ULDM and gravitational waves. Aghaie et al. [287] and Ghoshal & Strumia [288] constrained ULDM masses using NANOGrav data, but did not incorporate self-interactions in the soliton profile. Alonso-Álvarez et al. [289] demonstrated that self-interacting dark matter can assist SMBH binary hardening to address the final parsec problem, but worked within a particle dark matter framework rather than the wavelike, soliton-forming regime of ULDM. Glennon et al. [290] and Koo et al. [291] studied dynamical friction in self-interacting ULDM but did not connect their results to the observable SGWB spectrum or derive constraints from PTA data. The present work fills this gap by:

- (i) numerically solving the GPP equations with a quartic self-interaction term to obtain physically motivated solitonic density profiles for a range of both attractive and repulsive couplings $\hat{\lambda} \in [-1, +1]$;
- (ii) computing the resulting modification to the SGWB strain spectrum through soliton-induced dynamical friction on inspiralling SMBH binaries; and
- (iii) mapping the observable suppression features in the nanohertz band in the (m, λ) parameter space of the ULDM particles using current PTA observations from NANOGrav, EPTA, and PPTA.

Unlike approaches based on the particle scattering cross-section ratio σ/m , our framework begins from a fundamental scalar field action with a ϕ^4 self-interaction, providing a self-consistent connection between the microphysical coupling λ and the macroscopic solitonic dark matter density profile. Throughout this work, we assume that the ULDM particles carry no charge under Standard Model gauge groups and interact with ordinary matter exclusively through gravity. This makes the nanohertz SGWB a uniquely sensitive probe of ULDM self-coupling in a regime that is complementary to both galactic dynamics and gravitational lensing constraints.

The structure of this chapter is as follows. [Section 4.1](#) introduces the formation of ULDM solitons around SMBHs and outlines the method for solving the Gross–Pitaevskii–Poisson equations to obtain self-consistent solitonic profiles. [Section 4.2](#) examines the effect of these solitons on the gravitational wave emission from supermassive black hole binaries. The resulting implications for the SGWB are presented in [Section 4.3](#), and the main conclusions are summarized in [Section 4.4](#).

Throughout this work, natural units ($c = \hbar = 1$) are employed, though c and $\hbar$ are retained explicitly where clarity is required. The reduced Planck mass is defined as $M_{\text{Pl}} = \sqrt{\hbar c / 8\pi G_N} \simeq 2.4 \times 10^{18} \text{ GeV}$, and the solar mass is taken to be $M_\odot \simeq 2 \times 10^{30} \text{ kg}$.

4.1 ULDM Soliton around central SMBH

One of the characteristic predictions of ULDM models is the formation of a dense, macroscopic core at the centers of galaxies, commonly referred to as a soliton [190, 292, 293, 294, 295]. Owing to its extremely high occupation number, this form of dark matter behaves as a classical scalar field and is described by a wave equation [296, 297, 298]. The classical field theory action for this scalar field ϕ , is given by

$$S = \int d^4x \sqrt{-g} \left(\frac{M_{pl}^2}{2} R - \frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - U(\phi) \right), \quad (4.1)$$

where the potential has the form

$$U(\phi) = \frac{m^2 \phi^2}{2} + \frac{\lambda \phi^4}{4!}. \quad (4.2)$$

The coefficient λ here denotes the self-coupling parameter of the ULDM particles.

By varying the action with respect to ϕ , we obtain the equation of motion:

$$\partial_\alpha (\sqrt{-g} g^{\alpha\beta} \partial_\beta \phi) = \sqrt{-g} U'(\phi), \quad (4.3)$$

where $U'(\phi)$ denotes the derivative of the potential with respect to ϕ . R is the Ricci scalar, and g denotes $\det(g_{\mu\nu})$.

Strong evidence suggests that all large galaxies, and even dwarf galaxies, harbor a central SMBH [299, 300, 301]. ULDM particles can host this SMBH, feeding from it and evolving into a macroscopic soliton [302, 303, 304, 305, 306]. The system of an SMBH embedded within a scalar field dark matter halo is often described using a spherically symmetric metric [307, 308, 309, 310], with the Schwarzschild metric being the simplest example.

Alternatively, the cosmological ULDM scalar field can be analyzed by adopting the perturbed FLRW metric, as discussed in [292, 134, 311]. In galactic dynamics, the effects of cosmic expansion are negligible. Therefore, we can set the scale factor to unity and the Hubble parameter to zero, leading to the following metric:

$$ds^2 = -(1 + 2\Phi)(dx^0)^2 + (1 - 2\Phi)\delta_{ij} dx^i dx^j, \quad (4.4)$$

within the weak gravity limit, $\Phi \ll 1$, where the Newtonian potential Φ dominates the gravitational influence^b.

Under this metric, the equation of motion simplifies to:

$$\partial_0^2 \phi - \nabla^2 \phi + U'(\phi) = 2\Phi(2\partial_0^2 \phi + U'(\phi)). \quad (4.5)$$

The massive scalar field ϕ , which satisfies the Klein-Gordon equation and is minimally coupled to gravity, can be expressed in the non-relativistic limit as [292, 190],

^bAnisotropy effects were neglected in this analysis for simplicity.

$$\phi(t, \vec{x}) = \frac{1}{\sqrt{2m}} (e^{-imt} \Psi(t, \vec{x}) + \text{c.c.}), \quad (4.6)$$

where m represents the mass of the scalar field, and $\Psi(t, \vec{x})$ encapsulates its dynamical behavior. Here, the complex field Ψ varies slowly in both space and time, satisfying the conditions $|\nabla \Psi| \ll m|\Psi|$, $|\dot{\Psi}| \ll m|\Psi|$ and $|\ddot{\Psi}| \ll m|\dot{\Psi}|$.

In the regime of weak gravity and slowly varying fields, Ψ satisfies the Gross-Pitaevskii-Poisson (GPP) equation[190, 194, 195, 196, 197, 294, 295, 312]:

$$i \frac{\partial \Psi}{\partial t} = -\frac{\nabla^2 \Psi}{2m} + m\Phi\Psi + \frac{\lambda}{8m^3} |\Psi|^2 \Psi. \quad (4.7)$$

$$\nabla^2 \Phi = 4\pi G_N |\Psi|^2. \quad (4.8)$$

We seek spherically symmetric steady-state solutions, assuming an ansatz for^c $\Psi(t, r) = \chi(r)e^{-i\gamma t}$, where the ULDM density is given by $\rho(r) = \chi^2(r)$. The solution $\chi(r)$ is real, spherically symmetric, spatially localized, and without nodes. These solutions are known as solitons. The influence of a central black hole is incorporated via the term $m\Phi\Psi = m(V + \Phi_{BH})\Psi$, where V includes the self-potential and the black hole potential is given by, $\Phi_{BH} = -\frac{G_N M_{BH}}{r}$. Here, M_{BH} is the black hole mass, and G_N is Newton's constant. In the absence of self-interactions, the system is usually referred to as Schrödinger-Poisson system.

To solve the coupled equations numerically, we introduce dimensionless variables[294]: distance $\hat{r} = \frac{r}{\hbar/mc}$, wave function $\hat{\chi} = \frac{\sqrt{4\pi G_N \hbar}}{mc^2} \chi$, potential $\hat{V} = V/c^2$, eigen energy $\hat{\gamma} = \gamma/mc^2$. We also define dimensionless parameters for the black hole,

$$\hat{\alpha} = \frac{G_N M_{BH} m}{c\hbar} \quad (4.9)$$

and for the self interaction,

$$\hat{\lambda} = \frac{\lambda}{8} \left(\frac{m}{M_{\text{pl}}} \right)^{-2} = \left(\frac{\lambda}{1.35 \times 10^{-96}} \right) \left(\frac{m}{10^{-21} \text{eV}} \right)^{-2}. \quad (4.10)$$

Although the quartic self-coupling, λ of ultra-light particles is extremely small, it can become dynamically relevant in regions of high phase-space density. Even such a tiny interaction may significantly affect the stability and structure of ULDM halos, such as axion halos, as shown in several works [313, 314, 315, 316, 317, 318, 319, 320].

Using the dimensionless variables, the GPP equation takes the form

$$\begin{aligned} \hat{\gamma} \hat{\chi} &= \left(-\frac{1}{2} \hat{\nabla}^2 + \hat{V} - \frac{\hat{\alpha}}{\hat{r}} + 2\hat{\lambda} \hat{\chi}^2 \right) \hat{\chi}, \\ \hat{\nabla}^2 \hat{V} &= \hat{\chi}^2. \end{aligned} \quad (4.11)$$

Here, the parameters $\hat{\alpha}$ and $\hat{\lambda}$ are free and chosen to solve the system. The parameter $\hat{\gamma}$ is determined using a shooting method by applying boundary conditions to find solutions without nodes[321]. For numerical work, we impose $\hat{\chi}(\hat{r} = 0) \sim 1$

^c $\Psi(t, \vec{x}) = \chi(\vec{x})e^{-i\gamma t}$ and for spherical symmetry $\chi(\vec{x}) = \chi(r)$

along with the boundary conditions: $\hat{\chi}(\hat{r} = \infty) = 0$, $\hat{\chi}'(\hat{r} = 0) = 0$, $\hat{V}(\hat{r} = 0) = 0$, $\hat{V}'(\hat{r} = 0) = 0$. Although $\hat{\chi}(\hat{r} = 0) = 0$ may also yield solutions, these are not realistic for modeling physical soliton profiles, since the other dimensionless parameters are $\mathcal{O}(1)$. Details of the numerical procedure are provided in the Appendix B.

For each unique solution, the total mass of the soliton is given by

$$M = \frac{\hbar c}{G_N m} \int_0^\infty \hat{\chi}^2 \hat{r}^2 d\hat{r} = 1.33 \times 10^{11} M_\odot \left(\frac{10^{-21} \text{ eV}}{m} \right) \int_0^\infty \hat{\chi}^2 \hat{r}^2 d\hat{r}. \quad (4.12)$$

In dimensionless units, the soliton mass is expressed as

$$\hat{M} \equiv \frac{G_N M m}{\hbar c} = \int_0^\infty \hat{\chi}^2 \hat{r}^2 d\hat{r}. \quad (4.13)$$

The central ULDM density is derived from $\chi = \frac{mc^2}{\sqrt{4\pi G_N \hbar}} \hat{\chi}$ which is given by,

$$\rho = 2(M_{\text{pl}} m)^2 \hat{\chi}^2 \sim 10^{16} \left(\frac{m}{10^{-21} \text{ eV}} \right)^2 \hat{\chi}^2 M_\odot / \text{pc}^3. \quad (4.14)$$

Here this value is estimated without taking self-interactions. This density corresponds to a very large mass within a volume of $(\text{pc})^3$. Consequently, solutions assuming $\hat{\chi}(\hat{r} = 0) \sim 1$ yield excessively high densities.

In the absence of a central SMBH, setting $\hat{\alpha} = 0$ in the GPP equation, the Poisson equation implies $\hat{V} \sim \hat{L}^2 \hat{\chi}^2$ for a system of size $\hat{L}$ (in units of $\hbar/mc$). This leads to $\hat{L} \sim \hat{\chi}^{-1/2}$. If $\hat{\chi} \leq 1$, then $\hat{L}$ remains $\mathcal{O}(1)$. In dimensionful units without the central BH, the size of the soliton will be of the order of ULDM Compton wavelength, $\hbar/mc$ (i.e. $\frac{\hbar}{mc} \times \hat{L}$).

When a central BH is present ($\hat{\alpha} > 0$), for $\hat{\alpha} \ll 1$, the $\hat{V} \hat{\chi}$ term dominates over $\frac{\hat{\alpha}}{\hat{r}} \hat{\chi}$ in the GPP equation, keeping the soliton size $\mathcal{O}(1)$. When $\frac{\hat{\alpha}}{\hat{r}} \hat{\chi}$ dominates, the size becomes $\hat{L} \sim 1/\hat{\alpha}$, which requires $\hat{\alpha}$ to be at least $\mathcal{O}(1)$. Thus, the dimensionless size $\hat{L}$ remains $\mathcal{O}(1)$ in either case.

The soliton mass, determined from Eq. (4.13), is $\mathcal{O}(1)$ in dimensionless units, corresponding to a physical mass of $10^{11} M_\odot$, which exceeds the typical mass of a soliton in galaxies within the mass range of $M_{\text{halo}} \sim (10^{12} - 10^{13}) M_\odot$. The gravitational radius of the soliton is

$$R_G = \frac{G_N M}{c^2} = \hat{M} \frac{\hbar}{mc} \sim \frac{\hbar}{mc}, \quad (4.15)$$

comparable to the soliton size. Real solitons are expected to be larger, less dense, and much lighter. Mapping these estimated soliton densities and sizes to physical values requires scaling of the solutions and parameters in the GPP equation, which is discussed in the next section.

4.1.1 Scaling relation

The GPP system of equations has the scaling symmetry [190, 294, 312]

$$\begin{aligned} \hat{\chi} &\rightarrow \frac{1}{s^2} \hat{\chi}, & \hat{V} &\rightarrow \frac{1}{s^2} \hat{V}, & \hat{\alpha} &\rightarrow \frac{1}{s} \hat{\alpha}, & \hat{\gamma} &\rightarrow \frac{1}{s^2} \hat{\gamma}, \\ \hat{r} &\rightarrow s \hat{r}, & \hat{\lambda} &\rightarrow s^2 \hat{\lambda}, & \hat{M} &\rightarrow \frac{1}{s} \hat{M} \end{aligned}$$

s being a scaling parameter, which can be used to transform the dimensionless numerical solutions and parameters to their dimensionful physical values. Each of the term in GPP equation as well as the density scales as $1/s^4$. The ratio of the soliton mass and the black hole mass, M/M_{BH} is invariant under the scaling and is the primary parameter which stabilizes the shape of the soliton. In the limit $M/M_{BH} \lesssim 0.3$, we will get exponential solution just like a hydrogen atom when soliton profile is dominated by the BH gravity[294] .

Now we have to model the density profile of the soliton for a given halo mass as a function of ULDM mass using the numerical solutions as discussed in Appendix B.

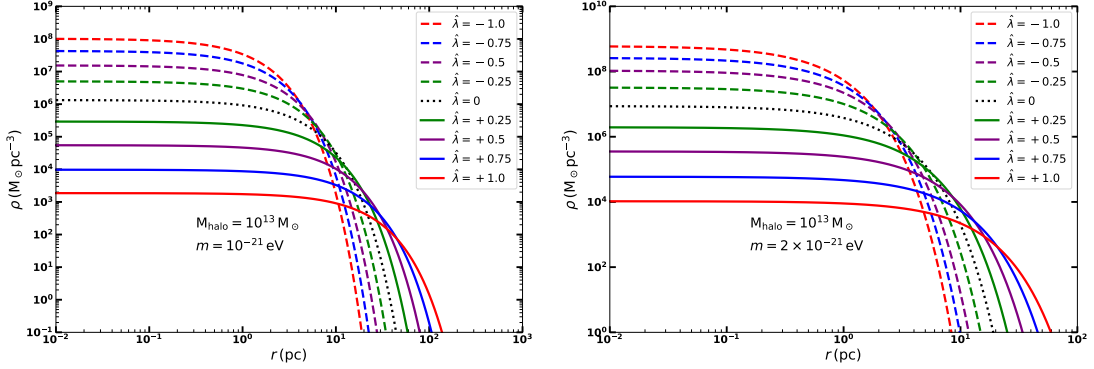


Figure 4.1: Numerical solutions to the GPP equation are presented for a halo mass of $10^{13}M_{\odot}$ and ULDM masses of 10^{-21}eV (Left) and $2 \times 10^{-21}\text{eV}$ (Right). The DM density profiles, shown in different colors, illustrate the effect of varying the dimensionless self-coupling parameter $\hat{\lambda}$, each with distinct characteristic signatures. The GPP equation was solved in dimensionful units, incorporating the gravitational influence of the SMBH through the appropriate values of the $\hat{\alpha}$ parameter corresponding to the given halo mass.

Over cosmic timescales, the stability of the solitonic core at the center of a galaxy depends on whether it can resist being accreted by the central SMBH. For ULDM particles, the timescale for this accretion is given by [322, 323],

$$t_{accr} = 5.6 \times 10^{12} \text{yr} \left(\frac{M_{BH}}{10^8 M_{\odot}} \right)^{-5} \left(\frac{m}{10^{-21} \text{eV}} \right)^{-6}. \quad (4.16)$$

To ensure that the solitonic core remains intact today, this accretion timescale is expected to be sufficiently long over cosmological durations. Since $t_{accr} \propto m^{-6}$, increasing the mass of the ULDM significantly shortens the accretion time. Therefore, requiring that the soliton has not yet been absorbed by the SMBH leads to an upper bound on the ULDM particle mass.

Once the SMBH completes its growth, it may dominate the central mass of the halo, exerting a perturbing force on the system's center. For a DM-only simulations, the mass of the soliton M_{sol} is related to the galactic halo mass via the relation[324]

$$M_{sol} = 2.67 \times 10^8 \left(\frac{M_{halo}}{10^{13} M_{\odot}} \right)^{1/3} \left(\frac{m}{10^{-21} \text{eV}} \right)^{-1} M_{\odot}. \quad (4.17)$$

Nevertheless, this relation was obtained in the absence of SMBHs: the simulation was performed by fixing the ULDM mass at $m = 10^{-22}$ eV, with halo masses in the range $M_{\text{halo}} = (10^9 - 5 \times 10^{11})M_{\odot}$. For $\hat{\lambda} = 0$, our treatment reproduces the SMBH-modified equilibrium solutions of [294] through their tabulated values of the dimensionless parameter $\hat{\alpha}$, and we extend this framework by introducing bosonic self-interactions into the GPP equation. To the best of our knowledge, no numerical calibration currently exists for systems including both an SMBH and self-interactions; we therefore retain the standard soliton–halo relation of Eq. (4.17) solely as a benchmark normalization between the halo and soliton masses, and the effects of attractive or repulsive self-interactions are then determined self-consistently from the modified equilibrium solutions. This provides a controlled extension of the existing SMBH–soliton framework while preserving the established SMBH treatment.

As discussed in Appendix B, we took the parameter $\hat{\alpha}$ that produces a solution consistent with the predicted soliton mass for the given halo mass. Next, we convert the dimensionless density profile into physical units and apply the scaling symmetry to adjust the soliton solution, so that its total mass aligns with the soliton-halo mass relation. This is done by comparing the mass of the scaled numerical solution to the soliton-halo mass relation in Eq. (4.17) for the given halo mass. The corresponding scaling factor is given by

$$s = \frac{c\hbar}{G_N m M_{\text{sol}}} \hat{M} = \frac{1.33 \times 10^{11}}{2.67 \times 10^8} \left(\frac{M_{\text{halo}}}{10^{13} M_{\odot}} \right)^{-1/3} \hat{M}. \quad (4.18)$$

Then using this s , we can scale the numerical solutions and other relevant quantities discussed above. For every initial value of $\hat{\lambda}$, we will get the s parameter for chosen $\hat{\alpha}$. The variation of scaling, s with respect to $\hat{\lambda}$ is shown in Section B.1. For attractive self interactions, s parameter gets larger in comparison to repulsive self interaction. This is obvious as the attractive self interaction makes the soliton core more dense and less in size as shown in Fig. [4.1].

Our study is confined to solitons composed of ULDM particles with masses in the range $\sim (5 \times 10^{-22} \text{ eV} - 2 \times 10^{-21} \text{ eV})$. We further restrict our analysis to galaxies with halo masses of $M_{\text{halo}} \sim 10^{13} M_{\odot}$, a mass range characteristic of massive early-type galaxies which constitute SMBHs at their centers.

4.1.2 Density Profiles

In addition to the soliton-halo mass relation, the SMBH mass is also correlated with the halo mass according to [325], dictated as follows:

$$\log_{10}(M_{\text{BH}}/M_{\odot}) = 8.18 + 1.55[\log_{10}(M_{\text{halo}}/M_{\odot}) - 13.0], \quad (4.19)$$

allowing us to predict the mass of the SMBH and the soliton core for a given halo mass. Assuming the soliton-halo mass relation holds true in the presence of an SMBH—treated as a perturber to the soliton-halo system, the gravitational potential of the SMBH modifies the soliton density profile.

We compute the density profiles in the presence of the SMBH, incorporating its influence through the parameter α . The coupled GPP equation, with fixed

halo mass, M_{halo} and the parameter α , are solved numerically using appropriate boundary conditions. These calculations yield density profiles for various values of the λ parameter while keeping α constant. From these solutions, the density is obtained by squaring the soliton solution χ . The ULDM density in physical units is then given by

$$\rho = 4.1 \times 10^{16} \left(\frac{m}{10^{-21}} \right)^2 \hat{\chi}^2 M_{\odot}/pc^3. \quad (4.20)$$

Finally, we apply the scaling symmetry relations detailed in the [Section B.1](#) to rescale the soliton density so that its total mass matches the value predicted by the soliton-halo mass relation.

[Fig. \[4.1\]](#) displays the density profiles for different values of $\hat{\lambda}$ while keeping the halo mass fixed at $10^{13}M_{\odot}$. The left panel corresponds to a ULDM mass of 10^{-21} eV while the right panel represents profiles for a ULDM mass of 2×10^{-21} eV. As shown in the figure, an attractive self-coupling ($\hat{\lambda} < 0$) results in more compact solitons—denser and smaller in size (dashed curves). In contrast, a repulsive self-coupling ($\hat{\lambda} > 0$) leads to solitons that are less dense and larger in size (solid curves). The black-dotted lines represent solitons with no self-coupling between ULDM particles.

4.2 DM influence on Gravitational Wave Background

Thus far, we have characterized solitonic DM profiles composed of ULDM with self-interactions, concentrated at the centers of galaxies. In this section, we study binary mergers of SMBH that occur within such solitonic environments. Our treatment assumes that the solitonic core is gravitationally perturbed by the central SMBH. This assumption is valid in scenarios where the soliton mass is subdominant compared to the black hole, specifically when the mass ratio $M_{sol}/M_{BH} \lesssim 0.3$ [[294](#), [194](#)]. In this regime, the SMBH dominates the gravitational potential.

In our analysis of the SGWB from SMBH inspirals embedded in ULDM solitons, we neglect the effects of tidal disruption or deformation of the solitonic cores. This is justified on several grounds. Tidal interactions between a solitonic DM core with the binary SMBH are generally suppressed due to the small gravitational coupling parameter, $M_{\bullet}m \sim 10^{-3}$ (where $M_{\bullet}$ is SMBH mass in the range $\sim (10^8 - 10^9)M_{\odot}$ and m is ULDM mass in the range ($m \sim 10^{-21}$ eV) introduced in Ref.[[326](#)].

In many astrophysical scenarios, the soliton is not fully self-gravitating but is instead stabilized by its host SMBH and the soliton behaves as a gravitationally intact bound state susceptible against the strong tidal disruption. The companion SMBH in the binary introduces only a weak tidal field at typical inspiral separations, and the deformation remains subdominant. Numerical and analytical studies (e.g., [[326](#), [327](#)]) further support that tidal effects are suppressed unless the binary reaches the very late stage of merger. Therefore, throughout most of the inspiral epoch relevant to the nHz GW spectrum observed by PTAs, solitonic cores remain gravitationally intact, and tidal disruption can be safely neglected in our present framework. Nonetheless, such effects could become relevant in high-precision studies of late-stage binary evolution and merit further investigation and beyond the scope of present work.

To model the influence of the dark matter environment on the binary evolution, we include the effect of dynamical friction from the ULDM distribution. In particular, the outer regions of the soliton contribute most significantly to the gravitational drag experienced by the SMBH. Before summing over all merger events for a fixed galaxy mass across cosmological distances, we first analyze a binary system consisting of two equal-mass SMBHs embedded within solitonic cores [288, 287, 328].

We consider a binary system of black holes with masses M_1 and M_2 , moving in circular orbits about their common center of mass, with a binary separation r . The angular orbital frequency ω is given by Kepler's third law,

$$\omega = \sqrt{\frac{G_N(M_1 + M_2)}{r^3}}. \quad (4.21)$$

In the nHz frequency band probed by PTAs, SMBH binaries are typically widely separated compared to their Schwarzschild radii. For one of the black holes with mass M_1 , the Schwarzschild radius is given by $R_{\text{Sch}} = 2G_N M_1$. At such separations, the binary dynamics remain well within the Newtonian regime. The corresponding orbital velocity is

$$v_{\text{orb}} = \omega r \sim \sqrt{\frac{R_{\text{Sch}}}{r}} \ll 1, \quad (4.22)$$

assuming equal-mass binaries.

The dominant mechanism for energy loss depends on the binary separation: at smaller separations, GW emission is the primary driver of orbital decay, whereas at larger separations, interactions with the surrounding environment from DM (through dynamical friction) play a more significant role. The frequency of the emitted GWs in the source frame is

$$f_s = \frac{\omega}{\pi}, \quad (4.23)$$

where the absence of a factor of 2 reflects the spin-2 nature of gravitational waves. For a GW source at redshift z , the observed frequency, f is related through

$$f = \frac{f_s}{1+z}. \quad (4.24)$$

Two primary mechanisms contribute to the orbital energy loss which are :

Gravitational Wave Emission The power radiated via GW emission is given by [329]

$$\mathcal{W}_{\text{GW}} = \frac{32}{5} G_N \mu^2 \omega^6 r^4, \quad (4.25)$$

where $\mu = \frac{M_1 M_2}{M_1 + M_2}$ represents the reduced mass. This follows from the quadrupole radiation of gravity waves. Systems with larger black holes (considering $M_1 \sim M_2$) dominate this radiation.

Dynamical Friction(DF) from ULDM Another process of energy loss is the dynamical friction(DF) which occurs when a mass moves in a background of ULDM particles. According to Refs. [330, 213, 331, 137], the dynamical friction force exerted on a test mass—here, the reduced mass μ of the binary black hole system, by

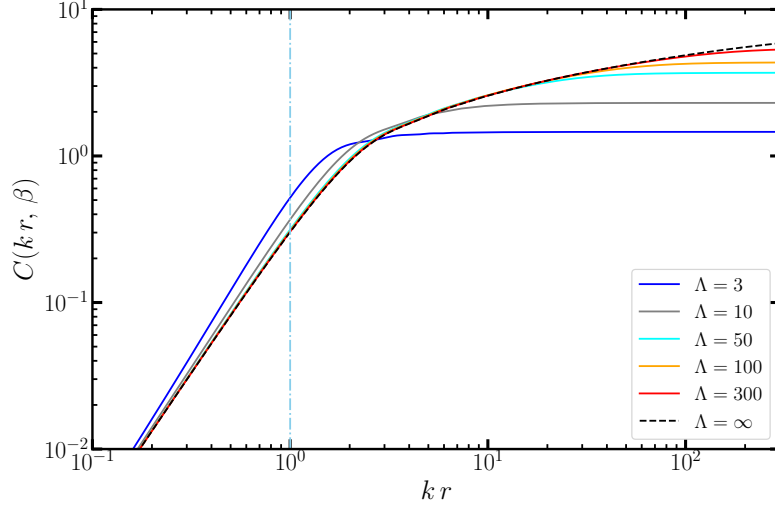


Figure 4.2: This figure illustrates the dimensionless function $C(kr, \beta)$, which enters the expression for the dynamical friction force experienced by a test object of mass μ moving with relative velocity v_{rel} through the ULDM [134, 330]. The cyan dot-dashed line is drawn at $kr = 1$, marking the transition between particle-like and wave-like behavior.

the ULDM background through which it moves, is given by

$$\vec{F}_{\text{DF}} = \frac{4\pi G_N^2 \mu^2 \rho}{v_{\text{rel}}^3} C(kr, \beta). \quad (4.26)$$

Here, v_{rel} denotes the relative velocity between the approaching black hole and ULDM particle, and ρ is given by Eq. (4.20).

In our case, the relative velocity v_{rel} between the approaching SMBH and the ULDM particles is comparable to the orbital velocity of the SMBH, i.e., $v_{\text{rel}} \approx \omega r$, for SMBH masses $\lesssim 10^{9.5} M_\odot$ and ULDM masses around 10^{-21} eV, corresponding to the nHz frequency range. With these into consideration, the energy dissipation due to this dynamical friction is

$$\mathcal{W}_{\text{DF}} = \vec{F}_{\text{DF}} \cdot \vec{v}_{\text{rel}} = \frac{4\pi G_N^2 \mu^2 \rho}{\omega r} C(kr, \beta). \quad (4.27)$$

The quantity $C(kr, \beta)$ accounts for gravitational feedback effects arising from infrared divergences due to the long-range nature of gravitational interactions. Depending on the values of kr and β , either the wave-like or particle-like character of ULDM becomes relevant. As thoroughly discussed in Appendix D of [134], the general behavior of this correction factor is illustrated in Fig. [4.2].

In our analysis, we consider the relevant physical scales that determine the value of $C(kr, \beta)$. The parameter β is defined as $\beta = 2\pi G\mu/(v^2\lambda)$, where the de Broglie wavelength of ULDM, λ is given by

$$\lambda = 2\pi \times 192 \text{ pc} \left(\frac{10^{-21} \text{ eV}}{m} \right) \left(\frac{10 \text{ km s}^{-1}}{v} \right), \quad (4.28)$$

while the dimensionless quantity Λ is given by $\Lambda = v^2 r / (G\mu)$. The product kr can then be estimated as $kr = \Lambda\beta$. In order to observe significant wave effects, the condition $kr \lesssim 1$ must be satisfied. As evident from this analysis, stronger wave effects correspond to smaller values of C , which in turn suppress the efficiency of dynamical friction.

In addition to the dynamical friction induced by an ULDM background, gaseous or dusty environments can also exert significant drag on inspiralling SMBHs [332, 333, 334]. However, in galactic centers and during SMBH inspiral, the motion often becomes transonic or subsonic (i.e. SMBH's velocity relative to the medium is $\lesssim 1$) owing to the high sound speed of the nuclear gas [335, 336, 334], leading to a reduced dynamical friction efficiency. Recent cosmological hydrodynamical simulations predict and calibrate DF models in realistic environments, testing both gas and collisionless components self-consistently [337]. These studies show that, while gas drag can assist orbital decay, DF from collisionless matter (stars and dark matter) generally dominates beyond the first Gyr of evolution, especially for black holes with masses $M_\bullet \gtrsim 10^7 M_\odot$. Additionally the environmental effects such as the eccentricity of the orbital motion, gaseous circumbinary disk and loss-cone hardening effects onto the SGWB have been briefly discussed in Appendix Section B.2.

We now proceed to calculate the GW strain spectrum generated by SMBH merger events occurring across different galaxies and redshifts within ULDM environments. The GW spectrum is characterized by the dimensionless strain amplitude $h_c(f)$, which is related to the GW energy density spectrum $\Omega_{\text{GW}}(f)$ via [210]:

$$\begin{aligned} \rho_c \Omega_{\text{GW}}(f) &= \frac{\pi}{4G_N} f^2 h_c^2(f) \\ &= \int \frac{dz}{1+z} \int dX \frac{dn_s}{dz dX} \left(f_s \frac{dE_{\text{GW}}}{df_s} \right) \bigg|_{f_s=f(1+z)}, \end{aligned} \quad (4.29)$$

where $dn_s/dz dX$ is the number of merger events per unit comoving volume between redshifts z and $z + dz$. X collectively denotes all SMBH sources distributed over that redshifts. Computation related to the SMBH merger rate are given in the Appendix Section B.3.

The strain can be expressed as a power law:

$$h_c^2(f) = \left[A_{\text{GW}} \left(\frac{f}{f_{\text{ref}}} \right)^\beta \right]^2. \quad (4.30)$$

Here, $\rho_c = 3H_0^2/8\pi G_N$ is the critical density of the universe, and A_{GW} is the amplitude and f_{ref} is the reference frequency at which the PTA measurements provide the best accuracy.

Including energy losses from ULDM dynamical friction and GW emission, the GW energy spectrum is [338]:

$$\frac{dE_{\text{GW}}}{df_s} = \frac{\mu}{3} \left[\pi G_N (M_1 + M_2) \right]^{2/3} f_s^{-1/3} \frac{1}{1 + \frac{\mathcal{W}_{\text{DF}}}{\mathcal{W}_{\text{GW}}}}. \quad (4.31)$$

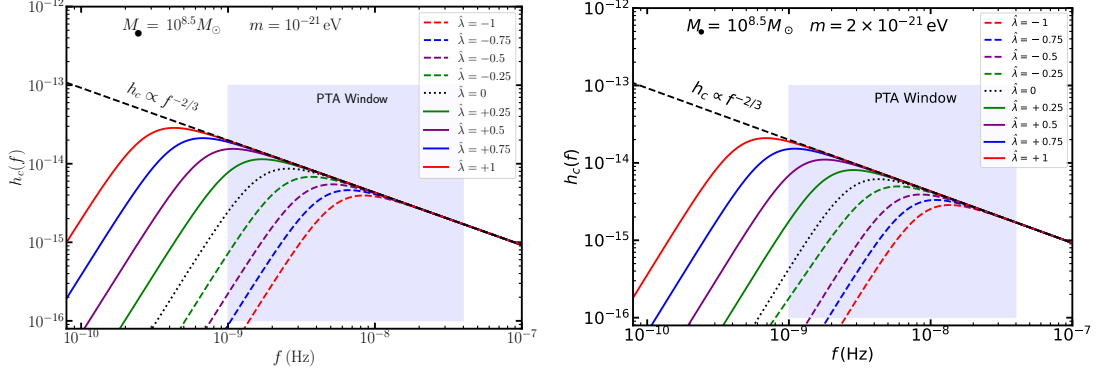


Figure 4.3: The GW strains corresponding to different DM density profiles, characterized by distinct self-coupling parameters, are displayed. **Left:** For a ULDM particle mass of 10^{-21} eV and an SMBH binary mass of $10^{8.5} M_\odot$, the strains corresponding to different values of the self-coupling parameter $\hat{\lambda}$ are shown, exhibiting deviations from the power-law behavior ($h_c \propto f^{-2/3}$). **Right:** The corresponding strains for a ULDM particle mass of 2×10^{-21} eV are presented. The blue-shaded region represents the PTA observation window for the SGWB generated by black hole mergers in a solitonic dark matter environment with self-interacting ULDM particles.

The density ρ is determined using the soliton density profiles outlined in [Section 4.1.2](#). We assume the mass ratio of the merging SMBHs is $q_\bullet$, such that $M_1 + M_2 = M_\bullet(1 + q_\bullet)$, where $M_1 \equiv M_\bullet$ and $q_\bullet \lesssim 1$ in this analysis. Now we identify the distance scale at which the dynamical friction becomes the dominant energy loss mechanism. We define the critical radius r_{cr} of the soliton core as the distance where the two dissipative forces are equal, i.e., $\mathcal{W}_{GW} = \mathcal{W}_{DF}$. Assuming the central soliton density $\rho_0 = \rho(r = 0)$, the critical radius is expressed as:

$$r_{cr} = \sqrt[11]{\frac{64 G_N^5 M_\bullet^7 (1 + q_\bullet)^7}{25 \pi^2 \rho_0^2 C(k, r, \beta)^2}}. \quad (4.32)$$

This expression depends on the soliton density, which itself is a function of the ULDM mass, the self-coupling parameter λ , and the mass of the merging SMBH. For a benchmark case where $M_\bullet = 10^{8.5} M_\odot$, $m = 10^{-21}$ eV and dimensionless self coupling $\hat{\lambda} = -1.0$, the critical radius evaluates to approximately 8×10^{-3} pc. Using Kepler's law, the corresponding radial frequency f_{cr} is 3.72 nHz, well within the detection range of current PTA experiments. Putting the loss of powers $\mathcal{W}_{GW}$ and $\mathcal{W}_{DF}$, we get the frequency spectra,

$$\frac{dE_{GW}}{df_s} = \frac{\pi^{2/3} q_\bullet G_N^{2/3} M_\bullet^{5/3} f_s^{-1/3}}{3(1 + q_\bullet)^{1/3}} \frac{1}{1 + \left(\frac{f_{cr}}{f_s}\right)^{11/3} e^{-\left(\frac{f_{sol}}{f_s}\right)^{2/3}}}. \quad (4.33)$$

From the above expression, it is evident that in the absence of $\mathcal{W}_{DF}$, the energy loss follows the standard power-law behaviour, $dE_{GW}/df_s \propto f_s^{-1/3}$. However, the presence of dynamical friction, influenced by the soliton dark matter density, distorts the strain amplitude. The radial frequency f_{sol} corresponds to the soliton radius r_{sol} of the dark matter density. When $f_{cr} < f_{sol}$, i.e. the critical radius is greater

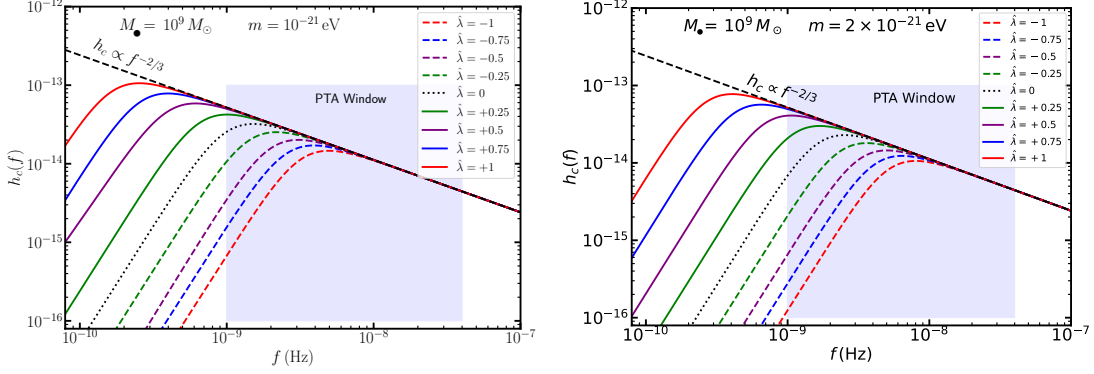


Figure 4.4: The GW strains corresponding to different DM density profiles, characterized by distinct self-coupling parameters, are displayed. **Left:** For a ULDM mass of 10^{-21}eV and an SMBH mass of $10^9 M_\odot$ of the binary, the strains associated with varying $\hat{\lambda}$ parameters are shown, demonstrating deviations from the power-law behavior ($h_c \propto f^{-2/3}$). **Right:** The strains are shown for a ULDM mass of $2 \times 10^{-21}\text{eV}$. The blue-shaded region marks the PTA observation window for the SGWB.

than the soliton radius the energy loss due to GW emission is more prominent than the dynamical friction. In this case the spectrum follows the standard power-law behaviour. On the other hand, when the critical radius is inside the soliton core ($f_{cr} > f_{sol}$), the dynamical friction is dominant and the frequency spectrum follows a completely different frequency dependence. Due to the soliton core made of the ULDM, the strain of the spectrum shows a sudden dip.

The depth of this dip depends on the relative location of the two characteristic frequencies entering Eq. (4.33): f_{cr} , at which the dynamical-friction and GW-emission powers balance ($W_{DF} = W_{GW}$), and f_{sol} , which marks the frequency corresponding to the outer edge of the soliton core beyond which the ULDM density falls off exponentially. When f_{cr} and f_{sol} lie close to one another, the onset of dynamical-friction domination coincides with the region where the available soliton density is itself declining steeply, so the two suppression effects compound rather than acting over separate frequency ranges, producing a deeper and narrower dip in the strain spectrum.

4.3 Results

4.3.1 The GW spectrum

Using Eq. (4.29) and substituting the energy-loss power from Eq. (4.33), the GW strain is derived. The soliton density profile influences this spectrum through dynamical friction, as discussed in the previous section. The resulting strains are shown in Fig. [4.3] and Fig. [4.4] for merging SMBH masses of $10^{8.5} M_\odot$ and $10^9 M_\odot$, respectively, with the subfigures corresponding to two different ULDM masses.

In the absence of self-interactions among ULDM particles, represented by $\hat{\lambda} = 0$, the GW spectrum follows the black dotted line. Non-zero values of $\hat{\lambda}$, which denote self-interactions, modify the spectrum. Specifically, positive $\hat{\lambda}$, corresponding to

repulsive soliton cores, are depicted with solid lines, while negative $\hat{\lambda}$, representing attractive soliton cores, are shown with dashed lines. If GWs are generated purely by mergers without incorporating the effects of dynamical friction (i.e., ignoring dark matter densities), the GW strain follows a simple power-law behavior ($h_c \propto f^{-2/3}$), represented by the black dashed line. The slope of the spectrum deviates from this power-law due to the influence of dark matter density, where the soliton serves as the dark matter profile in this analysis.

The PTA experiments operate in the frequency range of 1–30 nHz and can probe features of the ULDM, such as self-interactions. Within this window, PTA observations can place qualitative limits on the value of $\hat{\lambda}$. This qualitative limits would carry characteristic signatures detectable through the GW spectrum.

4.3.2 Typical values of self coupling $\hat{\lambda}$

The interplay between the strength of gravity and the self-interaction of ULDM particles plays a crucial role in the formation of soliton cores. From the GPP equations, the ratio of the gravitational potential energy, $m\Phi\Psi$, to the self-interaction energy, $\frac{\lambda}{8m^3}|\Psi|^3$, is given by:

$$\frac{\text{gravity}}{\text{self interaction}} = \frac{8m^4\Phi}{\lambda|\Psi|^2} = \frac{4}{\lambda} \left(\frac{M_{\text{pl}}}{m} \right)^{-2} \left(\frac{L}{m^{-1}} \right)^2, \quad (4.34)$$

L being the physical size of the soliton. This shows that gravity becomes more significant than self-interactions as the soliton size increases. Using the definition of the dimensionless self-coupling parameter $\hat{\lambda}$, this ratio can be rewritten as:

$$\frac{\text{gravity}}{\text{self interaction}} = \frac{1}{2\hat{\lambda}}. \quad (4.35)$$

For gravity to dominate over self-interactions, we require:

$$-\frac{1}{2} \leq \hat{\lambda} \leq \frac{1}{2}. \quad (4.36)$$

Here, a negative $\hat{\lambda}$ corresponds to attractive self-interactions, while a positive $\hat{\lambda}$ corresponds to repulsive self-interactions. In this study, we do not focus on cases where self-interactions dominate gravity, as the GPP equations enter into a highly non-linear regime, introducing additional complexities.

Eq. (4.36) gives a dimensionless limit of the self-coupling parameter $\hat{\lambda}$, as long as gravity dominates over the self interaction of ULDM particles and that the soliton cores at the galactic centers remain stable.

4.3.3 Allowed values of λ

The typical frequency band accessible to PTA experiments for the SGWB lies in the 1–30 nHz range. Within this window, a sharp dip in the strain spectrum can constrain the self-coupling parameter $\hat{\lambda}$, which sets the dimensionful coupling strength λ . For an equal-mass binary with a black hole mass of $10^{8.5}M_{\odot}$, attractive self-coupling values (0 to -1) fall inside the PTA band when the ULDM mass is 10^{-21} eV. For

lighter ULDM masses, the spectral peak shifts to lower frequencies, leaving only negative $\hat{\lambda}$ values within the probed region, as illustrated in the top panel of Fig. [4.5].

From theoretical considerations—namely that black hole gravity dominates over soliton self-gravity—the maximum allowed value of $\hat{\lambda}$ is restricted to ± 0.5 . This theoretical bound, shown as the red dashed lines in the top panel of Fig. [4.5], limits the extent to which the SGWB can constrain the coupling. In addition, the purple shaded region marks the parameter space excluded by the accretion timescale constraint according to Eq. (4.16).

From Eq. (4.10), the strength of the coupling for a ULDM mass of 10^{-21} eV is approximately $\mathcal{O}(10^{-96})$ for both attractive and repulsive self-couplings. The $\hat{\lambda}$ parameter scales with the s^2 factor, increasing the coupling strength by several orders of magnitude. For instance, for a ULDM mass of 10^{-21} eV and $\hat{\lambda} = -0.5$, the coupling strength λ increases to approximately -3.4×10^{-92} . Similarly, for a ULDM mass of 2×10^{-21} eV and $\hat{\lambda} = -0.5$, λ is approximately -1.33×10^{-91} .

As the scaling s increases, the strength of λ grows even larger for positive $\hat{\lambda}$ values. However, repulsive self-coupling ($+\hat{\lambda}$) in the range $\hat{\lambda} \in (0, 0.4)$ falls within the observational window for a ULDM mass of 10^{-21} eV. The entire range, i.e. $\hat{\lambda} \in (-1, 0.6)$, lies within the observational window for a ULDM mass of 2×10^{-21} eV, while for lighter ULDM masses, the repulsive coupling parameters shift outside the observational window, as seen in Figs. Fig. [4.3] and Fig. [4.4]. For higher black hole masses of the binary populations, the amplitude of the SGWB changes without affecting the spectral slope.

For large strengths of attractive self-interactions, the balance between gravity, self-interactions, and quantum pressure cannot be maintained, resulting in the absence of stable soliton solutions. A stable soliton exists only up to a maximum possible soliton mass, M_c , given by [313, 317]:

$$M_c \approx 10.2 \frac{M_{\text{pl}}}{(-\lambda)^{1/2}}, \quad (4.37)$$

where the negative sign indicates an attractive self-interaction. For a galaxy with a mass $M_{\text{halo}} = 10^{13} M_{\odot}$, the soliton mass can be determined using Eq. (4.13) and then incorporating an appropriate scaling. From this, an upper limit is established to ensure the soliton mass does not exceed M_c , given by:

$$|-\lambda| \leq 1.77 \times 10^{-91} \left(\frac{m}{10^{-21}} \right)^2. \quad (4.38)$$

In the bottom panel of Fig. [4.5], the maximum possible coupling is represented by dashed lines corresponding to different masses of ULDM and SMBH mass of $10^{8.5} M_{\odot}$ in binary populations having equal masses. The cyan shading over the blue and red plots delineates the range of self-coupling limits which can be probed from the SGWB spectrum amplitude.

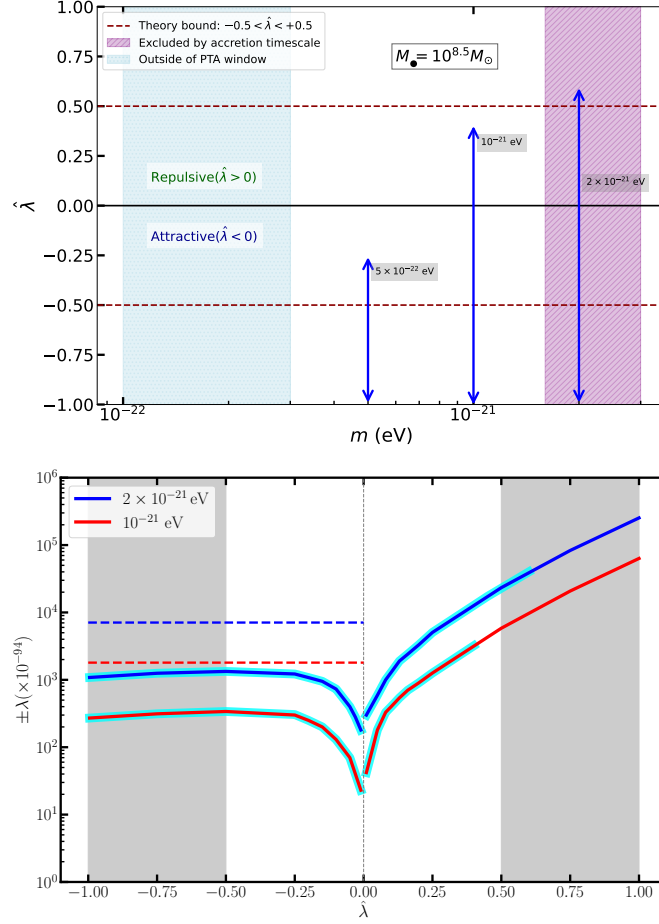


Figure 4.5: *Top*: Qualitative limits on the parameter space $(m, \hat{\lambda})$. The sign of $\hat{\lambda}$ indicates whether attractive ($\hat{\lambda} < 0$) or a repulsive coupling ($\hat{\lambda} > 0$). Horizontal red dashed lines show the bound $-0.5 < \hat{\lambda} < +0.5$ based on theoretical considerations discussed in [Section 4.3.2](#). The region $m > 1.6 \times 10^{-21}$ eV is excluded by accretion timescale limits (using [Eq. \(4.16\)](#)), while $m < 3 \times 10^{-22}$ eV lies outside the PTA sensitivity window. The blue arrows signify the region over which PTA can probe the $\hat{\lambda}$ parameter for 3 different ULDM masses. *Bottom*: The estimated values of the λ parameter are shown along with the dimensionless $\hat{\lambda}$, considering both positive and negative values (-ve estimate of λ is for the -ve values of the $\hat{\lambda}$ and positive λ is for the positive $\hat{\lambda}$ values). The plots are presented for two different ULDM masses but keeping the SMBH mass to $10^{8.5} M_{\odot}$. The cyan shading over the plots represents the range that can be probed through PTA observations, while the gray shaded region indicates the excluded region based on theoretical considerations. Dashed lines (blue and red), corresponding to the attractive self-coupling parameters derived using [Eq. \(4.38\)](#), illustrate the upper limit on the strength of λ .

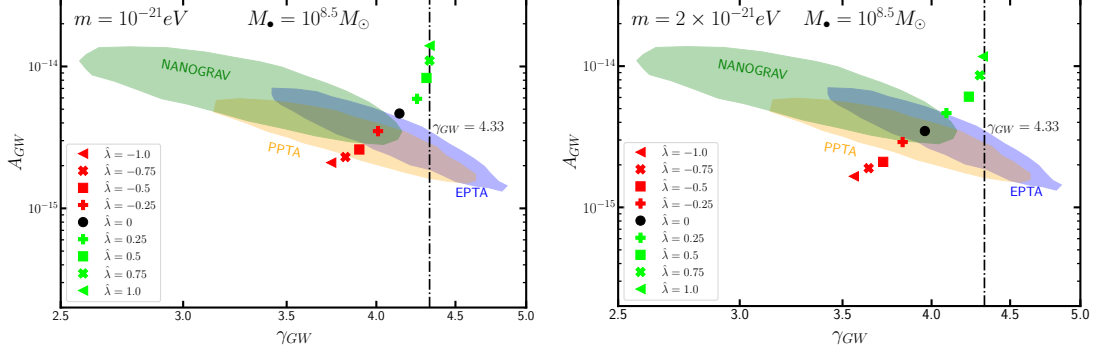


Figure 4.6: The amplitude of the GW strain with change of γ_{GW} parameter for PTA observations by NANOGrav, EPTA, and PPTA are represented by the shaded regions[339] within 2σ confidence interval for the GW signal. For a reference frequency (f_{PTA}) of 1 nHz, the amplitude and γ_{GW} values corresponding to different $\hat{\lambda}$ parameters are depicted using various legends. For a equal mass non-eccentric binary systems, keeping the SMBH mass fixed at $10^{8.5} M_{\odot}$, the correlation between A_{GW} and γ_{GW} for a ULDM mass of 10^{-21} eV is shown in the left figure, while for a ULDM mass of 2×10^{-21} eV, it is displayed in the right figure. The dashed vertical line at the $\gamma_{\text{GW}} = 4.33$ is the corresponding value for a GW strain that exactly follows the power-law $h_c \propto f^{-2/3}$.

4.3.4 Relation between the GW amplitude and the γ_{GW} parameter

PTAs probe the SGWB generated by SMBH mergers in the nHz frequency range. The PTA data analyzes the strain spectrum, typically parameterized as a power law: $h_c(f) = A_{\text{GW}}(f/f_{\text{PTA}})^{\beta}$ where f_{PTA} is a reference frequency. The GW spectrum is characterized by the amplitude A_{GW} and the slope parameter $\gamma_{\text{GW}} = 3 - 2\beta$.

Astrophysical environments in the vicinity of SMBHs can alter the spectral slope, causing it to deviate from that expected solely from gravitational wave radiation losses. In circular binaries driven primarily by GW radiation losses, the spectral index is $\beta = -2/3$, corresponding to $\gamma_{\text{GW}} = 13/3$. However, if dynamical friction dominates, deviations in γ_{GW} arise, providing a pathway to constrain the self-coupling parameter of ULDM.

Tentative SGWB signal found at frequencies above 1 nHz as fit by NANOGrav, EPTA, and PPTA often expressed through correlation of the GW amplitude A_{GW} and the γ_{GW} parameter[339]. So we set the reference frequency at $f_{\text{PTA}} = 1$ nHz for this analysis.

The Fig. [4.6] describes the relation between the GW amplitude and the slope parameter γ_{GW} for the two cases : the left one is for ULDM mass $m = 10^{-21}$ eV and the right one is for $m = 2 \times 10^{-21}$ eV and keeping the SMBH mass fixed at $10^{8.5} M_{\odot}$. The effect of the self-coupling which are specified by different legends are shown in the figures. As depicted, the range of the $\hat{\lambda}$ parameter that lies within the correlated regions defined by the current PTA observations is relatively narrow.

4.4 Discussion and Conclusion

In this study, we have explored the possibility of placing qualitative limits on the ULDM self-coupling λ using the recent PTA observation. The ULDM with masses $\sim (10^{-22} - 10^{-20})$ eV, can form solitonic cores at the centers of galaxies. The inspiral of black holes within the dense dark matter environments such as these soliton cores can impact the stochastic gravitational wave background. The presence of the self interaction of the ULDM particles would modify the GW strain through dynamical friction. We have inspected which part of the SMBH population can contribute to the GW signal. For non-eccentric equal mass SMBH binaries, the SMBH masses around $\sim 10^{8.5} M_\odot$ mostly contribute to the strain h_c at the nHz frequencies of the spectrum.

The empirical relations of the soliton mass and central SMBH mass infer that the GPP equation solved numerically for the DM halo mass $\sim 10^{13} M_\odot$ would give the best results on the GW spectrum. We numerically solve the GPP equations by fixing the central black hole parameter α , corresponding to a halo mass of $10^{13} M_\odot$. For each solution, the self-coupling parameter λ is also fixed, as shown in Fig. [B.1] (presented in Appendix B). The corresponding dark matter densities are then obtained by applying the appropriate scaling to convert the results into physical units. The presence of self coupling λ with characteristic signature will change the DM profiles as shown in Fig. [4.1]. These individual ULDM density profiles (through the solitonic core formation) leads to various GW strains, which are presented in Fig. [4.3] and Fig. [4.4] corresponds to different self-coupling parameter λ .

The nHz window of the spectrum is particularly sensitive to recent observations from EPTA, PPTA, and NANOGRV. Deviations in the strain caused by the λ parameter from the usual power-law dependence ($h_c \propto f^{-2/3}$) can help identify the range and characteristic signature of λ that falls within this frequency band. Based on this analysis, we can place constraints on λ for specific ULDM masses, as shown in Fig. [4.5] for a fixed SMBH mass. Our results indicate that attractive self-coupling ($\lambda < 0$) can be best probed through GW strains when ULDM masses are in the range $\sim (3 \times 10^{-22} - 2 \times 10^{-21})$ eV, given an SMBH mass of $10^{8.5} M_\odot$. In contrast, repulsive self-coupling ($\lambda > 0$) can be probed within a specific range of the dimensionless $\hat{\lambda}$ parameter for ULDM masses around $\sim (10^{-21} - 2 \times 10^{-21})$ eV. The soliton cores made out of the ULDM particles with masses exceeding $\gtrsim 1.6 \times 10^{-21}$ eV, are excluded due to accretion time constraints caused by the gravitational attraction of central SMBHs in galaxies. Therefore, accounting for the self-coupling effects of ULDM particles requires precise tuning of both the ULDM mass and the masses of the merging SMBHs.

The novelty of this work lies in modeling to probe ULDM self-coupling through soliton formation and its impact on the SGWB, in contrast to conventional studies that constrain dark matter interactions via particle scattering characterized by σ/m , σ being scattering cross section of the ULDM particles. Our analysis connects the self-interacting scalar field framework, starting from a fundamental ϕ^4 interaction, to the evolution of physical dark matter density profiles given the halo mass of galaxies through solving GPP equation. By linking soliton-induced dynamical friction to observable suppression features in the nHz SGWB spectrum, this study provides a complementary and physically motivated approach to understanding ULDM phe-

nomenology. Advanced data-driven techniques can be employed in the future for a more thorough investigation.

Upcoming PTA experiments, such as IPTA30 [340], are expected to reach frequencies up to $\mathcal{O}(100)$ nHz and analyze SMBH populations with masses as low as $M_{\bullet} \simeq 10^6 M_{\odot}$ [341], providing a more rigorous probe of self-interactions of ULDM particles. In addition to PTA detection of GWs from SMBHs orbiting soliton cores, the planned LISA mission, operating in the milli-Hz frequency band, may also provide detection capabilities[342].

Chapter 5

Conclusions and Outlook

We three.

We're not a crowd;

We're not even company —

My echo, My shadow, And me ...

Ruskin Bond, Whistling in the Dark

Axion-like particles (ALPs) constitute a well-motivated class of candidates for physics beyond the Standard Model, with origins in global symmetry breaking, string compactifications, and solutions to the strong CP problem. Their feeble interactions with Standard Model particles and wide mass range make them elusive in terrestrial experiments, but at the same time render them particularly amenable to investigation through astrophysical and cosmological observations. This thesis has explored how diverse astrophysical systems—black holes, galaxies, and the stochastic gravitational-wave background—can serve as complementary laboratories for probing ALP properties across multiple mass and coupling regimes.

The first part of this work focused on the interaction between photons and ALPs in the strong gravitational and magnetic environments surrounding black holes. Through the photon–ALP coupling, photons propagating in magnetized plasma near black holes can convert into ALPs, leading to observable consequences such as a dimming of electromagnetic signals. By deriving a general expression for photon–ALP conversion in spherically symmetric black hole spacetimes, this study demonstrated that supermassive black holes provide a natural setting where conversion probabilities can become appreciable over astrophysically relevant length scales. In particular, for ALP masses below $\mathcal{O}(100)$ neV and photon energies in the X-ray and gamma-ray bands, conversion can occur over distances comparable to the Schwarzschild radius of a supermassive black hole.

Applying this framework to realistic black hole environments revealed that objects such as M87* are promising targets for probing photon–ALP conversion, especially in the presence of moderately strong magnetic fields and low plasma densities.

While current observations, such as those by the Event Horizon Telescope, are limited to radio frequencies, future high-resolution observations in higher-energy bands may enable direct tests of this mechanism. More broadly, this analysis illustrates how black hole spacetimes can act as precision probes of ALP couplings through their influence on photon propagation.

The second component of this thesis investigated the decay of ALP dark matter into two photons and its observational signatures in galactic environments. Assuming that ALPs constitute the dominant component of dark matter, their decay in dark-matter-rich regions can produce narrow spectral features in the infrared, optical, and ultraviolet bands. Using multiwavelength observations of the central region of the M87 galaxy, including recent data from space- and ground-based telescopes, this work derived robust constraints on the ALP–photon coupling for ALP masses in the eV range.

The analysis demonstrated that current galactic observations already surpass existing bounds in certain mass windows, highlighting the power of astrophysical searches for decaying dark matter. Importantly, this approach relies minimally on assumptions about astrophysical backgrounds and can be systematically improved with enhanced instrumental sensitivity and future missions. These results establish galaxies as effective spectroscopic laboratories for testing ALP scenarios that are otherwise inaccessible to laboratory experiments.

The third and final part of this thesis addressed the role of ultralight ALPs as fuzzy ultralight dark matter and explored how their wave-like nature manifests in gravitational wave observables. Ultralight ALPs can form solitonic cores at the centers of dark matter halos, whose structure is governed by the coupled Gross–Pitaevskii–Poisson (GPP) equations. In the presence of supermassive black holes, these solitonic configurations modify the dynamics of inspiraling black hole binaries through dynamical friction, thereby altering the SGWB.

By numerically solving the GPP equations and incorporating the resulting dark matter profiles into models of black hole binary evolution, this work demonstrated that PTA observations provide a sensitive probe of ultralight dark matter properties. Deviations from the canonical gravitational-wave power-law spectrum in the nHz frequency band encode information about the underlying dark matter environment, including the mass of the ultralight scalar field and the structure of solitonic cores. This analysis establishes a novel connection between fundamental scalar field dynamics and gravitational-wave phenomenology, complementing traditional particle-physics approaches to dark matter constraints.

Taken together, the three studies presented in this thesis illustrate the breadth and versatility of astrophysical probes of axion-like particles. Black holes probe ALP interactions through photon propagation in extreme spacetimes; galaxies constrain ALP decays through multiwavelength spectroscopy; and gravitational-wave observations offer a new window into the wave-like nature of ultralight ALPs and their impact on compact-object dynamics. Each probe accesses a distinct region of ALP parameter space, yet all are united by the common theme that astrophysical systems amplify otherwise feeble effects into observable signatures.

Looking ahead, the prospects for ALP searches through astrophysical observa-

tions are highly promising. Advances in high-energy imaging of black hole environments, improved spectroscopic sensitivity in galactic observations, and the rapidly evolving landscape of gravitational-wave astronomy—particularly with next-generation PTA datasets and space-based detectors such as LISA—will substantially enhance our ability to test ALP scenarios. Moreover, combining multiple astrophysical probes within a unified framework will be essential for disentangling ALP signatures from conventional astrophysical processes and for building a coherent picture of physics beyond the Standard Model.

Bibliography

- [1] S. L. Glashow, *Nucl. Phys.* **22**, 579 (1961).
- [2] S. Weinberg, *Phys. Rev. Lett.* **19**, 1264 (1967).
- [3] A. Salam, *Conf. Proc. C* **680519**, 367 (1968).
- [4] D. J. Gross and F. Wilczek, *Phys. Rev. Lett.* **30**, 1343 (1973).
- [5] H. D. Politzer, *Phys. Rev. Lett.* **30**, 1346 (1973).
- [6] H. Fritzsch, M. Gell-Mann, and H. Leutwyler, *Phys. Lett. B* **47**, 365 (1973).
- [7] P. W. Higgs, *Phys. Rev. Lett.* **13**, 508 (1964).
- [8] F. Englert and R. Brout, *Phys. Rev. Lett.* **13**, 321 (1964).
- [9] G. S. Guralnik, C. R. Hagen, and T. W. B. Kibble, *Phys. Rev. Lett.* **13**, 585 (1964).
- [10] G. Aad *et al.* (ATLAS), *Phys. Lett. B* **716**, 1 (2012), [arXiv:1207.7214 \[hep-ex\]](#) .
- [11] S. Chatrchyan *et al.* (CMS), *Phys. Lett. B* **716**, 30 (2012), [arXiv:1207.7235 \[hep-ex\]](#) .
- [12] T. D. Lee and C. N. Yang, *Phys. Rev.* **104**, 254 (1956).
- [13] G. 't Hooft, *Nucl. Phys. B* **35**, 167 (1971).
- [14] G. 't Hooft, *Phys. Rev. Lett.* **37**, 8 (1976).
- [15] V. C. Rubin and W. K. Ford, Jr, *Astrophys. J.* **159**: 379-403(Feb 1970). (1970), [10.1086/150317](#).
- [16] V. C. Rubin, N. Thonnard, and W. K. Ford, Jr., *Astrophys. J.* **238**, 471 (1980).
- [17] N. Aghanim *et al.* (Planck), *Astron. Astrophys.* **641**, A6 (2020), [Erratum: *Astron. Astrophys.* 652, C4 (2021)], [arXiv:1807.06209 \[astro-ph.CO\]](#) .
- [18] R. H. Cyburt, B. D. Fields, K. A. Olive, and T.-H. Yeh, *Rev. Mod. Phys.* **88**, 015004 (2016), [arXiv:1505.01076 \[astro-ph.CO\]](#) .
- [19] B. D. Fields, K. A. Olive, T.-H. Yeh, and C. Young, *JCAP* **03**, 010 (2020), [Erratum: *JCAP* 11, E02 (2020)], [arXiv:1912.01132 \[astro-ph.CO\]](#) .
- [20] F. Zwicky, *Helvetica Physica Acta* **6**, 110 (1933).
- [21] D. Clowe, M. Bradac, A. H. Gonzalez, M. Markevitch, S. W. Randall, C. Jones, and D. Zaritsky, *Astrophys. J. Lett.* **648**, L109 (2006), [arXiv:astro-ph/0608407](#)

- [22] A. D. Sakharov, *JETP Lett.* **5**, 24 (1967).
- [23] M. Dine and A. Kusenko, *Rev. Mod. Phys.* **76**, 1 (2003), [arXiv:hep-ph/0303065](#).
- [24] L. Canetti, M. Drewes, and M. Shaposhnikov, *New J. Phys.* **14**, 095012 (2012), [arXiv:1204.4186 \[hep-ph\]](#).
- [25] Y. Fukuda *et al.* (Super-Kamiokande), *Phys. Rev. Lett.* **81**, 1562 (1998), [arXiv:hep-ex/9807003](#).
- [26] Q. R. Ahmad *et al.* (SNO), *Phys. Rev. Lett.* **89**, 011301 (2002), [arXiv:nucl-ex/0204008](#).
- [27] C. A. Baker *et al.*, *Phys. Rev. Lett.* **97**, 131801 (2006), [arXiv:hep-ex/0602020](#).
- [28] R. D. Peccei and H. R. Quinn, *Phys. Rev. Lett.* **38**, 1440 (1977).
- [29] R. D. Peccei and H. R. Quinn, *Phys. Rev. D* **16**, 1791 (1977).
- [30] S. Weinberg, *Phys. Rev. Lett.* **40**, 223 (1978).
- [31] F. Wilczek, *Phys. Rev. Lett.* **40**, 279 (1978).
- [32] E. Witten, *Phys. Lett. B* **149**, 351 (1984).
- [33] P. Svrcek and E. Witten, *JHEP* **06**, 051 (2006).
- [34] S. Weinberg, *Phys. Rev. D* **11**, 3583 (1975).
- [35] A. A. Belavin, A. M. Polyakov, A. S. Schwartz, and Y. S. Tyupkin, *Phys. Lett. B* **59**, 85 (1975).
- [36] C. G. Callan, Jr., R. F. Dashen, and D. J. Gross, *Phys. Lett. B* **63**, 334 (1976).
- [37] R. Jackiw and C. Rebbi, *Phys. Rev. Lett.* **37**, 172 (1976).
- [38] M. Dine and W. Fischler, *Phys. Lett. B* **120**, 137 (1983).
- [39] L. F. Abbott and P. Sikivie, *Phys. Lett. B* **120**, 133 (1983).
- [40] J. Preskill, M. B. Wise, and F. Wilczek, *Phys. Lett. B* **120**, 127 (1983).
- [41] S. L. Adler, *Phys. Rev.* **177**, 2426 (1969).
- [42] W. A. Bardeen, *Phys. Rev.* **184**, 1848 (1969).
- [43] W. A. Bardeen, *Nuclear Physics B* **75**, 246 (1974).
- [44] G. 't Hooft, *Phys. Rev. Lett.* **37**, 8 (1976).
- [45] G. 't Hooft, *Phys. Rev. D* **14**, 3432 (1976).
- [46] R. J. Crewther, *Acta Phys. Austriaca Suppl.* **19**, 47 (1978).
- [47] M. Tanabashi *et al.* (Particle Data Group), *Phys. Rev. D* **98**, 030001 (2018).
- [48] C. Alexandrou, J. Finkenrath, L. Funcke, K. Jansen, B. Kostrzewa, F. Pittler, and C. Urbach, *Phys. Rev. Lett.* **125**, 232001 (2020), [arXiv:2002.07802 \[hep-lat\]](#).

-
- [49] M. Dine, W. Fischler, and M. Srednicki, *Phys. Lett. B* **104**, 199 (1981).
 - [50] F. Wilczek, *Phys. Rev. Lett.* **40**, 279 (1978).
 - [51] J. E. Kim, *Phys. Rev. Lett.* **43**, 103 (1979).
 - [52] M. A. Shifman, A. I. Vainshtein, and V. I. Zakharov, *Nucl. Phys. B* **166**, 493 (1980).
 - [53] A. R. Zhitnitsky, *Sov. J. Nucl. Phys.* **31**, 260 (1980).
 - [54] H. Georgi, D. B. Kaplan, and L. Randall, *Phys. Lett. B* **169**, 73 (1986).
 - [55] W. A. Bardeen and S. H. H. Tye, *Phys. Lett. B* **74**, 229 (1978).
 - [56] D. B. Kaplan, *Nucl. Phys. B* **260**, 215 (1985).
 - [57] M. Srednicki, *Nucl. Phys. B* **260**, 689 (1985).
 - [58] G. G. Raffelt, *Lect. Notes Phys.* **741**, 51 (2008), [arXiv:hep-ph/0611350](#) .
 - [59] P. A. R. Ade *et al.* (BICEP/Keck), *Phys. Rev. D* **103**, 042002 (2021), [arXiv:2011.03483 \[astro-ph.CO\]](#) .
 - [60] M. S. Turner, F. Wilczek, and A. Zee, *Phys. Lett. B* **125**, 35 (1983), [Erratum: *Phys.Lett.B* 125, 519 (1983)].
 - [61] M. S. Turner, *Phys. Rev. D* **33**, 889 (1986).
 - [62] K. J. Bae, J.-H. Huh, and J. E. Kim, *JCAP* **09**, 005 (2008), [arXiv:0806.0497 \[hep-ph\]](#) .
 - [63] L. Visinelli and P. Gondolo, *Phys. Rev. D* **80**, 035024 (2009), [arXiv:0903.4377 \[astro-ph.CO\]](#) .
 - [64] O. Wantz and E. P. S. Shellard, *Phys. Rev. D* **82**, 123508 (2010), [arXiv:0910.1066 \[astro-ph.CO\]](#) .
 - [65] P. Fox, A. Pierce, and S. D. Thomas, (2004), [arXiv:hep-th/0409059](#) .
 - [66] M. Beltran, J. Garcia-Bellido, and J. Lesgourgues, *Phys. Rev. D* **75**, 103507 (2007), [arXiv:hep-ph/0606107](#) .
 - [67] M. P. Hertzberg, M. Tegmark, and F. Wilczek, *Phys. Rev. D* **78**, 083507 (2008), [arXiv:0807.1726 \[astro-ph\]](#) .
 - [68] J. Hamann, S. Hannestad, G. G. Raffelt, and Y. Y. Y. Wong, *JCAP* **06**, 022 (2009), [arXiv:0904.0647 \[hep-ph\]](#) .
 - [69] P. A. R. Ade *et al.* (Planck), *Astron. Astrophys.* **571**, A22 (2014), [arXiv:1303.5082 \[astro-ph.CO\]](#) .
 - [70] T. Higaki, K. S. Jeong, and F. Takahashi, *Phys. Lett. B* **734**, 21 (2014), [arXiv:1403.4186 \[hep-ph\]](#) .
 - [71] D. J. E. Marsh, D. Grin, R. Hlozek, and P. G. Ferreira, *Phys. Rev. Lett.* **113**, 011801 (2014), [arXiv:1403.4216 \[astro-ph.CO\]](#) .
 - [72] L. Visinelli and P. Gondolo, *Phys. Rev. Lett.* **113**, 011802 (2014), [arXiv:1403.4594 \[hep-ph\]](#) .
 - [73] S. Andriamonje *et al.* (CAST), *JCAP* **04**, 010 (2007), [arXiv:hep-ex/0702006](#) .
-

- [74] V. Anastassopoulos *et al.* (CAST), *Nature Phys.* **13**, 584 (2017), [arXiv:1705.02290 \[hep-ex\]](#) .
- [75] K. Altenmüller *et al.* (CAST), *Phys. Rev. Lett.* **133**, 221005 (2024), [arXiv:2406.16840 \[hep-ex\]](#) .
- [76] E. Armengaud *et al.* (IAXO), *JCAP* **06**, 047 (2019), [arXiv:1904.09155 \[hep-ph\]](#) .
- [77] S. J. Asztalos, G. Carosi, C. Hagmann, D. Kinion, K. van Bibber, M. Hotz, L. J. Rosenberg, G. Rybka, J. Hoskins, J. Hwang, P. Sikivie, D. B. Tanner, R. Bradley, J. Clarke, and ADMX Collaboration, *Phys. Rev. Lett.* **104**, 041301 (2010), [arXiv:0910.5914 \[astro-ph.CO\]](#) .
- [78] N. Du *et al.* (ADMX), *Phys. Rev. Lett.* **120**, 151301 (2018), [arXiv:1804.05750 \[hep-ex\]](#) .
- [79] T. Braine *et al.* (ADMX), *Phys. Rev. Lett.* **124**, 101303 (2020), [arXiv:1910.08638 \[hep-ex\]](#) .
- [80] C. Bartram *et al.* (ADMX), *Phys. Rev. Lett.* **127**, 261803 (2021), [arXiv:2110.06096 \[hep-ex\]](#) .
- [81] G. Carosi *et al.* (ADMX), *Phys. Rev. Lett.* **135**, 191001 (2025), [arXiv:2504.07279 \[hep-ex\]](#) .
- [82] C. M. Adair *et al.*, *Nature Commun.* **13**, 6180 (2022), [arXiv:2211.02902 \[hep-ex\]](#) .
- [83] J. Jaeckel and A. Ringwald, *Ann. Rev. Nucl. Part. Sci.* **60**, 405 (2010), [arXiv:1002.0329 \[hep-ph\]](#) .
- [84] A. Ringwald, *Phys. Dark Univ.* **1**, 116 (2012), [arXiv:1210.5081 \[hep-ph\]](#) .
- [85] A. Ringwald, *J. Phys. Conf. Ser.* **485**, 012013 (2014), [arXiv:1209.2299 \[hep-ph\]](#) .
- [86] Y. Chikashige, R. N. Mohapatra, and R. D. Peccei, *Phys. Lett. B* **98**, 265 (1981).
- [87] G. B. Gelmini and M. Roncadelli, *Phys. Lett. B* **99**, 411 (1981).
- [88] F. Wilczek, *Phys. Rev. Lett.* **49**, 1549 (1982).
- [89] Z. G. Berezhiani and M. Y. Khlopov, *Sov. J. Nucl. Phys.* **51**, 739 (1990).
- [90] J. Jaeckel, *Phys. Lett. B* **732**, 1 (2014), [arXiv:1311.0880 \[hep-ph\]](#) .
- [91] E. Witten, *Phys. Lett. B* **149**, 351 (1984).
- [92] J. P. Conlon, *JHEP* **05**, 078 (2006), [arXiv:hep-th/0602233](#) .
- [93] P. Svrcek and E. Witten, *JHEP* **06**, 051 (2006), [arXiv:hep-th/0605206](#) .
- [94] A. Arvanitaki, S. Dimopoulos, S. Dubovsky, N. Kaloper, and J. March-Russell, *Phys. Rev. D* **81**, 123530 (2010), [arXiv:0905.4720 \[hep-th\]](#) .
- [95] B. S. Acharya, K. Bobkov, and P. Kumar, *JHEP* **11**, 105 (2010), [arXiv:1004.5138 \[hep-th\]](#) .
- [96] M. Cicoli, M. Goodsell, and A. Ringwald, *JHEP* **10**, 146 (2012),

- arXiv:1206.0819 [hep-th] .
- [97] G. Lazarides, C. Panagiotakopoulos, and Q. Shafi, *Phys. Rev. Lett.* **56**, 432 (1986).
 - [98] K.-S. Choi, I.-W. Kim, and J. E. Kim, *JHEP* **03**, 116 (2007), arXiv:hep-ph/0612107 .
 - [99] K.-S. Choi, H. P. Nilles, S. Ramos-Sanchez, and P. K. S. Vaudrevange, *Phys. Lett. B* **675**, 381 (2009), arXiv:0902.3070 [hep-th] .
 - [100] C. O'Hare, <https://github.com/cajohare/AxionLimits>.
 - [101] P. Arias, D. Cadamuro, M. Goodsell, J. Jaeckel, J. Redondo, and A. Ringwald, *JCAP* **06**, 013 (2012), arXiv:1201.5902 [hep-ph] .
 - [102] D. J. E. Marsh and J. Silk, *Mon. Not. Roy. Astron. Soc.* **437**, 2652 (2014), arXiv:1307.1705 [astro-ph.CO] .
 - [103] K. Kadota, Y. Mao, K. Ichiki, and J. Silk, *JCAP* **06**, 011 (2014), arXiv:1312.1898 [hep-ph] .
 - [104] N. Blinov, M. J. Dolan, P. Draper, and J. Kozaczuk, *Phys. Rev. D* **100**, 015049 (2019), arXiv:1905.06952 [hep-ph] .
 - [105] R. Ballou *et al.* (OSQAR), *Phys. Rev. D* **92**, 092002 (2015), arXiv:1506.08082 [hep-ex] .
 - [106] F. Della Valle, A. Ejlli, U. Gastaldi, G. Messineo, E. Milotti, R. Pengo, G. Ruoso, and G. Zavattini, *Eur. Phys. J. C* **76**, 24 (2016), arXiv:1510.08052 [physics.optics] .
 - [107] Y. Kirita *et al.* (SAPPHIRES), *JHEP* **06**, 138 (2025), arXiv:2409.01805 [hep-ex] .
 - [108] J. M. Batllori, Y. Gu, D. Horns, M. Maroudas, and J. Ulrichs, *Phys. Rev. D* **109**, 123001 (2024), arXiv:2305.12969 [hep-ex] .
 - [109] D. Banerjee *et al.* (NA64), *Phys. Rev. Lett.* **125**, 081801 (2020), arXiv:2005.02710 [hep-ex] .
 - [110] F. Abudinén *et al.* (Belle-II), *Phys. Rev. Lett.* **125**, 161806 (2020), arXiv:2007.13071 [hep-ex] .
 - [111] M. Ablikim *et al.* (BESIII), *Phys. Lett. B* **838**, 137698 (2023), arXiv:2211.12699 [hep-ex] .
 - [112] F. Capozzi, B. Dutta, G. Gurung, W. Jang, I. M. Shoemaker, A. Thompson, and J. Yu, *Phys. Rev. D* **108**, 075019 (2023), arXiv:2307.03878 [hep-ph] .
 - [113] S. Knapen, T. Lin, H. K. Lou, and T. Melia, *Phys. Rev. Lett.* **118**, 171801 (2017), arXiv:1607.06083 [hep-ph] .
 - [114] J. Jaeckel and M. Spannowsky, *Phys. Lett. B* **753**, 482 (2016), arXiv:1509.00476 [hep-ph] .
 - [115] A. M. Sirunyan *et al.* (CMS), *Phys. Lett. B* **797**, 134826 (2019), arXiv:1810.04602 [hep-ex] .
 - [116] G. Aad *et al.* (ATLAS), *JHEP* **03**, 243 (2021), [Erratum: *JHEP* 11, 050

- (2021)], [arXiv:2008.05355 \[hep-ex\]](#) .
- [117] D. Bauman, *Lectures on Cosmology* (lecture notes).
- [118] D. Horns and M. Meyer, *JCAP* **02**, 033 (2012), [arXiv:1201.4711 \[astro-ph.CO\]](#) .
- [119] L. Di Luzio, M. Giannotti, E. Nardi, and L. Visinelli, *Phys. Rept.* **870**, 1 (2020), [arXiv:2003.01100 \[hep-ph\]](#) .
- [120] A. Friedland, M. Giannotti, and M. Wise, *Phys. Rev. Lett.* **110**, 061101 (2013), [arXiv:1210.1271 \[hep-ph\]](#) .
- [121] V. L. Fish, K. Akiyama, K. L. Bouman, A. A. Chael, M. D. Johnson, S. S. Doeleman, L. Blackburn, J. F. C. Wardle, and W. T. Freeman (Event Horizon Telescope), *Galaxies* **4**, 54 (2016), [arXiv:1607.03034 \[astro-ph.IM\]](#) .
- [122] K. Akiyama *et al.* (Event Horizon Telescope), *Astrophys. J. Lett.* **875**, L1 (2019), [arXiv:1906.11238 \[astro-ph.GA\]](#) .
- [123] K. Akiyama *et al.* (Event Horizon Telescope), *Astrophys. J. Lett.* **875**, L2 (2019), [arXiv:1906.11239 \[astro-ph.IM\]](#) .
- [124] K. Akiyama *et al.* (Event Horizon Telescope), *Astrophys. J. Lett.* **875**, L3 (2019), [arXiv:1906.11240 \[astro-ph.GA\]](#) .
- [125] K. Akiyama *et al.* (Event Horizon Telescope), *Astrophys. J. Lett.* **875**, L4 (2019), [arXiv:1906.11241 \[astro-ph.GA\]](#) .
- [126] K. Akiyama *et al.* (Event Horizon Telescope), *Astrophys. J. Lett.* **875**, L5 (2019), [arXiv:1906.11242 \[astro-ph.GA\]](#) .
- [127] K. Akiyama *et al.* (Event Horizon Telescope), *Astrophys. J. Lett.* **875**, L6 (2019), [arXiv:1906.11243 \[astro-ph.GA\]](#) .
- [128] K. Nomura, K. Saito, and J. Soda, *Phys. Rev. D* **107**, 123505 (2023), [arXiv:2212.03020 \[hep-ph\]](#) .
- [129] C. Csaki, N. Kaloper, and J. Terning, *Phys. Rev. Lett.* **88**, 161302 (2002), [arXiv:hep-ph/0111311](#) .
- [130] Y. Grossman, S. Roy, and J. Zupan, *Phys. Lett. B* **543**, 23 (2002), [arXiv:hep-ph/0204216](#) .
- [131] A. Mirizzi, G. G. Raffelt, and P. D. Serpico, *Phys. Rev. D* **72**, 023501 (2005), [arXiv:astro-ph/0506078](#) .
- [132] Y.-S. Song and W. Hu, *Phys. Rev. D* **73**, 023003 (2006), [arXiv:astro-ph/0508002](#) .
- [133] L. D. Duffy and K. van Bibber, *New J. Phys.* **11**, 105008 (2009), [arXiv:0904.3346 \[hep-ph\]](#) .
- [134] L. Hui, J. P. Ostriker, S. Tremaine, and E. Witten, *Phys. Rev. D* **95**, 043541 (2017), [arXiv:1610.08297 \[astro-ph.CO\]](#) .
- [135] F. Chadha-Day, J. Ellis, and D. J. E. Marsh, *Sci. Adv.* **8**, abj3618 (2022), [arXiv:2105.01406 \[hep-ph\]](#) .
- [136] C. B. Adams *et al.*, in *Snowmass 2021* (2022) [arXiv:2203.14923 \[hep-ex\]](#) .

-
- [137] M. Cirelli, A. Strumia, and J. Zupan, (2024), [arXiv:2406.01705 \[hep-ph\]](#) .
 - [138] B. M. Roach, S. Rossland, K. C. Y. Ng, K. Perez, J. F. Beacom, B. W. Grefenstette, S. Horiuchi, R. Krivonos, and D. R. Wik, *Phys. Rev. D* **107**, 023009 (2023), [arXiv:2207.04572 \[astro-ph.HE\]](#) .
 - [139] J. L. Bernal, A. Caputo, G. Sato-Polito, J. Mirocha, and M. Kamionkowski, *Phys. Rev. D* **107**, 103046 (2023), [arXiv:2208.13794 \[astro-ph.CO\]](#) .
 - [140] T. Bessho, Y. Ikeda, and W. Yin, *Phys. Rev. D* **106**, 095025 (2022), [arXiv:2208.05975 \[hep-ph\]](#) .
 - [141] S. Roy, C. Blanco, C. Dessert, A. Prabhu, and T. Temim, (2023), [arXiv:2311.04987 \[hep-ph\]](#) .
 - [142] A. Dekker, E. Peerbooms, F. Zimmer, K. C. Y. Ng, and S. Ando, *Phys. Rev. D* **104**, 023021 (2021).
 - [143] H. Wang *et al.*, *Phys. Rev. D* **110**, 103007 (2024), [arXiv:2311.05476 \[astro-ph.CO\]](#) .
 - [144] P. Carenza, G. Lucente, and E. Vitagliano, *Phys. Rev. D* **107**, 083032 (2023), [arXiv:2301.06560 \[hep-ph\]](#) .
 - [145] S. Porras-Bedmar, M. Meyer, and D. Horns, *Phys. Rev. D* **110**, 103501 (2024), [arXiv:2407.10618 \[astro-ph.CO\]](#) .
 - [146] D. Grin, G. Covone, J.-P. Kneib, M. Kamionkowski, A. Blain, and E. Jullo, *Phys. Rev. D* **75**, 105018 (2007), [arXiv:astro-ph/0611502](#) .
 - [147] M. Regis, M. Taoso, D. Vaz, J. Brinchmann, S. L. Zoutendijk, N. F. Bouché, and M. Steinmetz, *Phys. Lett. B* **814**, 136075 (2021), [arXiv:2009.01310 \[astro-ph.CO\]](#) .
 - [148] A. Caputo, M. Regis, M. Taoso, and S. J. Witte, *JCAP* **03**, 027 (2019), [arXiv:1811.08436 \[hep-ph\]](#) .
 - [149] D. Wadekar and Z. Wang, *Phys. Rev. D* **106**, 075007 (2022), [arXiv:2111.08025 \[hep-ph\]](#) .
 - [150] E. Todarello, M. Regis, J. Reynoso-Cordova, M. Taoso, D. Vaz, J. Brinchmann, M. Steinmetz, and S. L. Zoutendijke, *JCAP* **05**, 043 (2024), [arXiv:2307.07403 \[astro-ph.CO\]](#) .
 - [151] R. Janish and E. Pinetti, (2023), [arXiv:2310.15395 \[hep-ph\]](#) .
 - [152] W. Yin *et al.*, (2024), [arXiv:2402.07976 \[astro-ph.CO\]](#) .
 - [153] M. Regis, M. Taoso, and J. Terol Calvo, *JCAP* **05**, 008 (2025), [arXiv:2412.12286 \[astro-ph.CO\]](#) .
 - [154] E. Pinetti, (2025), [arXiv:2503.11753 \[hep-ph\]](#) .
 - [155] A. K. Saha, S. Bouri, A. Das, A. Dubey, and R. Laha, (2025), [arXiv:2503.14582 \[hep-ph\]](#) .
 - [156] E. Todarello and M. Regis, (2024), [arXiv:2412.02543 \[hep-ph\]](#) .
 - [157] E. Todarello, (2025), [arXiv:2506.19962 \[astro-ph.CO\]](#) .
-

- [158] G. C. Perola and M. Tarenghi, *The Astrophysics Journal* **240**, 447 (1980).
- [159] J. A. Biretta, C. P. Stern, and D. E. Harris, *The Astronomical Journal* **101**, 1632 (1991).
- [160] V. Despringre, D. Fraix-Burnet, and E. Davoust, *Astronomy and Astrophysics* **309**, 375 (1996).
- [161] J. A. Biretta, W. B. Sparks, and F. Macchetto, *The Astrophysics Journal* **520**, 621 (1999).
- [162] H. L. Marshall, B. P. Miller, D. S. Davis, E. S. Perlman, M. W. Wise, C. R. Canizares, and D. E. Harris, *Astrophys. J.* **564**, 683 (2002), [arXiv:astro-ph/0109160](#) .
- [163] E. S. Perlman, D. E. Harris, J. A. Biretta, W. B. Sparks, and F. D. Macchetto, *The Astrophysical Journal* **599**, L65 (2003).
- [164] H. Beuther, E. B. Churchwell, and C. F. McKee, in *Protostars and Planets V* (2006) [arXiv:astro-ph/0602012](#) .
- [165] Y. Shi, G. H. Rieke, D. C. Hines, K. D. Gordon, and E. Egami, *Astrophys. J.* **655**, 781 (2007), [arXiv:astro-ph/0610494](#) .
- [166] J. C. Tan, H. Beuther, and F. Walter, *Astrophys. J.* **689**, 775 (2008), [arXiv:astro-ph/0610488](#) .
- [167] E. S. Perlman, R. E. Mason, C. Packham, N. A. Levenson, M. Elitzur, J. J. Schaefer, M. Imanishi, W. B. Sparks, and J. Radomski, *Astrophys. J.* **663**, 808 (2007), [arXiv:0704.1156 \[astro-ph\]](#) .
- [168] S. K. Ghosh, P. Joseph, A. Kumar, J. Postma, C. S. Stalin, A. Subramaniam, S. N. Tandon, I. V. Barve, A. Devaraj, K. George, V. Girish, J. B. Hutchings, P. U. Kamath, S. Kathiravan, J. P. Lancelot, D. Leahy, P. K. Mahesh, R. Mohan, S. Nagabhushana, A. K. Pati, N. K. Rao, K. Sankarasubramanian, P. Sreekumar, and S. Sriram, *Journal of Astrophysics and Astronomy* **42**, 20 (2021), [arXiv:2012.13525 \[astro-ph.IM\]](#) .
- [169] J. C. Algaba *et al.* (Event Horizon Telescope, Fermi-LAT, H.E.S.S., MAGIC, VERITAS, EAVN), *Astrophys. J. Lett.* **911**, L11 (2021), [arXiv:2104.06855 \[astro-ph.HE\]](#) .
- [170] J. C. Algaba *et al.* (Event Horizon Telescope - Multi-wavelength science working group, Event Horizon Telescope, Fermi Large Area Telescope, H.E.S.S., MAGIC, VERITAS, EAVN, Event Horizon Telescope Multi-wavelength Science), *Astronomy and Astrophysics* (2024), [10.1051/0004-6361/202450497](#), [arXiv:2404.17623 \[astro-ph.HE\]](#) .
- [171] E. Corbelli and P. Salucci, *Mon. Not. Roy. Astron. Soc.* **311**, 441 (2000), [arXiv:astro-ph/9909252](#) .
- [172] A. Refregier, *Ann. Rev. Astron. Astrophys.* **41**, 645 (2003), [arXiv:astro-ph/0307212](#) .
- [173] S. W. Allen, A. E. Evrard, and A. B. Mantz, *Ann. Rev. Astron. Astrophys.* **49**, 409 (2011), [arXiv:1103.4829 \[astro-ph.CO\]](#) .
- [174] V. A. Rubakov, *CERN Yellow Rep. School Proc.* **5**, 129 (2022),

- [arXiv:1912.04727 \[hep-ph\]](#) .
- [175] R. A. Flores and J. R. Primack, *Astrophys. J. Lett.* **427**, L1 (1994), [arXiv:astro-ph/9402004](#) .
 - [176] B. Moore, *Nature* **370**, 629 (1994).
 - [177] J. S. Bullock and M. Boylan-Kolchin, *Ann. Rev. Astron. Astrophys.* **55**, 343 (2017), [arXiv:1707.04256 \[astro-ph.CO\]](#) .
 - [178] J. Zavala and C. S. Frenk, *Galaxies* **7**, 81 (2019), [arXiv:1907.11775 \[astro-ph.CO\]](#) .
 - [179] P. Bode, J. P. Ostriker, and N. Turok, *Astrophys. J.* **556**, 93 (2001), [arXiv:astro-ph/0010389](#) .
 - [180] M. Viel, J. Lesgourgues, M. G. Haehnelt, S. Matarrese, and A. Riotto, *Phys. Rev. D* **71**, 063534 (2005), [arXiv:astro-ph/0501562](#) .
 - [181] D. N. Spergel and P. J. Steinhardt, *Phys. Rev. Lett.* **84**, 3760 (2000), [arXiv:astro-ph/9909386](#) .
 - [182] S. Tulin, H.-B. Yu, and K. M. Zurek, *Phys. Rev. Lett.* **110**, 111301 (2013), [arXiv:1210.0900 \[hep-ph\]](#) .
 - [183] S. Tulin and H.-B. Yu, *Phys. Rept.* **730**, 1 (2018), [arXiv:1705.02358 \[hep-ph\]](#) .
 - [184] W. Hu, R. Barkana, and A. Gruzinov, *Phys. Rev. Lett.* **85**, 1158 (2000), [arXiv:astro-ph/0003365](#) .
 - [185] A. Del Popolo and M. Le Delliou, *Galaxies* **5**, 17 (2017), [arXiv:1606.07790 \[astro-ph.CO\]](#) .
 - [186] V. Delgado and A. Muñoz Mateo, *Mon. Not. Roy. Astron. Soc.* **518**, 4064 (2022), [arXiv:2201.12418 \[astro-ph.CO\]](#) .
 - [187] D. Budker, J. Eby, M. Gorghetto, M. Jiang, and G. Perez, *JCAP* **12**, 021 (2023), [arXiv:2306.12477 \[hep-ph\]](#) .
 - [188] D. Chowdhury, A. Hait, S. Mohanty, and S. Prakash, *Phys. Rev. D* **110**, 083023 (2024), [arXiv:2311.10148 \[hep-ph\]](#) .
 - [189] B. Dave and G. Goswami, *JCAP* **07**, 015 (2023), [arXiv:2304.04463 \[astro-ph.CO\]](#) .
 - [190] D. J. E. Marsh and A.-R. Pop, *Mon. Not. Roy. Astron. Soc.* **451**, 2479 (2015), [arXiv:1502.03456 \[astro-ph.CO\]](#) .
 - [191] R. G. García, P. Brax, and P. Valageas, *Phys. Rev. D* **109**, 043516 (2024), [arXiv:2304.10221 \[astro-ph.CO\]](#) .
 - [192] P. Brax, J. A. R. Cembranos, and P. Valageas, *Phys. Rev. D* **100**, 023526 (2019), [arXiv:1906.00730 \[astro-ph.CO\]](#) .
 - [193] P. Brax, J. A. R. Cembranos, and P. Valageas, *Phys. Rev. D* **101**, 023521 (2020), [arXiv:1909.02614 \[astro-ph.CO\]](#) .
 - [194] N. Bar, K. Blum, T. Lacroix, and P. Panci, *JCAP* **07**, 045 (2019), [arXiv:1905.11745 \[astro-ph.CO\]](#) .

- [195] E. Gross, *Annals of Physics* **4**, 57 (1958).
- [196] E. P. Gross, *Il Nuovo Cimento* **20**, 454 (1961).
- [197] L. P. PITAEVSKII, *Journal of Experimental and Theoretical Physics*, 13, 451-454. **13**, 451 (1961).
- [198] C. Barenghi and N. G. Parker, *A Primer on Quantum Fluids* (2016).
- [199] J. Antoniadis *et al.* (EPTA, InPTA:), *Astron. Astrophys.* **678**, A50 (2023), [arXiv:2306.16214 \[astro-ph.HE\]](#) .
- [200] J. Antoniadis *et al.* (EPTA, InPTA), *Astron. Astrophys.* **685**, A94 (2024), [arXiv:2306.16227 \[astro-ph.CO\]](#) .
- [201] D. J. Reardon *et al.*, *Astrophys. J. Lett.* **951**, L6 (2023), [arXiv:2306.16215 \[astro-ph.HE\]](#) .
- [202] A. Zic *et al.*, *Publ. Astron. Soc. Austral.* **40**, e049 (2023), [arXiv:2306.16230 \[astro-ph.HE\]](#) .
- [203] D. J. Reardon *et al.*, *Astrophys. J. Lett.* **951**, L7 (2023), [arXiv:2306.16229 \[astro-ph.HE\]](#) .
- [204] G. Agazie *et al.* (NANOGrav), *Astrophys. J. Lett.* **952**, L37 (2023), [arXiv:2306.16220 \[astro-ph.HE\]](#) .
- [205] A. Afzal *et al.* (NANOGrav), *Astrophys. J. Lett.* **951**, L11 (2023), [arXiv:2306.16219 \[astro-ph.HE\]](#) .
- [206] H. Xu *et al.*, *Res. Astron. Astrophys.* **23**, 075024 (2023), [arXiv:2306.16216 \[astro-ph.HE\]](#) .
- [207] G. Agazie *et al.* (International Pulsar Timing Array), *Astrophys. J.* **966**, 105 (2024), [arXiv:2309.00693 \[astro-ph.HE\]](#) .
- [208] M. C. Begelman, R. D. Blandford, and M. J. Rees, *Nature* **287**, 307 (1980).
- [209] N. J. McConnell and C.-P. Ma, *Astrophys. J.* **764**, 184 (2013), [arXiv:1211.2816](#) .
- [210] E. S. Phinney, [arXiv: astro-ph/0108028](#) (2001).
- [211] L. Hu, R.-G. Cai, and S.-J. Wang, *JCAP* **02**, 067 (2025), [arXiv:2312.14041 \[gr-qc\]](#) .
- [212] Y. Chen *et al.* (NANOGrav), (2024), [arXiv:2411.05906 \[astro-ph.HE\]](#) .
- [213] S. Chandrasekhar, *Astrophys. J.* **97**, 255 (1943).
- [214] S. Roy, P. Sarkar, S. Sau, and S. SenGupta, *JCAP* **11**, 099 (2023), [arXiv:2310.05908 \[gr-qc\]](#) .
- [215] A. Kar, S. Roy, and P. Sarkar, *JCAP* **05**, 100 (2025), [arXiv:2501.01860 \[hep-ph\]](#) .
- [216] P. Sarkar, (2025), [arXiv:2504.19505 \[hep-ph\]](#) .
- [217] B. P. Abbott *et al.* (LIGO Scientific, Virgo), *Phys. Rev. Lett.* **116**, 061102 (2016), [arXiv:1602.03837 \[gr-qc\]](#) .
- [218] B. P. Abbott *et al.* (LIGO Scientific, Virgo), *Phys. Rev. Lett.* **119**, 161101

- (2017), [arXiv:1710.05832 \[gr-qc\]](#) .
- [219] T. E. H. T. Collaboration and K. A. et al., *The Astrophysical Journal Letters* **910**, L12 (2021).
 - [220] K. Akiyama *et al.* (Event Horizon Telescope), *Astrophys. J. Lett.* **910**, L13 (2021), [arXiv:2105.01173 \[astro-ph.HE\]](#) .
 - [221] Y. Chen, J. Shu, X. Xue, Q. Yuan, and Y. Zhao, *Phys. Rev. Lett.* **124**, 061102 (2020).
 - [222] Y. Chen, C. Li, Y. Mizuno, J. Shu, X. Xue, Q. Yuan, Y. Zhao, and Z. Zhou, *Journal of Cosmology and Astroparticle Physics* **2022**, 073 (2022).
 - [223] Y. Chen, X. Xue, R. Brito, and V. Cardoso, *Phys. Rev. Lett.* **130**, 111401 (2023).
 - [224] P.-C. Li, M. Guo, and B. Chen, *Phys. Rev. D* **101**, 084041 (2020).
 - [225] L. Maiani, R. Petronzio, and E. Zavattini, *Phys. Lett. B* **175**, 359 (1986).
 - [226] G. Raffelt and L. Stodolsky, *Phys. Rev. D* **37**, 1237 (1988).
 - [227] E. Masaki, A. Aoki, and J. Soda, *Phys. Rev. D* **96**, 043519 (2017), [arXiv:1702.08843 \[astro-ph.CO\]](#) .
 - [228] K. A. Hochmuth and G. Sigl, *Phys. Rev. D* **76**, 123011 (2007), [arXiv:0708.1144 \[astro-ph\]](#) .
 - [229] A. Mirizzi, G. G. Raffelt, and P. D. Serpico, *Phys. Rev. D* **72**, 023501 (2005).
 - [230] B. Mukhopadhyaya, S. Sen, and S. SenGupta, *Phys. Rev. Lett.* **89**, 121101 (2002).
 - [231] D. Maity and S. SenGupta, *Class. Quant. Grav.* **21**, 3379 (2004), [arXiv:hep-th/0311142](#) .
 - [232] I. Antoniadis, A. Boyarsky, and O. Ruchayskiy, (2006), [arXiv:hep-ph/0606306](#) .
 - [233] D. Maity, S. Roy, and S. SenGupta, *Phys. Rev. D* **77**, 015010 (2008), [arXiv:0709.3940 \[hep-ph\]](#) .
 - [234] A. Mirizzi, G. G. Raffelt, and P. D. Serpico, *Lect. Notes Phys.* **741**, 115 (2008), [arXiv:astro-ph/0607415](#) .
 - [235] K. Gebhardt, J. Adams, D. Richstone, T. R. Lauer, S. Faber, K. Gültekin, J. Murphy, and S. Tremaine, *The Astrophysical Journal* **729**, 119 (2011).
 - [236] S. E. Gralla, D. E. Holz, and R. M. Wald, *Phys. Rev. D* **100**, 024018 (2019), [arXiv:1906.00873 \[astro-ph.HE\]](#) .
 - [237] G. Rybicki, *Radiative processes in Astrophysics* (Wiley-VCH, 2004).
 - [238] E. Quataert, *Astrophys. J.* **575**, 855 (2002), [arXiv:astro-ph/0201395](#) .
 - [239] F. Yuan and R. Narayan, *Astrophys J* **52**, 529 (2014), [arXiv:1401.0586 \[astro-ph.HE\]](#) .
 - [240] T. Shiromizu, K.-i. Maeda, and M. Sasaki, *Phys. Rev. D* **62**, 024012 (2000), [arXiv:gr-qc/9910076](#) .

- [241] T. Harko and M. K. Mak, *Phys. Rev. D* **69**, 064020 (2004), [arXiv:gr-qc/0401049](#) .
- [242] S. Chakraborty and S. SenGupta, *Eur. Phys. J. C* **75**, 11 (2015), [arXiv:1409.4115 \[gr-qc\]](#) .
- [243] A. N. Aliev and A. E. Gumrukcuoglu, *Phys. Rev. D* **71**, 104027 (2005), [arXiv:hep-th/0502223](#) .
- [244] N. Dadhich, R. Maartens, P. Papadopoulos, and V. Rezanian, *Phys. Lett. B* **487**, 1 (2000), [arXiv:hep-th/0003061](#) .
- [245] P. Uttley *et al.*, *Exper. Astron.* **51**, 1081 (2021), [arXiv:1908.03144 \[astro-ph.HE\]](#) .
- [246] A. Capolupo, I. De Martino, G. Lambiase, and A. Stabile, *Phys. Lett. B* **790**, 427 (2019), [arXiv:1901.10473 \[hep-ph\]](#) .
- [247] T. Kawashima, M. Kino, and K. Akiyama, *Astrophys. J.* **878**, 27 (2019), [arXiv:1905.10717 \[astro-ph.HE\]](#) .
- [248] J. M. Bardeen, W. H. Press, and S. A. Teukolsky, *Astrophys. J.* **178**, 347 (1972).
- [249] S. Chandrasekhar, *The Mathematical Theory of Black Holes*, International series of monographs on physics (Oxford University Press, 1992).
- [250] S. E. Gralla and A. Lupsasca, *Phys. Rev. D* **101**, 044032 (2020).
- [251] C. Dessert, D. Dunsy, and B. R. Safdi, *Phys. Rev. D* **105**, 103034 (2022), [arXiv:2203.04319 \[hep-ph\]](#) .
- [252] J. A. Dror, H. Murayama, and N. L. Rodd, *Phys. Rev. D* **103**, 115004 (2021), [Erratum: *Phys.Rev.D* 106, 119902 (2022)], [arXiv:2101.09287 \[hep-ph\]](#) .
- [253] A. Kar, T. Kumar, S. Roy, and J. Zupan, *JCAP* **08**, 056 (2023), [arXiv:2212.04647 \[hep-ph\]](#) .
- [254] R. Brito, V. Cardoso, and P. Pani, *Lect. Notes Phys.* **906**, pp.1 (2015), [arXiv:1501.06570 \[gr-qc\]](#) .
- [255] F. V. Day and J. I. McDonald, *JCAP* **10**, 051 (2019), [arXiv:1904.08341 \[hep-ph\]](#) .
- [256] I. S. McLean, S. K. Ramsay, and H. Takami, eds., *Ground-based and Airborne Instrumentation for Astronomy IV*, Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series, Vol. 8446 (2012).
- [257] S. N. Tandon, J. B. Hutchings, S. K. Ghosh, A. Subramaniam, G. Koshy, V. Girish, P. U. Kamath, S. Kathiravan, A. Kumar, J. P. Lancelot, P. K. Mahesh, R. Mohan, J. Murthy, S. Nagabhushana, A. K. Pati, J. Postma, N. K. Rao, K. Sankarasubramanian, P. Sreekumar, S. Sriram, C. S. Stalin, F. Sutaria, Y. H. Sreedhar, I. V. Barve, C. Mondal, and S. Sahu, *Journal of Astrophysics and Astronomy* **38** (2017), [10.1007/s12036-017-9445-x](#).
- [258] J. F. Navarro, C. S. Frenk, and S. D. M. White, *Astrophys. J.* **462**, 563 (1996), [arXiv:astro-ph/9508025](#) .
- [259] A. Albert *et al.* (HAWC), *Phys. Rev. D* **109**, 043034 (2024), [arXiv:2309.03973](#)

- [astro-ph.HE] .
- [260] T. Lacroix, C. Boehm, and J. Silk, *Phys. Rev. D* **92**, 043510 (2015), [arXiv:1505.00785 \[astro-ph.GA\]](#) .
- [261] M. De Laurentis and P. Salucci, *Astrophys. J.* **929**, 17 (2022), [arXiv:2206.01997 \[astro-ph.CO\]](#) .
- [262] M. Phoroutan-Mehr and H.-B. Yu, *Phys. Rev. D* **111**, 115033 (2025), [arXiv:2411.18751 \[hep-ph\]](#) .
- [263] T. D. P. Edwards, M. Chianese, B. J. Kavanagh, S. M. Nissanke, and C. Weniger, *Phys. Rev. Lett.* **124**, 161101 (2020), [arXiv:1905.04686 \[hep-ph\]](#) .
- [264] P. Gondolo and J. Silk, *Phys. Rev. Lett.* **83**, 1719 (1999), [arXiv:astro-ph/9906391](#) .
- [265] S. Balaji, D. Sachdeva, F. Sala, and J. Silk, *JCAP* **08**, 063 (2023), [arXiv:2303.12107 \[hep-ph\]](#) .
- [266] <https://eventhorizontelescope.org/for-astronomers/data>.
- [267] M. L. Lister, H. D. Aller, M. F. Aller, M. H. Cohen, D. C. Homan, M. Kadler, K. I. Kellermann, Y. Y. Kovalev, E. Ros, T. Savolainen, J. A. Zensus, and R. C. Vermeulen, *The Astronomical Journal* **137**, 3718 (2009), [arXiv:0812.3947 \[astro-ph\]](#) .
- [268] D. E. Harris, C. C. Cheung, L. Stawarz, J. A. Biretta, and E. S. Perlman, *The Astrophysics Journal* **699**, 305 (2009), [arXiv:0904.3925 \[astro-ph.CO\]](#) .
- [269] A. et al, *The Astrophysics Journal* **707**, 55 (2009), [arXiv:0910.3565 \[astro-ph.HE\]](#) .
- [270] F. Aharonian *et al.* (H.E.S.S.), *Science* **309**, 746 (2005), [arXiv:astro-ph/0508298](#) .
- [271] V. A. Acciari *et al.*, *Astrophys. J.* **679**, 397 (2008), [arXiv:0802.1951 \[astro-ph\]](#) .
- [272] V. A. Acciari *et al.* (MAGIC), *Astron. Astrophys.* **670**, A145 (2023), [arXiv:2210.03321 \[astro-ph.HE\]](#) .
- [273] M. Ackermann *et al.* (Fermi-LAT), *Astrophys. J.* **840**, 43 (2017), [arXiv:1704.03910 \[astro-ph.HE\]](#) .
- [274] T. S. Poole *et al.*, *Mon. Not. Roy. Astron. Soc.* **383**, 627 (2008), [arXiv:0708.2259 \[astro-ph\]](#) .
- [275] A. A. Breeveld, W. Landsman, S. T. Holland, P. Roming, N. P. M. Kuin, and M. J. Page, *AIP Conf. Proc.* **1358**, 373 (2011), [arXiv:1102.4717 \[astro-ph.IM\]](#) .
- [276] H. Zhou, S. Covino, Z.-P. Jin, Y.-Z. Fan, D.-M. Wei, and N. P. Kuin, *Astrophys. J. Suppl.* **277**, 27 (2025), [arXiv:2502.10241 \[astro-ph.IM\]](#) .
- [277] S. N. Tandon, J. Postma, P. Joseph, A. Devaraj, A. Subramaniam, I. V. Barve, K. George, S. K. Ghosh, V. Girish, J. B. Hutchings, P. U. Kamath, S. Kathiravan, A. Kumar, J. P. Lancelot, D. Leahy, P. K. Mahesh, R. Mohan, S. Nagabhushana, A. K. Pati, N. K. Rao, K. Sankarasubramanian, S. Sriram,

- and C. S. Stalin, *The Astrophysics Journal* **159**, 158 (2020), [arXiv:2002.01159 \[astro-ph.IM\]](#) .
- [278] Y. Ikejiri *et al.*, *Publ. Astron. Soc. Jap.* **63**, 639 (2011), [arXiv:1105.0255 \[astro-ph.HE\]](#) .
- [279] M. Sasada *et al.*, *Publ. Astron. Soc. Jap.* **64**, 58 (2012), [arXiv:1112.4583 \[astro-ph.HE\]](#) .
- [280] R. Itoh *et al.*, *Publ. Astron. Soc. Jap.* **65**, 18 (2013), [arXiv:1301.2664 \[astro-ph.HE\]](#) .
- [281] R. C. Bohlin, K. D. Gordon, and P. E. Tremblay, *Publ. Astron. Soc. Pac.* **126**, 711 (2016), [arXiv:1406.1707 \[astro-ph.IM\]](#) .
- [282] Spitzer Science Center, “*Spitzer irac instrument handbook*,” (2024).
- [283] D. Massa and E. L. Fitzpatrick, *Astrophys. J. Suppl.* **126**, 517 (2000), [arXiv:astro-ph/9812152](#) .
- [284] R. C. Bohlin and L. Bianchi, *The Astrophysics Journal* **155**, 162 (2018), [arXiv:1802.03419 \[astro-ph.IM\]](#) .
- [285] <https://asd.gsfc.nasa.gov/luvoir/>.
- [286] S. R. Kulkarni *et al.*, (2021), [arXiv:2111.15608 \[astro-ph.GA\]](#) .
- [287] M. Aghaie, G. Armando, A. Dondarini, and P. Panci, *Phys. Rev. D* **109**, 103030 (2024), [arXiv:2308.04590 \[astro-ph.CO\]](#) .
- [288] A. Ghoshal and A. Strumia, *JCAP* **02**, 054 (2024), [arXiv:2306.17158 \[astro-ph.CO\]](#) .
- [289] G. Alonso-Álvarez, J. M. Cline, and C. Dewar, *Phys. Rev. Lett.* **133**, 021401 (2024), [arXiv:2401.14450 \[astro-ph.CO\]](#) .
- [290] N. Glennon, N. Musoke, E. O. Nadler, C. Prescod-Weinstein, and R. H. Wechsler, *Phys. Rev. D* **109**, 063501 (2024), [arXiv:2312.07684 \[astro-ph.CO\]](#) .
- [291] H. Koo, D. Bak, I. Park, S. E. Hong, and J.-W. Lee, *Physics Letters B* **856**, 138908 (2024), [arXiv:2311.03412 \[astro-ph.GA\]](#) .
- [292] L. M. Widrow and N. Kaiser, *Astrophysical J. Lett.* **416**, L71 (1993).
- [293] P. Coles and K. Spencer, *Mon. Not. Roy. Astron. Soc.* **342**, 176 (2003), [arXiv:astro-ph/0212433](#) .
- [294] E. Y. Davies and P. Mocz, *Mon. Not. Roy. Astron. Soc.* **492**, 5721 (2020), [arXiv:1908.04790 \[astro-ph.GA\]](#) .
- [295] N. Bar, D. Blas, K. Blum, and S. Sibiryakov, *Phys. Rev. D* **98**, 083027 (2018), [arXiv:1805.00122 \[astro-ph.CO\]](#) .
- [296] S. Tremaine and J. E. Gunn, *Phys. Rev. Lett.* **42**, 407 (1979).
- [297] M. Y. Khlopov, B. A. Malomed, I. B. Zeldovich, and Y. B. Zeldovich, *Mon. Not. Roy. Astron. Soc.* **215**, 575 (1985).
- [298] L. Hui, *Ann. Rev. Astron. Astrophys.* **59**, 247 (2021), [arXiv:2101.11735 \[astro-ph.CO\]](#) .

-
- [299] J. Kormendy and D. Richstone, *Ann. Rev. Astron. Astrophys.* **33**, 581 (1995).
 - [300] J. Kormendy and L. C. Ho, *Ann. Rev. Astron. Astrophys.* **51**, 511 (2013), [arXiv:1304.7762 \[astro-ph.CO\]](#) .
 - [301] A. E. Reines, *Nature Astron.* **6**, 26 (2022), [arXiv:2201.10569 \[astro-ph.GA\]](#) .
 - [302] R. Ruffini and J. A. Wheeler, *Phys. Today* **24**, 30 (1971).
 - [303] W. Israel, *Phys. Rev.* **164**, 1776 (1967).
 - [304] W. Israel, *Commun. Math. Phys.* **8**, 245 (1968).
 - [305] B. Carter, *Phys. Rev. Lett.* **26**, 331 (1971).
 - [306] J. D. Bekenstein, *Phys. Rev. Lett.* **28**, 452 (1972).
 - [307] V. Cardoso, T. Ikeda, R. Vicente, and M. Zilhão, *Phys. Rev. D* **106**, L121302 (2022), [arXiv:2207.09469 \[gr-qc\]](#) .
 - [308] E. Figueiredo, A. Maselli, and V. Cardoso, *Phys. Rev. D* **107**, 104033 (2023), [arXiv:2303.08183 \[gr-qc\]](#) .
 - [309] Z. Zhong, V. Cardoso, T. Ikeda, and M. Zilhão, *Phys. Rev. D* **108**, 084051 (2023), [arXiv:2307.02548 \[gr-qc\]](#) .
 - [310] S. Mitra, S. Chakraborty, R. Vicente, and J. C. Feng, *Phys. Rev. D* **110**, 084012 (2024), [arXiv:2312.06783 \[gr-qc\]](#) .
 - [311] J. Fan, *Phys. Dark Univ.* **14**, 84 (2016), [arXiv:1603.06580 \[hep-ph\]](#) .
 - [312] S. Chakrabarti, B. Dave, K. Dutta, and G. Goswami, *JCAP* **09**, 074 (2022), [arXiv:2202.11081 \[astro-ph.CO\]](#) .
 - [313] P.-H. Chavanis, *Phys. Rev. D* **84**, 043531 (2011), [arXiv:1103.2050 \[astro-ph.CO\]](#) .
 - [314] P. H. Chavanis and L. Delfini, *Phys. Rev. D* **84**, 043532 (2011), [arXiv:1103.2054 \[astro-ph.CO\]](#) .
 - [315] J. Eby, P. Suranyi, C. Vaz, and L. C. R. Wijewardhana, *JHEP* **03**, 080 (2015), [Erratum: *JHEP* 11, 134 (2016)], [arXiv:1412.3430 \[hep-th\]](#) .
 - [316] P.-H. Chavanis, *Phys. Rev. D* **94**, 083007 (2016), [arXiv:1604.05904 \[astro-ph.CO\]](#) .
 - [317] D. G. Levkov, A. G. Panin, and I. I. Tkachev, *Phys. Rev. Lett.* **118**, 011301 (2017), [arXiv:1609.03611 \[astro-ph.CO\]](#) .
 - [318] J. Eby, M. Ma, P. Suranyi, and L. C. R. Wijewardhana, *JHEP* **01**, 066 (2018), [arXiv:1705.05385 \[hep-ph\]](#) .
 - [319] T. Helfer, D. J. E. Marsh, K. Clough, M. Fairbairn, E. A. Lim, and R. Becerril, *JCAP* **03**, 055 (2017), [arXiv:1609.04724 \[astro-ph.CO\]](#) .
 - [320] V. Desjacques, A. Kehagias, and A. Riotto, *Phys. Rev. D* **97**, 023529 (2018), [arXiv:1709.07946 \[astro-ph.CO\]](#) .
 - [321] V. Lora, J. Magana, A. Bernal, F. J. Sanchez-Salcedo, and E. K. Grebel, *JCAP* **02**, 011 (2012), [arXiv:1110.2684 \[astro-ph.GA\]](#) .
 - [322] J. Barranco, A. Bernal, J. C. Degollado, A. Diez-Tejedor, M. Megevand,
-

- M. Alcubierre, D. Nunez, and O. Sarbach, *Phys. Rev. D* **84**, 083008 (2011), [arXiv:1108.0931 \[gr-qc\]](#) .
- [323] J. Barranco, A. Bernal, J. C. Degollado, A. Diez-Tejedor, M. Megevand, M. Alcubierre, D. Nunez, and O. Sarbach, *Phys. Rev. Lett.* **109**, 081102 (2012), [arXiv:1207.2153 \[gr-qc\]](#) .
- [324] H.-Y. Schive, M.-H. Liao, T.-P. Woo, S.-K. Wong, T. Chiueh, T. Broadhurst, and W. Y. P. Hwang, *Phys. Rev. Lett.* **113**, 261302 (2014), [arXiv:1407.7762 \[astro-ph.GA\]](#) .
- [325] K. Bandara, D. Crampton, and L. Simard, *Astrophys. J.* **704**, 1135 (2009), [arXiv:0909.0269 \[astro-ph.GA\]](#) .
- [326] V. Cardoso, F. Duque, and T. Ikeda, *Phys. Rev. D* **101**, 064054 (2020), [arXiv:2001.01729 \[gr-qc\]](#) .
- [327] D. Baumann, H. S. Chia, and R. A. Porto, *Phys. Rev. D* **99**, 044001 (2019), [arXiv:1804.03208 \[gr-qc\]](#) .
- [328] G. M. Tomaselli, *Phys. Rev. D* **111**, 063075 (2025), [arXiv:2501.00090 \[gr-qc\]](#) .
- [329] M. Maggiore, *Gravitational Waves. Vol. 1: Theory and Experiments* (Oxford University Press, 2007).
- [330] L. Lancaster, C. Giovanetti, P. Mocz, Y. Kahn, M. Lisanti, and D. N. Spergel, *JCAP* **01**, 001 (2020), [arXiv:1909.06381 \[astro-ph.CO\]](#) .
- [331] X.-J. Yue, W.-B. Han, and X. Chen, *Astrophys. J.* **874**, 34 (2019), [arXiv:1802.03739 \[gr-qc\]](#) .
- [332] E. C. Ostriker, *Astrophys. J.* **513**, 252 (1999), [arXiv:astro-ph/9810324](#) .
- [333] D. Chapon, L. Mayer, and R. Teyssier, *MNRAS* **429**, 3114 (2013), [arXiv:1110.6086 \[astro-ph.GA\]](#) .
- [334] S. Lescaudron, Y. Dubois, R. S. Beckmann, and M. Volonteri, *Astronomy and Astrophysics* **674**, A217 (2023), [arXiv:2209.13548 \[astro-ph.GA\]](#) .
- [335] S. Kazantzidis, L. Mayer, M. Colpi, P. Madau, V. P. Debattista, B. Moore, J. Wadsley, J. Stadel, and T. R. Quinn, *Astrophys. J. Lett.* **623**, L67 (2005), [arXiv:astro-ph/0407407](#) .
- [336] L. Mayer, S. Kazantzidis, P. Madau, M. Colpi, T. R. Quinn, and J. Wadsley, *Science* **316**, 1874 (2007), [arXiv:0706.1562 \[astro-ph\]](#) .
- [337] N. Chen, Y. Ni, M. Tremmel, T. Di Matteo, S. Bird, C. DeGraf, and Y. Feng, *MNRAS* **510**, 531 (2022), [arXiv:2104.00021 \[astro-ph.GA\]](#) .
- [338] J. A. Dror, B. V. Lehmann, H. H. Patel, and S. Profumo, *Phys. Rev. D* **104**, 083021 (2021), [arXiv:2105.04559 \[astro-ph.CO\]](#) .
- [339] W. DeRocco and J. A. Dror, *Phys. Rev. D* **108**, 103011 (2023), [arXiv:2304.13042 \[astro-ph.HE\]](#) .
- [340] A. R. Kaiser and S. T. McWilliams, *Class. Quant. Grav.* **38**, 055009 (2021), [arXiv:2010.02135 \[gr-qc\]](#) .
- [341] J. Ellis, M. Fairbairn, G. Hütsi, M. Raidal, J. Urrutia, V. Vaskonen, and

- H. Veermäe, *Astron. Astrophys.* **676**, A38 (2023), [arXiv:2301.13854 \[astro-ph.CO\]](#) .
- [342] M. Colpi *et al.* (LISA), (2024), [arXiv:2402.07571 \[astro-ph.CO\]](#) .
- [343] S. Weinberg, *The Quantum Theory of Fields* (Cambridge University Press, 1996).
- [344] A. Bilal, (2008), [arXiv:0802.0634 \[hep-th\]](#) .
- [345] H. Georgi, D. B. Kaplan, and L. Randall, *Physics Letters B* **169**, 73 (1986).
- [346] M. Bauer, M. Neubert, S. Renner, M. Schnubel, and A. Thamm, *JHEP* **04**, 063 (2021), [arXiv:2012.12272 \[hep-ph\]](#) .
- [347] M. Bauer, M. Neubert, and A. Thamm, *JHEP* **12**, 044 (2017), [arXiv:1708.00443 \[hep-ph\]](#) .
- [348] M. Chala, G. Guedes, M. Ramos, and J. Santiago, *Eur. Phys. J. C* **81**, 181 (2021), [arXiv:2012.09017 \[hep-ph\]](#) .
- [349] G. Passarino and M. Veltman, *Nuclear Physics B* **160**, 151 (1979).
- [350] G. 't Hooft and M. Veltman, *Nuclear Physics B* **153**, 365 (1979).
- [351] H. H. Patel, *Comput. Phys. Commun.* **218**, 66 (2017), [arXiv:1612.00009 \[hep-ph\]](#) .
- [352] S. R. Taylor, J. Simon, and L. Sampson, *Phys. Rev. Lett.* **118**, 181102 (2017), [arXiv:1612.02817 \[astro-ph.GA\]](#) .
- [353] L. Z. Kelley, L. Blecha, L. Hernquist, A. Sesana, and S. R. Taylor, *Mon. Not. Roy. Astron. Soc.* **471**, 4508 (2017), [arXiv:1702.02180 \[astro-ph.HE\]](#) .
- [354] A. Sesana, F. Haardt, and P. Madau, *Astrophys. J.* **651**, 392 (2006), [arXiv:astro-ph/0604299](#) .
- [355] M. Siwek, R. Weinberger, and L. Hernquist, *MNRAS* **522**, 2707 (2023), [arXiv:2302.01785 \[astro-ph.HE\]](#) .
- [356] D. Merritt, *Classical and Quantum Gravity* **30**, 244005 (2013), [arXiv:1307.3268 \[astro-ph.GA\]](#) .
- [357] C. J. Harris and K. Gültekin, *MNRAS* **528**, 1 (2024), [arXiv:2311.04877 \[astro-ph.GA\]](#) .
- [358] A. Sesana, *Class. Quant. Grav.* **30**, 224014 (2013), [arXiv:1307.2600 \[astro-ph.CO\]](#) .
- [359] A. Sesana, F. Shankar, M. Bernardi, and R. K. Sheth, *Mon. Not. Roy. Astron. Soc.* **463**, L6 (2016), [arXiv:1603.09348 \[astro-ph.GA\]](#) .
- [360] S. Chen, A. Sesana, and W. Del Pozzo, *Mon. Not. Roy. Astron. Soc.* **470**, 1738 (2017), [arXiv:1612.00455 \[astro-ph.CO\]](#) .
- [361] S. Chen, A. Sesana, and C. J. Conselice, *Mon. Not. Roy. Astron. Soc.* **488**, 401 (2019), [arXiv:1810.04184 \[astro-ph.GA\]](#) .
- [362] M. Bernardi, A. Meert, V. Vikram, M. Huertas-Company, S. Mei, F. Shankar, and R. K. Sheth (2012).

Appendix A

The Effective Field theory for the ALP

ALPs arise as (pseudo-)Nambu–Goldstone bosons associated with the spontaneous breaking of a global symmetry [34]. As a consequence of their Goldstone nature, ALPs are naturally light degrees of freedom and interact with SM fields through operators suppressed by the symmetry-breaking scale Λ . At energies well below this scale, details of the ultraviolet (UV) completion are irrelevant, and the dynamics of the ALP can be described by an effective field theory (EFT) containing the ALP and SM fields as the only propagating degrees of freedom. The parameters of this EFT therefore constitute the relevant quantities for phenomenological studies and experimental constraints.

The Goldstone origin of the ALP imposes several structural constraints on the EFT and restricts the allowed interactions with SM fields. The ALP is an electrically neutral pseudoscalar particle and can be described by a real scalar field, denoted by a , which carries no gauge charge. Consequently, gauge-covariant and ordinary derivatives coincide when acting on the ALP field, $D_\mu a = \partial_\mu a$. The EFT is invariant under continuous shift transformations of the ALP field, $a \rightarrow a + \text{const.}$, reflecting the underlying global symmetry. Deviations from this symmetry arise only through explicit symmetry-breaking effects, in which case the ALP becomes a pseudo-Nambu–Goldstone boson. Such explicit breaking generically induces a non-vanishing ALP mass m_a , which remains naturally small since radiative corrections to m_a must be proportional to the symmetry-breaking parameter itself [343].

A prominent source of explicit symmetry breaking is provided by quantum anomalies, which arise upon quantization of theories containing chiral fermions [344]. In particular, the $U(1)_{\text{PQ}} \times SU(3)_c \times SU(3)_c$ anomaly characterizes the QCD axion and underlies its ability to dynamically resolve the strong CP problem. In the research work included in this thesis, we assume that the scale of new physics satisfies $\Lambda \gg m, \omega$, where m and ω denote the characteristic masses and energies relevant to the processes under consideration. Under this assumption, the effective Lagrangian can be truncated at dimension five, since operators of higher dimension are suppressed by additional powers of $1/\Lambda$.

Finally, we assume that the ALP sector of the effective theory is invariant under

CP transformations. This assumption simplifies the structure of the EFT and is sufficient for the phenomenological applications considered in this thesis. The EFT describing an ALP and its interactions with SM fields can be written as [345, 346]

$$\begin{aligned} \mathcal{L} = & -\frac{1}{2} a (\Box + m_a^2) a + (\partial_\mu a) \sum_F \sum_{i,j=1}^3 (\hat{g}_{aF})_{ij} \bar{\Psi}_F^i \gamma^\mu \Psi_F^j \\ & + \frac{g_{aB}}{4} a B_{\mu\nu} \tilde{B}^{\mu\nu} + \frac{g_{aW}}{4} a W_{\mu\nu}^a \tilde{W}^{\mu\nu,a} + \frac{g_{aG}}{4} a G_{\mu\nu}^a \tilde{G}^{\mu\nu,a} + \mathcal{L}_{\text{SM}}, \end{aligned} \quad (\text{A.1})$$

where the first summation extends over all chiral fermion multiplets Ψ_F of the SM, and $i, j = 1, 2, 3$ label the fermion generations. For example, $\Psi_F^1 \in \{L_e = (\nu_L, e_L)^T, e_R, Q_L, u_R, d_R\}$ denotes the first generation. The fermionic coupling matrices are Hermitian, $(\hat{g}_{aF})_{ij} = (\hat{g}_{aF})_{ji}^*$, while the ALP couplings to gauge fields, g_{aB} , g_{aW} , and g_{aG} , are real parameters. Since all interaction terms are dimension-five operators, these couplings carry inverse mass dimension and are often expressed as $g_{ai} = c_{ai}/\Lambda$, where Λ denotes the EFT cutoff scale and the dimensionless coefficients c_{ai} are generically expected to be of order unity [347].

Here, $B_{\mu\nu}$, $W_{\mu\nu}^a$, and $G_{\mu\nu}^a$ represent the field strength tensors of the hypercharge, weak isospin, and color gauge groups, respectively. Their duals are defined as

$$\tilde{X}_{\mu\nu} \equiv \frac{1}{2} \epsilon_{\mu\nu\alpha\beta} X^{\alpha\beta}, \quad X \in \{B, W^a, G^a\}. \quad (\text{A.2})$$

No direct coupling between the ALP and the Higgs doublet appears in (Eq. (A.1)). The only possible dimension-five operator of the form $(\partial^\mu a) \phi^\dagger i D_\mu \phi + \text{h.c.}$ is redundant and can be eliminated through a non-anomalous field redefinition of the fermion fields [346, 348].

In general, the ALP–fermion interactions in (Eq. (A.1)) are chiral, meaning that the couplings to left- and right-handed fermions are independent. Expressing chiral fermions as projections of Dirac spinors, e.g. $e_{L,R} = P_{L,R} e = \frac{1}{2}(1 \mp \gamma_5)e$, one finds that the ALP couples both to vector currents $\bar{\psi}\gamma_\mu\psi$ and axial-vector currents $\bar{\psi}\gamma_\mu\gamma_5\psi$ constructed from these fields.

After electroweak symmetry breaking, the ALP interactions with the photon, Z boson, and charged weak bosons can be written as

$$\begin{aligned} \mathcal{L} \supset & \frac{g_{a\gamma}}{4} a F_{\mu\nu} \tilde{F}^{\mu\nu} + \frac{g_{aZ}}{4} a Z_{\mu\nu} \tilde{Z}^{\mu\nu} + \frac{g_{a\gamma Z}}{2} a F_{\mu\nu} \tilde{Z}^{\mu\nu} \\ & + \frac{g_{aW}}{2} a W_{\mu\nu}^+ \tilde{W}^{\mu\nu,-} + \mathcal{O}_4, \end{aligned} \quad (\text{2.2})$$

where $F_{\mu\nu}$ denotes the electromagnetic field-strength tensor, while $Z_{\mu\nu}$ corresponds to the field-strength tensor of the Z –boson arising after electroweak symmetry breaking from the mixing of the $SU(2)_L$ and $U(1)_Y$ gauge fields. The charged tensors $W_{\mu\nu}^\pm$ originate from the non-Abelian $SU(2)_L$ gauge sector.

The couplings appearing above are related to those in Eq. (A.1) according to

$$\begin{aligned} g_{a\gamma} &= s_W^2 g_{aW} + c_W^2 g_{aB}, \\ g_{aZ} &= c_W^2 g_{aW} + s_W^2 g_{aB}, \\ g_{a\gamma Z} &= s_W c_W (g_{aW} - g_{aB}), \end{aligned} \quad (\text{2.3})$$

where s_W and c_W denote the sine and cosine of the Weinberg angle.

At energies below the electroweak scale, the heavy $W^\pm$ and Z bosons, as well as the top quark, are integrated out. In this regime, vector currents of Dirac fermions are conserved and do not contribute to physical observables involving ALPs [346]. Moreover, for simplicity and motivated by phenomenological constraints, we neglect non-diagonal ALP–fermion couplings in flavor space (Flavor Changing Neutral Currents) in the following. The resulting low-energy effective Lagrangian, valid for energies $\omega < m_W \simeq 80$ GeV, takes the form

$$\mathcal{L}_{\text{LE}} = -\frac{1}{2} a(\Box + m_a^2) a + \sum_f \hat{g}_{af} (\partial_\mu a) \bar{\psi}_f \gamma^\mu \gamma_5 \psi_f + \frac{g_{a\gamma}}{4} a F_{\mu\nu} \tilde{F}^{\mu\nu} + \frac{g_{ag}}{4} a G_{\mu\nu}^a \tilde{G}^{\mu\nu, a} + \mathcal{L}_{\text{SM}}, \quad (2.5)$$

where $f \in \{u, d, s, c, b, e, \mu, \tau\}$, after integrating out the heavy top quark. The low-energy fermionic couplings $\hat{g}_{af}$ are related at tree level to the couplings defined before electroweak symmetry breaking via

$$\hat{g}_{af} = (\hat{g}_{aF'})_{ii} - (\hat{g}_{aF})_{ii}, \quad (2.6)$$

where the indices satisfy $P_L \psi_f^i = \Psi_F^i$ and $P_R \psi_f^i = \Psi_{F'}^i$ and $\psi_f = \Psi_F + \Psi_{F'}$. In the above, the vector-current component of the ALP–fermion interaction can be eliminated through an appropriate field redefinition, such that only the axial-vector current, $\bar{\psi}_f \gamma^\mu \gamma_5 \psi_f$, contributes to physical observables.

Finally, for later convenience, we introduce the axion QED (aQED) Lagrangian, which contains only ALPs, leptons (electrons and muons), and photons as dynamical degrees of freedom,

$$\begin{aligned} \mathcal{L}_{\text{aQED}} = & -\frac{1}{2} a(\Box + m_a^2) a - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \frac{g_{a\gamma}}{4} a F_{\mu\nu} \tilde{F}^{\mu\nu} \\ & + \bar{\psi}_\ell (i \not{D} - m_\ell) \psi_\ell + \hat{g}_{a\ell} (\partial_\mu a) \bar{\psi}_\ell \gamma^\mu \gamma_5 \psi_\ell, \quad \ell \in \{e, \mu\}, \end{aligned} \quad (2.7)$$

where $m_e \simeq 511$ KeV and $m_\mu \simeq 106$ MeV denote the electron and muon masses, respectively, and D_μ is the QED gauge-covariant derivative. This Lagrangian governs most of the ALP phenomenology relevant for this thesis, as it includes all light degrees of freedom with masses below $\mathcal{O}(100)$ MeV that are not charged under QCD. If the ALP couplings to quarks or gluons in Eq. (2.5) are nonvanishing, additional hadronic degrees of freedom such as pions and nucleons must be included in the low-energy description.

A.1 ALP-Photon Effective Couplings

The one-loop diagram shown in Fig. [A.1] generates an effective interaction between ALPs and photons at the quantum level. Evaluating this diagram, together with the contribution obtained by reversing the direction of the electron flow, and in the absence of a tree-level ALP–photon coupling, we obtain

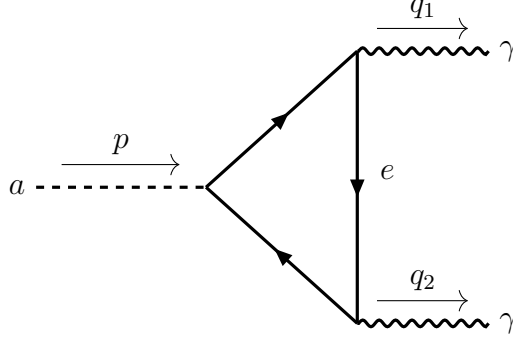


Figure A.1: Axion–photon interaction mediated by an electron loop.

$$\begin{aligned}
i\mathcal{M}_{a\gamma\gamma}^{\mu\nu} = & - \int \frac{d^4k}{(2\pi)^4} \text{Tr} \left\{ (-ie\gamma^\mu) \frac{i(\not{k} + m_e)}{k^2 - m_e^2 + i\epsilon} (-ie\gamma^\nu) \frac{i(\not{k} + \not{q}_2 + m_e)}{(k + q_2)^2 - m_e^2 + i\epsilon} \right. \\
& \times \left. (\hat{g}_{ae} \not{p} \gamma_5) \frac{i(\not{k} - \not{q}_1 + m_e)}{(k - q_1)^2 - m_e^2 + i\epsilon} \right\} + (\mu \leftrightarrow \nu, q_1 \leftrightarrow q_2)
\end{aligned} \tag{A.3}$$

$$\begin{aligned}
& = i \frac{2\alpha \hat{g}_{ae}}{\pi} q_1^\alpha q_2^\beta \epsilon^{\mu\nu\alpha\beta} \left\{ 1 - \int_0^1 dx \int_0^1 dy \frac{2m_e^2(1-x)}{m_e^2 - x(1-x) \left[\frac{y}{x}(1-y+xy)p^2 \right] q_1^2 + 2y p \cdot q_1 - i\epsilon} \right\} \\
& = i \frac{2\alpha \hat{g}_{ae}}{\pi} \left[1 + 2m_e^2 C_0(q_1^2, q_2^2, p^2, m_e^2) \right] q_1^\alpha q_2^\beta \epsilon^{\mu\nu\alpha\beta} \equiv i g_{a\gamma}^{\text{eff}}(q_1^2, q_2^2, p^2) q_1^\alpha q_2^\beta \epsilon^{\mu\nu\alpha\beta}. \tag{A.4}
\end{aligned}$$

where C_0 denotes the scalar Passarino–Veltman three-point function [349]. We employ the shorthand notation

$$C_0(q_1^2, q_2^2, p^2, m_e^2) \equiv C_0(q_1^2, q_2^2, p^2, m_e^2, m_e^2, m_e^2).$$

For generic kinematics, this function can be expressed as a linear combination of dilogarithmic functions [350]. The computation of this full scalar function C_0 can be numerically done using **FeynCalc** in combination with **Package-X** [351]. We define the quantity $g_{a\gamma}^{\text{eff}}$ appearing in Eq. (A.4) as the off-shell effective ALP–photon coupling. This triangle diagram, Fig. [A.1] is not the only contribution to the $a \rightarrow \gamma\gamma$ process. Additional one-loop contributions arise from diagrams involving fermion loops, charged $W^\pm$ bosons, their associated Goldstone fields Φ_W , as well as self-energy insertions $\Pi_{\gamma z}$. Collecting all such contributions, we define an effective $a\gamma\gamma$ vertex that consistently incorporates the full set of one-loop effects. The relevant one-loop diagrams are shown in Fig. [A.2].

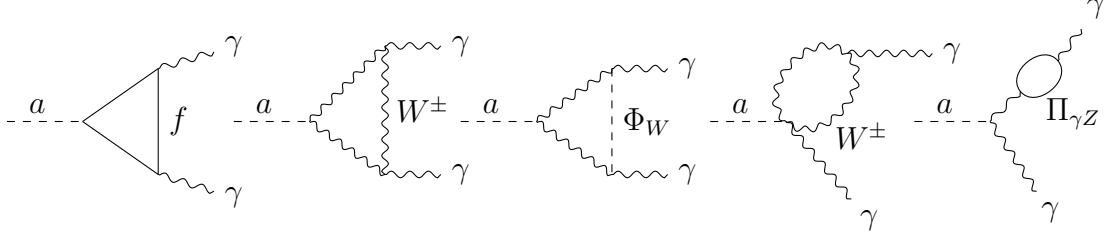


Figure A.2: Some of the one-loop contributions to the $a \rightarrow \gamma\gamma$ process mediated by fermions, charged gauge bosons, goldstone bosons, and self-energy insertions.

A.2 Axion decay into two photons

We start from the effective axion–photon–photon amplitude,

$$i\mathcal{M}_{a\gamma\gamma}^{\mu\nu} = i g_{a\gamma}^{\text{eff}}(q_1^2, q_2^2, p^2) q_1^\alpha q_2^\beta \epsilon^{\mu\nu\alpha\beta}, \quad (\text{A.5})$$

where p^μ is the axion four-momentum and q_1^μ, q_2^μ denote the outgoing photon momenta.

For the decay process

$$a(p) \rightarrow \gamma(q_1, \varepsilon_1^\mu) + \gamma(q_2, \varepsilon_2^\nu), \quad (\text{A.6})$$

the physical matrix element is

$$\mathcal{M} = g_{a\gamma}^{\text{eff}} \epsilon^{\mu\nu\alpha\beta} q_{1\alpha} q_{2\beta} \varepsilon_{1\mu}^*(q_1) \varepsilon_{2\nu}^*(q_2). \quad (\text{A.7})$$

We consider on-shell photons,

$$q_1^2 = q_2^2 = 0, \quad p^2 = m_a^2, \quad (\text{A.8})$$

and evaluate the effective coupling at these kinematics, $g_{a\gamma}^{\text{eff}} \equiv g_{a\gamma}^{\text{eff}}(0, 0, m_a^2)$.

Squared amplitude

Summing over the physical photon polarizations,

$$\sum_{\lambda} \varepsilon_{\mu}^{(\lambda)}(q) \varepsilon_{\nu}^{(\lambda)*}(q) = -g_{\mu\nu}, \quad (\text{A.9})$$

the squared matrix element becomes

$$|\mathcal{M}|^2 = (g_{a\gamma}^{\text{eff}})^2 \epsilon^{\mu\nu\alpha\beta} \epsilon^{\mu'\nu'\alpha'\beta'} q_{1\alpha} q_{2\beta} q_{1\alpha'} q_{2\beta'} (-g_{\mu\mu'}) (-g_{\nu\nu'}). \quad (\text{A.10})$$

Using the identity

$$\epsilon^{\mu\nu\alpha\beta} \epsilon_{\mu\nu}^{\alpha'\beta'} = -2 \left(g^{\alpha\alpha'} g^{\beta\beta'} - g^{\alpha\beta'} g^{\beta\alpha'} \right), \quad (\text{A.11})$$

we obtain

$$|\mathcal{M}|^2 = 2 (g_{a\gamma}^{\text{eff}})^2 [(q_1 \cdot q_2)^2 - q_1^2 q_2^2]. \quad (\text{A.12})$$

For massless photons, $q_1^2 = q_2^2 = 0$, and using

$$q_1 \cdot q_2 = \frac{m_a^2}{2}, \quad (\text{A.13})$$

the squared amplitude reduces to

$$|\mathcal{M}|^2 = \frac{1}{2} (g_{a\gamma}^{\text{eff}})^2 m_a^4. \quad (\text{A.14})$$

Decay width

The decay width for a two-body decay is given by

$$\Gamma(a \rightarrow \gamma\gamma) = \frac{1}{2m_a} \int d\Phi_2 |\mathcal{M}|^2, \quad (\text{A.15})$$

where the two-body phase space reads

$$\int d\Phi_2 = \frac{1}{8\pi} \frac{|\vec{q}|}{m_a}. \quad (\text{A.16})$$

For massless final states, $|\vec{q}| = m_a/2$, yielding

$$\int d\Phi_2 = \frac{1}{16\pi}. \quad (\text{A.17})$$

Substituting this into the decay width expression, we find

$$\Gamma(a \rightarrow \gamma\gamma) = \frac{1}{2m_a} \cdot \frac{1}{16\pi} \cdot \frac{1}{2} (g_{a\gamma}^{\text{eff}})^2 m_a^4. \quad (\text{A.18})$$

Therefore, the final result is

$$\boxed{\Gamma(a \rightarrow \gamma\gamma) = \frac{(g_{a\gamma}^{\text{eff}})^2}{64\pi} m_a^3}. \quad (\text{A.19})$$

Axion lifetime

The axion lifetime is given by

$$\tau_a = \frac{1}{\Gamma(a \rightarrow \gamma\gamma)}, \quad (\text{A.20})$$

which yields

$$\boxed{\tau_a = \frac{64\pi}{(g_{a\gamma}^{\text{eff}})^2 m_a^3}}. \quad (\text{A.21})$$

From the above expression, it is evident that the lifetime of the ALP decreases with increasing mass. In order for a decaying ALP to constitute dark matter and still be present today, its lifetime must exceed the current age of the Universe, approximately 10^{17} s. This requirement imposes an upper bound on the ALP mass, above which the particle can no longer account for the total dark matter abundance in the Universe.

Appendix B

Numerically produced solutions of Gross-Pitaevskii-Poisson equation

The Gross-Pitaevskii-Poisson equation in dimensionless form is given by

$$\frac{1}{2}\hat{\nabla}^2\hat{\chi} = \left(\hat{V} - \hat{\gamma} - \frac{\hat{\alpha}}{\hat{r}} + 2\hat{\lambda}\hat{\chi}^2\right)\hat{\chi}, \quad (\text{B.1})$$

$$\hat{\nabla}^2\hat{V} = \hat{\chi}^2, \quad (\text{B.2})$$

with the boundary conditions: $\hat{\chi}(\hat{r} = \infty) = 0$, $\hat{\chi}'(\hat{r} = 0) = 0$, $\hat{V}(\hat{r} = 0) = 0$, $\hat{V}'(\hat{r} =$

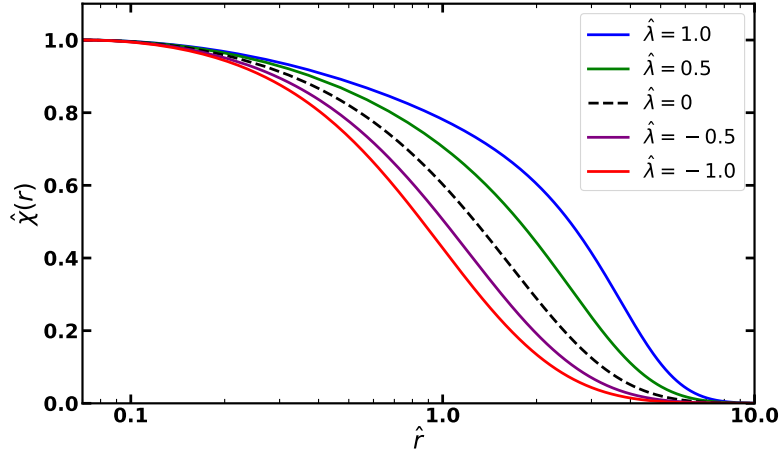


Figure B.1: Solutions of the coupled, dimensionless Gross-Pitaevskii-Poisson equation Eq. (B.1) for various values of $\hat{\lambda}$ keeping $\hat{\alpha}$ fixed at 0.48 for a ULDM mass of 10^{-21}eV .

$0) = 0$. We use the arbitrary normalization $\hat{\chi}(\hat{r} = 0) = 1$ to get the non zero value at the center. For a fixed SMBH mass parameter $\hat{\alpha}$ corresponding to halo mass $M_{\text{halo}} = 10^{13}M_{\odot}$ (shown in the table), we choose the $\hat{\lambda}$ parameter of $\mathcal{O}(1)$. To solve this system of differential equations, we apply a shooting method. This method integrates the solution from $\hat{r} = 0$, aiming to satisfy the asymptotic boundary condition $\hat{\chi}(\hat{r} = \infty) = 0$. To achieve this, an initial guess for $\hat{\gamma}$ is needed, as only quantized values meet the asymptotic boundary conditions. Using a standard root-finding technique,

we determine the smallest $\hat{\gamma}_0$ that satisfies the boundary conditions, giving the 0-node (ground state) solution. Fig. [B.1] illustrates the dimensionless solutions obtained for various values of $\hat{\lambda}$ with a specific $\hat{\alpha}$ value. In the limit of large $\hat{\alpha}$ (representing a large black hole mass), the solution approaches the exponential form characteristic of the hydrogen atom. As with the previous variables we define a dimensionless mass:

$$\hat{M} \equiv \frac{GMm}{\hbar c} = \int_0^\infty \hat{\chi}^2 \hat{r}^2 d\hat{r}. \quad (\text{B.3})$$

A relevant parameter in our analysis is the ratio of the soliton mass to the black hole mass. To ensure that the SMBH acts as the dominant perturbing potential in the system, this ratio has been kept at $\lesssim 0.3$ throughout the study[294], allowing us to recover the exponential solutions.

The numerical shooting procedure adopted here follows the standard approach used in solving the dimensionless GPP equations for ultralight dark matter solitons (see, e.g., Refs. [190, 294]). In particular, the choice of the central normalization $\hat{\chi}(0) = 1$ is a numerical convention in the dimensionless formulation, while the physical density profile is recovered after applying the scaling transformation discussed below.

B.1 Amount of scaling s

Fixing the halo mass, M_{halo} , there exists an empirical relationship between soliton mass M_{sol} and halo mass[324]:

$$M_{\text{sol}} = 1.25 \times 10^9 \left(\frac{M_{\text{halo}}}{10^{12} M_\odot} \right)^{1/3} \left(\frac{m}{10^{-22} \text{ eV}} \right)^{-1} M_\odot. \quad (\text{B.4})$$

This relation is derived from DM-only simulations without SMBHs and is valid for $M_{\text{halo}} \sim (10^9 - 5 \times 10^{11}) M_\odot$. For $M_{\text{halo}} \gtrsim 10^{12} M_\odot$ or different ULDM mass values, the relation weakens but remains useful for our goal of constraining the self-coupling parameter via the SGWB.

In this work, we assume $M_{\text{halo}} = 10^{13} M_\odot$ and use the $\hat{\alpha}$ values from [294] to construct density profiles. These values of $\hat{\alpha}$ were obtained using numerical simulation considering virialized halo with SMBH placed at the centers of the galaxies. Table B.1 summarizes $\hat{\alpha}$ and the corresponding dimensionless soliton mass $\hat{M}$ for different ULDM masses.

M_{halo}	$m \text{ (eV)}$	$\hat{\alpha}$	$\hat{M}$
$10^{13} M_\odot$	10^{-21}	0.48	0.84
$10^{13} M_\odot$	2×10^{-21}	0.55	0.75

Table B.1: Values of $\hat{\alpha}$ chosen to produce density profiles of halo mass $10^{13} M_\odot$ without the λ parameter, along with the dimensionless soliton mass $\hat{M}$ for different ULDM masses.

With fixed $\hat{\alpha}$ and nonzero $\hat{\lambda}$, we compute $\hat{M}$. For attractive self-interaction ($\hat{\lambda} < 0$) and repulsive ($\hat{\lambda} > 0$), $\hat{M}$ relates to M_{sol} by:

$$\hat{M} \equiv \frac{G_N M_{\text{sol}} m}{\hbar c}. \quad (\text{B.5})$$

Scaling modifies $\hat{M}$ as:

$$\hat{M} \rightarrow \frac{1}{s} \hat{M}. \quad (\text{B.6})$$

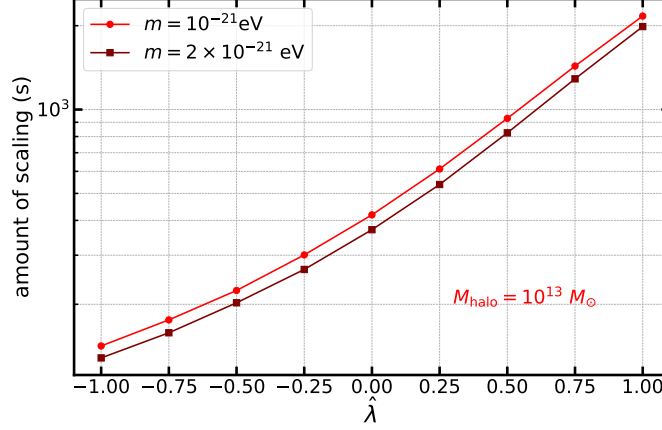


Figure B.2: Amount of scaling with different values of $\hat{\lambda}$ parameter for a fixed halo mass. Three cases are shown for three ULDM masses.

Using the relations above, we calculate the scaling factor s :

$$s = \frac{1.33 \times 10^{12}}{1.25 \times 10^9} \left(\frac{M_{\text{halo}}}{10^{12} M_{\odot}} \right)^{-1/3} \hat{M}. \quad (\text{B.7})$$

The dimensionless soliton mass $\hat{M}$ is calculated numerically through the integral Eq. (B.3) for each soliton solutions with the non zero $\hat{\lambda}$ parameters.

Fig. [B.2] illustrates how s changes with $\hat{\lambda}$. The scaling s increases with decreasing ULDM mass for a fixed $\hat{\lambda}$, and s also grows as $\hat{\lambda}$ shifts from negative to positive values.

B.2 Environmental degeneracy

Our analysis is restricted to circular, equal-mass SMBH binaries embedded in an otherwise pristine ULDM environment. In realistic systems, several standard astrophysical mechanisms can also reshape the SGWB spectrum away from the fiducial $h_c \propto f^{-2/3}$ power law, and we discuss below a few such environmental effects.

Orbital eccentricity redistributes GW power into higher harmonics and can flatten or invert the spectrum at low frequencies [352, 353]. However, detailed population studies find that the stellar-scattering hardening rate itself depends only weakly on eccentricity ($\sim 10\%$ variation between $e = 0$ and $e = 0.9$), and that binaries evolving

through gas-poor mergers tend to circularize substantially by the time they enter the PTA band, while gas-coupled binaries settle toward a modest equilibrium eccentricity of order $e \sim 0.5$ [354, 353, 355]. Such eccentricity-driven effects reshape the spectrum smoothly and gradually, in contrast to the sharp, localized dip predicted by soliton-induced dynamical friction, whose location is fixed by the parameters such as m and λ .

Loss-cone hardening is the phase of SMBH binary inspiral during which stars scattered into the binary’s loss cone extract orbital energy and angular momentum through three-body interactions, causing the binary to shrink. Stellar loss-cone hardening and gaseous circumbinary disk interactions are the two leading candidate mechanisms invoked to explain the observed low-frequency turnover in the SGWB [356]. Notably, a recent multi-messenger analysis combining core-scouring observations in elliptical galaxies with the NANOGrav 15-year dataset finds that stellar scattering alone – even under a full loss-cone and a hardening rate calibrated to be $\sim 60\%$ faster than standard N-body predictions – is insufficient to reproduce the observed turnover, strongly implicating gas dynamics as a necessary additional ingredient [357]. It demonstrates that the dominant standard astrophysical explanation for a low-frequency SGWB turnover is, on its own, already in tension with observations, so any additional suppression feature attributed to the standard hardening channels alone cannot trivially reproduce or mimic the localized signature we predict from ULDM dynamical friction. It underscores that robustly disentangling our proposed feature from gas-driven or eccentricity-driven modifications will ultimately require a joint, multi-mechanism fit to PTA data that includes ULDM dynamical friction alongside circumbinary disk torques, loss-cone hardening, and orbital eccentricity as simultaneous ingredients. We identify such a joint analysis as an important direction for future work.

B.3 SMBH binary merger rate

Here we will shortly describe the parameters needed to calculate the number of merger events between SMBH binaries from the galaxy merger rate, available in the literature [358, 359, 360, 361, 212]. As detailed in [361], the differential galaxy merger rate per unit redshift, mass and mass ratio, can be written as,

$$\frac{d^3 n_G}{dz dM_G dq} = \frac{\phi(M_G, z)}{M_G \log 10} \frac{\Upsilon(M_G, z, q)}{\tau(M_G, z, q)} \frac{dt}{dz}, \quad (\text{B.8})$$

where $\phi(M_G, z)$ is galaxy stellar mass function (GSMF), $\Upsilon(M_G, z, q)$ is the differential pair function and $\tau(M_G, z, q)$ is the merger timescale. Here M_G is the primary galaxy mass, q is the mass ratio of galaxy pair and z is the redshift of the galaxy pair. This galaxy merger rate can be converted into the SMBH binary merger rate through

$$\frac{dn_s}{dz dX} \equiv \frac{d^3 n}{dz dM_{BH} dq_{BH}} = \frac{d^3 n_G}{dz dM_G dq} \frac{dM_G}{dM_{BH}} \frac{dq}{dq_{BH}}. \quad (\text{B.9})$$

The galaxy stellar masses needs to be translated into SMBH masses. The total mass of a galaxy M_G can be converted into its bulge mass M_{bulge} using phenomenological fitting function [362, 359],

$$\frac{M_{bulge}}{M_G} = \begin{cases} \frac{\sqrt{6.9}}{(\log M_G - 10)^{1.5}} \exp\left(\frac{-3.45}{\log M_G - 10}\right) + 0.615, & \text{if } \log M_G > 10, \\ 0.615, & \text{if } \log M_G \leq 10. \end{cases} \quad (\text{B.10})$$

Now according to [300], the scaling between the galaxy bulge mass M_{bulge} and black hole mass M_{BH} is of the form

$$\log_{10} \left(\frac{M_{BH}}{M_\odot} \right) = \mu + \alpha_* \log_{10} \left(\frac{M_{bulge}}{10^{11} M_\odot} \right) + \mathcal{N}(0, \epsilon). \quad (\text{B.11})$$

Here $\mathcal{N}$ is the normal distribution with mean 0 and standard deviation ϵ , to translate galaxy mass into BH mass. The black hole mass ratio q_{BH} is related to galaxy mass ratio q through $q_{BH} = q^{\alpha_*}$. The galaxy stellar mass function is written as

$$\phi(M_G, z) = 10^{\phi_0 + \phi_I z} \left(\frac{M_G}{M_0} \right)^{1 + \alpha_0 + \alpha_I z} \exp \left(\frac{-M_G}{M_0} \right) \ln 10. \quad (\text{B.12})$$

The five parameters $\phi_0, \phi_I, M_0, \alpha_0, \alpha_I$ are sufficient to describe the GSMF. The differential galaxy pair function is given by,

$$\Upsilon(M_G M_G, z, q) = f'_0 \left(\frac{M_G}{10^{11} M_\odot} \right)^{\alpha_f} (1 + z)^{\beta_f} q^{\gamma_f}. \quad (\text{B.13})$$

The quantity f_0 is related to f'_0 through $f_0 = f'_0 \int q^{\gamma_f} dq$. So there are four model parameters $f'_0, \alpha_f, \beta_f, \gamma_f$. The merger timescale is written as,

$$\tau(M_G, z, q) = \tau_0 \left(\frac{h_0 \times M_G}{0.4 \times 10^{11} M_\odot} \right)^{\alpha_\tau} (1 + z)^{\beta_\tau} q^{\gamma_\tau}. \quad (\text{B.14})$$

Here, $\tau_0, \alpha_\tau, \beta_\tau, \gamma_\tau$ are another four model parameters. The quantity $\frac{dt}{dz}$ is given assuming a flat Lambda CDM model as,

$$\frac{dt}{dz} = \frac{1}{H_0(1+z)\sqrt{\Omega_M(1+z)^3 + \Omega_k(1+z)^2 + \Omega_\Lambda}}. \quad (\text{B.15})$$

Here, Hubble parameter is $h_0 = 0.7$ and Hubble constant $H_0 = 70 \text{ km Mpc}^{-1} \text{s}^{-1}$ and energy density ratios $\Omega_M = 0.3, \Omega_k = 0$, and $\Omega_\Lambda = 0.7$.

Summarizing, the SMBH binary merger rate is described using 18 empirical parameters that are estimated through astrophysical observations (See [361] for details). The SGWB signal is most sensitive for primary blackhole masses in between $10^8 M_\odot$ and $10^9 M_\odot$ and redshift parameter $z \lesssim 0.3$. The SMBH binary merger rate is typically estimated using the Holodeck library [204].